I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL

A Capstone Project Presented to the Faculty, College of Computer Studies
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In partial fulfillment of the requirements for the degree
Bachelor of Science in Information Technology
Specialized in Web and Mobile Application Development

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The Capstone Project entitled "**I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL"**, prepared and submitted by **Sean D. Pacanan, Vince Ardrei Pecayo,** and **Del Johnver C. Pascual** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development, is hereby recommended for approval and acceptance.

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To God be the glory!

S. D. P
DJ. C. P
VA. P

ABSTRACT

Monitoring student attendance is essential for academic achievement, but many schools still depend on manual methods that are time-consuming, error-prone, and susceptible to proxy attendance. At Cabuyao Integrated National High School, attendance is traditionally logged through roll calls and written records, which can cause delays and hinder accurate record-keeping or timely parent notification. To improve this, the study introduced I AM PRESENT, a web-based attendance system that uses group-shot facial recognition to automatically track attendance. Teachers take a single group photo during class, and faces are detected and identified using MTCNN and FaceNet. Attendance is confirmed in real-time and stored securely in the cloud. The system also automatically emails parents or guardians to notify them of their child's attendance. Additionally, reports are easily accessible and exportable, streamlining record management for teachers and administrators. Implemented and tested at Cabuyao Integrated National High School, the system was evaluated according to ISO/IEC 25010 standards and the Technology Acceptance Model (TAM). Results indicated that it is reliable, user-friendly, and effective in enhancing attendance monitoring. Users appreciated its ability to reduce manual effort, cut errors, and improve communication between the school and families. Overall, this study shows that a web-based facial recognition attendance system offers a practical and efficient solution for secondary schools to improve attendance management.

Keywords: *Facial recognition, Student attendance monitoring, Group shot-based recognition, MTCNN, FaceNet, Web-based system*

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	v
Abstract	vi
Table of Contents	vii
List of Figures	ix
List of Tables	x
Definition of Terms	xii
CHAPTER	
I INTRODUCTION	
Project Context	1
Project Objectives	3
Project Purpose	3
Scope and Limitations	4
Conceptual Framework	6
II REVIEW OF RELATED LITERATURE AND STUDIES	
Related Literature	8
Synthesis and Relevance of the Study	11
Literature Matrix	13
III METHODOLOGY	
Project Design	14
System Architecture	16
Block Diagram	17
Project Locale	18
Model Development	19
Software Development	24
Materials	24
Software	25
Project Development	29
Agile Scrum Methodology	30
Scope	31
Design	32
Product Backlogs	33
Sprint Planning for Teacher	35
Sprint Planning for Admin	37
Daily Scrum	38
Sprint Execution	39
Sprint Review Meeting	39
Sprint Retrospective	40
Project Testing and Evaluation	40
Software Testing	41

Unit Testing 42
Integration Testing 44
System Testing 44
Acceptance Testing 45
System Flowchart 46
Use Case Diagram 52
Evaluation Procedure 53
Population of the Study 54

IV RESULTS AND DISCUSSIONS

Attendance Monitoring and Email Notification 56
Group Facial Attendance 57
Generated Report on Attendance 58
Absent Notification 58
Test and Evaluate the Software using ISO 25010 59
Results of Survey Answered by the Respondents 72
Result of Software Testing 80
Integration Testing Result 82
Load Testing Result 83
Compatibility Testing Result 84
Result of Stress Test 85

V SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary 86
Conclusion 87
Recommendations 89

REFERENCES

APPENDICES

Appendix A. Output Screenshot
Appendix B. Figures
Appendix C. Relevant Source Code
Appendix D. Communications Letters and Approvals
Appendix E. Plagiarism Results
Appendix F. Documentation
Appendix G. Publishable Paper
Appendix H. Curriculum Vitae

LIST OF FIGURES

No.	Title	Page
1	Conceptual Framework	7
2	System Architecture	16
3	Block Diagram	17
4	Project Locale	18
5	MTCNN Model	21
6	Agile Scrum Methodology	30
7	Levels of Testing	42
8	Unit Testing	43
9	Integration Testing	44
10	System Testing	45
11	User Acceptance Testing	46
12	System Flowchart of Teacher	48
13	System Flowchart of Admin	49
14	System Flowchart of Super Admin	51
15	Used Case Diagram	52
16	Quality Characteristics of ISO 25010 Standard	53
17	Attendance Monitoring and Email Notification	56
18	Group Facial Attendance	57
19	Generated Report of Attendance	58
20	Absent Notification	59

LIST OF TABLES

No.	Title	Page
1	Manual vs. Automated Attendance Monitoring Method	12
2	Literature matrix	13
3	MTCNN Model Parameters	23
4	Hardware Requirements Used for Laptop/Computer	25
5	Product Backlog for Admin, Teachers, Super Admin	34
6	Sprint planning for teacher	36
7	Sprint Planning for Admin Features	37
8	Sprint Planning for Super Admin Features	38
9	Respondents of the Research Project	54
10	Functional Suitability based on IT Specialist Response	60
11	Performance Efficiency based on IT Specialist Response	61
12	Compatibility based on IT Specialist Response	62
13	Interaction Capability based on IT Specialist Response	64
14	Reliability based on IT Specialist Response	65
15	Security based on IT Specialist Response	67
16	Maintainability based on IT Specialist Response	68
17	Flexibility based on IT Specialist Response	69
18	Safety based on IT Specialist Response	70
19	Result of ISO 25010 based on IT Specialist Response	71
20	Perceived Usefulness based on Respondents' Response	72
21	Perceived Ease of Use based on Respondents' Response	73
22	Personal Innovation based on Respondents' Response	74
23	Social Influence based on Respondents' Response	75
24	Information Awareness based on Respondents' Response	76
25	User Satisfaction based on Respondents' Response	77
26	Willingness to Adopt based on Respondents' Response	78
27	Result of Technology Acceptance Model based on Response	79

28	Unit Testing for Web-Based Module	80
29	Integration Testing for Web-Based Module	81
30	Load Testing Result for Web-Based Module	82
31	Compatibility Testing Result in each Web-Browser Platform	83
32	Stress Testing Result	84

DEFINITION OF TERMS

Attendance Monitoring Dashboard	The process of recording student presence in class using facial recognition technology.
CSV Export	A function that allows attendance records to be saved in a spreadsheet-compatible file format.
FaceNet	A deep learning model that converts detected faces into numerical embeddings for accurate identity matching.
Facial Recognition	A biometric technology that identifies students by analyzing and matching their facial features from captured images.
Group Shot-Based Facial Recognition	A method that detects and recognizes multiple student faces from a single group photo to record attendance at the same time.
ISO 25010	ISO 25010 is an international standard that defines software quality in terms of functionality, usability, and reliability. It was used in this project to assess the quality of the developed system.
Logs	Time-stamped records that store system activities and attendance transactions.
MTCNN	A deep learning face detection algorithm is used to locate and align faces in images before recognition.
Real-Time Recognition	The system's ability to detect and identify student faces instantly, without delay.
Technology Acceptance Model (TAM)	A model used to measure user acceptance based on perceived usefulness and ease of use.

CHAPTER 1

INTRODUCTION

Project Context

Face detection is a critical area of research in computer vision and artificial intelligence (AI). It involves identifying and localizing human faces in digital images or video streams. This technology has numerous applications, ranging from security and surveillance systems to user interaction in mobile devices and social media platforms. The evolution of face detection techniques has been marked by significant advancements, particularly with the advent of deep learning methodologies.

One of the foremost challenges in face detection is the variability in facial appearances due to factors such as age, gender, ethnicity, and individual facial features. According to Zhang et al. (2018), the diversity in facial structures and expressions can lead to significant discrepancies in detection accuracy. The authors emphasize that algorithms must be robust enough to handle this variability to ensure reliable performance across different demographics. Hence, in the late 1990s and early 2000s, there was a paradigm shift with the introduction of machine learning approaches. Viola and Jones (2001) revolutionized face detection with their cascade classifier, which utilized Haar-like features and boosted classifiers. This method significantly improved the speed and accuracy of face detection, paving the way for real-time applications.

In addition, face detection for attendance systems is a modern technology that automatically records attendance by scanning and recognizing a person’s face, making the process faster and more efficient. One significant advantage is accuracy, as it reduces the risk of cheating, such as proxy

attendance, and eliminates the need for manual sign-ins. It also improves security by ensuring that only authorized individuals can enter certain areas.

According to DepEd, there is a challenge in monitoring student attendance, as some students skip classes without permission, affecting their academic performance. A significant concern is that parents are often unaware of their child's whereabouts after school, increasing the risk of safety issues and truancy. Without an efficient tracking system, it becomes difficult for schools to ensure that students attend their classes regularly and go home safely.

To address this issue, schools can implement stricter attendance monitoring, such as digital attendance systems or face recognition technology, to accurately track students in real-time. Additionally, improving communication between schools and parents through mobile apps or text alerts can help inform parents if their child is absent or has not yet left school. Strengthening school policies and collaboration with local authorities can also help prevent students from loitering or engaging in risky activities after school hours. Addressing these challenges requires joint efforts from DepEd, schools, parents, and the community to ensure student safety and academic success.

Student attendance plays a crucial role in academic success, yet many schools struggle with monitoring it effectively. At Cabuyao Integrated National High School, traditional attendance methods, such as manual roll calls and sign-in sheets, are time-consuming and error-prone, leading to inaccurate records and potential cases of truancy. Additionally, some students skip classes without permission, making it difficult for teachers and parents to track their whereabouts. To address these challenges, there is a need for a more efficient and reliable attendance system. This study explores the implementation of a

group shot-based facial recognition system to optimize student attendance monitoring. By automating the process, this technology aims to improve accuracy, reduce time spent on attendance-taking, and enhance student accountability, ultimately ensuring a more secure and organized learning environment.

Project Objectives

The main objective of this capstone project is to develop a real-time group face detection for an attendance system that provides efficient, accurate, and secure student attendance monitoring at Cabuyao Integrated National High School. Specifically, it sought to achieve the following:

1. To integrate web-based development for accessible attendance monitoring and provide real-time notification to parents/guardians.
2. To implement an MTCNN–FaceNet–based group facial recognition system for accurate and efficient identification of students in group photos.
3. To generate and export attendance reports.
4. To evaluate the system based on ISO/IEC 25010 software quality standards.

Project Purpose

The purpose of this project is to create I AM PRESENT, a web-based application that helps make student attendance checking at Cabuyao Integrated National High School more convenient and reliable. Instead of the usual manual roll call, the system uses group-shot-based facial recognition to automatically identify students in a single group photo, allowing attendance to be recorded quickly and accurately.

The system is designed to help teachers save time and reduce errors in attendance monitoring while giving school administrators an easier way to manage records. It also sends real-time notifications to parents or guardians to keep them informed about their child's attendance. In addition, the application can generate attendance reports to support daily monitoring and school documentation. To ensure that the system works well and meets quality standards, it will be evaluated using the ISO/IEC 25010 software quality model, focusing on usability, performance, security, and overall reliability.

Scope and Limitations

This study focuses on the development of I AM PRESENT, a web-based attendance-monitoring system designed to improve the recording of student attendance at Cabuyao Integrated National High School. The system uses group shot-based facial recognition technology to make attendance checking faster, more accurate, and less disruptive to class time. It is intended for use by students, teachers, and school administrators within the school environment.

Login Authentication Module. This module manages user registration and login to ensure that only authorized users can access the system. Users are assigned roles as students, teachers, or administrators, which determine what features they can use. Security measures such as encrypted passwords, password recovery, and automatic logout after inactivity are included to protect user information and attendance records.

Dashboard Module. This serves as the primary interface, offering users a well-organized overview of attendance data. Upon login, teachers and administrators can access real-time attendance summaries, class records, and

notifications all in one view. Visual tools like tables and status icons enable quick monitoring of student attendance and detection of absences or anomalies. The dashboard also provides straightforward navigation to other system features, allowing users to manage attendance records efficiently and make prompt decisions based on current information.

Class Management Module. This module allows teachers and administrators to create and manage class information, including subjects, sections, and enrolled students. By organizing class data in one place, teachers can easily monitor attendance and track student participation. This feature supports accurate attendance recording through group facial recognition.

Student Registration Module. This module is responsible for registering students into the system by capturing their facial data for attendance recognition. It also collects basic student information and guardian contact details. These records help ensure accurate attendance detection while allowing the school to communicate with parents or guardians when necessary.

Facial Recognition Module. The facial recognition module is the core feature of the system. It detects and identifies students from group photos taken during class using pre-registered facial data. Once students are recognized, their attendance is automatically recorded in real time. This reduces manual effort, minimizes errors, and allows teachers to conduct attendance without interrupting lessons.

Notification System Module. This module automatically sends email notifications to parents or guardians once a student's attendance is confirmed.

The real-time alerts help keep families informed and strengthen communication between the school and the home.

Reporting and Export Module. The system provides organized attendance records that show student presence and absence by date. Teachers and administrators can export these records in CSV format for documentation, reporting, or backup purposes.

Settings & Configuration Module. This module allows users to view and update their profile information, including personal and contact details. Administrators have additional privileges to manage user accounts, including updating or removing profiles when needed.

Conceptual Framework

The conceptual framework of the facial recognition–based attendance monitoring system illustrates how attendance data are captured, processed, and presented to improve classroom attendance management. The process begins with the collection of student information, registered facial data, class details, and a group photo taken during class hours. These inputs are supported by facial detection and recognition models, system settings, and a centralized database, which work together to identify students and record attendance accurately.

This system allows teachers to monitor attendance automatically without disrupting class activities. By using facial recognition technology, the need for manual checking is reduced, making attendance recording faster and more reliable while preventing common issues such as proxy attendance and recording errors. The system is flexible and scalable, allowing it to adapt to different class sizes and school settings. Attendance records are securely

stored and can be reviewed over time to help track student participation and attendance patterns.

Security measures ensure that only authorized users can access the system and student data, while regular updates help maintain accuracy and performance. The Processing Module manages key functions, including user authentication, image capture, face detection, and facial matching, followed by attendance logging and report generation. Through the Output Module, the system produces automated attendance records, sends real-time notifications to parents or guardians, and generates downloadable attendance reports. Overall, the framework presents a practical and efficient approach to attendance monitoring that supports accuracy, accountability, and a more organized learning environment.

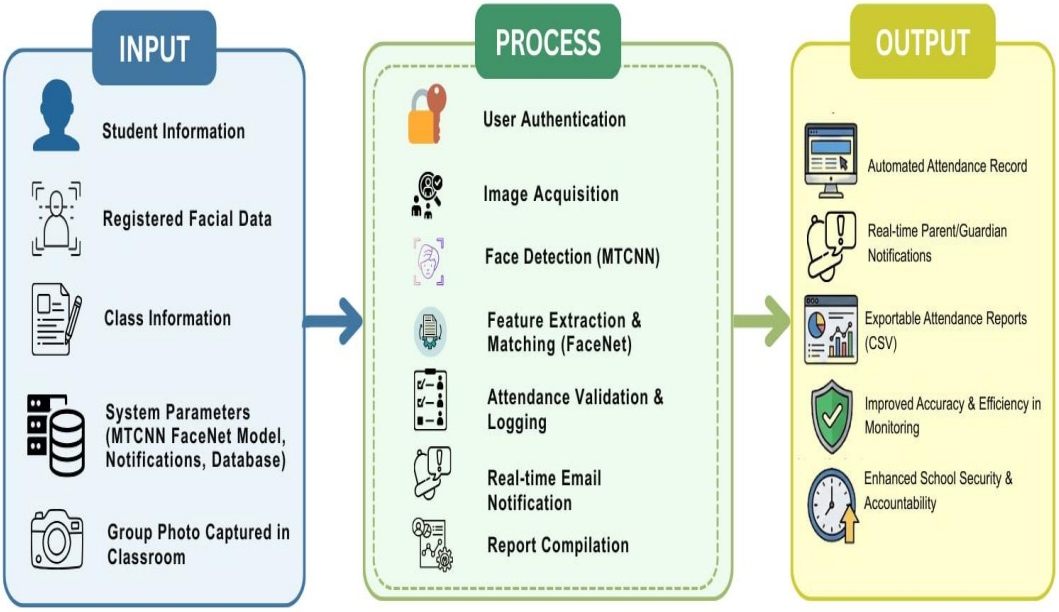


Figure 1. Conceptual Framework

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents a comprehensive review of related literature, studies, and systems relevant to the development of a facial recognition–based attendance monitoring system. Sources include local and foreign studies obtained from reliable materials such as published theses, academic journals, conference papers, online articles, and other credible references. This review serves as a guide for researchers to understand existing concepts, technologies, and issues related to facial recognition and attendance monitoring. Furthermore, it examines prior work that supports the significance and direction of the present study.

Facial Recognition Technology

The journey of face recognition began in the 1960s with the development of early algorithms that relied on geometric features of the face (Kanade, 1973). Initial systems were rudimentary and often struggled with variations in lighting, pose, and facial expressions. However, the advent of machine learning in the late 1990s marked a turning point, leading to more robust systems. Traditional face recognition methods include Eigenfaces (Turk & Pentland, 1991) and Fisher faces (Belhumeur et al., 1997). These techniques rely on linear algebra and statistical methods to represent and classify facial features. While effective in controlled environments, they demonstrated limitations under varying conditions. On the other hand, the introduction of deep learning has revolutionized face recognition. Convolutional Neural Networks (CNNs) have

become the backbone of modern face recognition systems. Notable architectures such as VGGFace (Parkhi et al., 2015) and FaceNet (Schroff et al., 2015) leverage deep learning to achieve remarkable accuracy. These models learn hierarchical representations of facial features, enabling them to generalize better across diverse datasets.

Applications of Facial Recognition

Facial recognition technology is widely used across different sectors. In security and surveillance, it is employed for identity verification, access control, and monitoring of restricted areas (Liu et al., 2019). Social media platforms utilize facial recognition to automatically identify and tag individuals in images (Xu et al., 2018). In the retail industry, facial recognition supports customer analysis and personalized marketing strategies (Zhao et al., 2020).

In the educational sector, facial recognition has increasingly been adopted for campus security and attendance monitoring. Several universities in the Philippines have integrated facial recognition systems to enhance security and improve administrative efficiency. San Beda University implemented facial recognition for controlled campus access and automated attendance tracking. Similarly, De La Salle University and Ateneo de Manila University have piloted facial recognition systems to streamline student verification and monitoring processes (Latonero, 2024). These implementations demonstrate the growing acceptance of facial recognition technology in academic institutions.

Advancements in Facial Recognition

Recent advancements in facial recognition have focused on improving accuracy and robustness. Margin-based loss functions, such as ArcFace,

enhance class separation and improve recognition performance (Deng et al., 2019). Generative Adversarial Networks (GANs) have also been introduced to generate synthetic facial images, expanding training datasets and improving model generalization (Karras et al., 2020). Moreover, transformer-based models, initially designed for natural language processing, are now being explored in computer vision tasks, including facial recognition. Vision Transformers have shown promising results in feature representation and recognition accuracy (Dosovitskiy et al., 2021), suggesting new directions for future research.

Challenges in Facial Recognition Systems

Despite technological advancements, facial recognition systems still face several challenges. Variations in lighting conditions, facial expressions, pose, and occlusions can significantly affect recognition accuracy (Masi et al., 2016). Ethical and privacy concerns have also emerged, notably regarding consent, surveillance, and data protection (Binns, 2018).

Studies have revealed biases in facial recognition systems, with higher error rates observed among certain racial and gender groups (Buolamwini & Gebru, 2018). In addition, facial recognition systems are vulnerable to adversarial attacks, where manipulated images are used to deceive recognition models (Goodfellow et al., 2015). These issues highlight the importance of fairness, security, and ethical considerations in system design and deployment.

Future Directions in Facial Recognition Research

Future research in facial recognition aims to improve system fairness, security, and efficiency. Hybrid architectures combining CNNs and

Transformers are being explored to enhance recognition performance while reducing computational costs (Dosovitskiy et al., 2020). Privacy-preserving machine learning techniques are also gaining attention for protecting biometric data and addressing security concerns.

Another emerging direction involves the development of real-time and low-power facial recognition systems for mobile and embedded devices. These advancements are expected to support wider adoption while maintaining accuracy and ethical standards.

Synthesis and Relevance of the Study

The reviewed literature shows that facial recognition technology has progressed from early geometric-based methods to advanced deep learning models such as FaceNet and ArcFace, significantly improving recognition accuracy and reliability. Its application in education, particularly in attendance monitoring and campus security, demonstrates its practical value in improving efficiency and reducing manual processes. However, challenges related to accuracy, security, bias, and privacy remain significant concerns. Addressing these issues is essential for the responsible implementation of facial recognition systems. This study builds on existing research by applying facial recognition technology to an attendance-monitoring system while considering accuracy, security, and ethical implications. By doing so, the study contributes to the development of a reliable and efficient attendance monitoring solution suitable for educational institutions.

Comparison of Manual vs. Automated Attendance Monitoring Methods

The integration of facial recognition technology has improved traditional classroom attendance monitoring. Automated systems such as I AM PRESENT use group-shot facial recognition with MTCNN to quickly and accurately record attendance, reducing manual effort and human error. In contrast, traditional methods like roll calls and sign-in sheets are time-consuming and prone to inaccuracies, including proxy attendance. Automated attendance systems provide real-time records, improved security, and better data management, supporting more efficient classroom management.

Table 1. Comparison of Manual vs. Automated Attendance Monitoring Methods

Criteria	Manual Process	Automated Process Value
Attendance Method	Roll call or manual sign-in sheet	Group-shot facial recognition using MTCNN–FaceNet
Average Class Size	40 students	40 students
Attendance Capture	One student at a time	Multiple students captured simultaneously
Time per Student	15–30 seconds	8-10 seconds
Total Time per Class	5–7 minutes	1–2 minutes
Time Efficiency	Low	High (saves ~3–5 minutes)
Accuracy Level	Moderate; prone to proxy attendance	High; minimizes errors and proxies
Human Intervention	High	Low
Classroom Disruption	Noticeable	Minimal
Data Storage	Paper-based or manual encoding	Digital with real-time updates

Literature Matrix

Table 2. Literature Matrix

Author(s) & Year	Title	Purpose	Methodology	Key Findings	Relevance to Capstone
Koc, M. (2019)	*Technology Facial Recognition for Attendance Management in Schools*	To examine the effectiveness of facial recognition technology in improving attendance management in educational institutions.	Analyzed existing attendance systems and evaluated the application of facial recognition technology in school environments.	Facial recognition significantly improves attendance accuracy and efficiency compared to traditional manual methods, which are prone to errors and time consumption.	Supports the capstone by highlighting how facial recognition can improve attendance accuracy and reduce human error in school settings.
Patel, S., & Patel, R. (2021)	*Time-Saving Benefits of Facial Recognition in Education*	To evaluate how facial recognition systems reduce time spent on attendance checking in classrooms.	Implemented a facial recognition-based attendance system that captures group images and automatically records attendance.	Attendance can be recorded within seconds, eliminating roll calls and manual checks, allowing more class time for instruction.	Reinforces the capstone's goal of creating a fast and efficient attendance monitoring system using facial recognition.
Smith, J., et al. (2019)	*Ethical Concerns in Implementing Facial Recognition in Schools*	To analyze privacy and ethical issues associated with the use of facial recognition in educational institutions.	Conducted a review of ethical policies, privacy laws, and case studies related to facial recognition usage in schools.	The study highlights concern regarding data privacy, consent, and security when storing facial data of students.	Guides the capstone in addressing privacy, data protection, and ethical considerations in system design.
Zhao, W., & Li, Z. (2020)	*Enhancing School Attendance with Facial Recognition: A Case Study*	To assess the impact of facial recognition on attendance accuracy and school security.	Conducted a case study on the implementation of an automated facial recognition attendance system in a school.	Automated attendance reduced fraudulent check-ins and increased accountability and security within the school environment.	Aligns with the capstone's objective of improving attendance authenticity and preventing proxy attendance.
Huang, R., & Zhou, L. (2020)	*Integrating Facial Recognition with School Management Systems: A New Approach to Attendance Tracking*	To explore the integration of facial recognition with existing school management systems.	Designed and tested a system that integrates facial recognition attendance data with academic and administrative databases.	Automated attendance updates improved monitoring of student participation and enhanced overall school management efficiency.	Supports the capstone by demonstrating the benefits of integrating facial recognition with centralized attendance and reporting systems.

CHAPTER 3

METHODOLOGY

This study employed a quantitative research approach and the Agile software development methodology to develop I AM PRESENT: A Web-Based Application Optimizing Students' Attendance Monitoring Through Group Shot-Based Facial Recognition at Cabuyao Integrated National High School. Agile was chosen to enable iterative development through sprints, enabling continuous testing and refinement of system features.

The system was developed as a web-based application featuring capabilities for capturing group photos, recognizing student faces, and automatically recording attendance. Quantitative data were gathered to assess the system's accuracy, response time, and reliability in identifying students during live class sessions. Testing took place in both controlled settings and actual classrooms to verify the application's effectiveness and its suitability for teachers and students.

Project Design

The proposed I AM PRESENT: Web-Based Application Optimizing Students' Attendance Monitoring Through Group Shot-Based Facial Recognition adopts a modular system architecture that integrates web-based interfaces, machine learning, cloud storage, and automated notifications to optimize student attendance monitoring. Figure 2 illustrates the overall system design, which comprises the End Users Layer, Front-End Interface, Application Server, Machine Learning Processing Component, Supabase Cloud Database, and Email Notification Server.

The End Users Layer includes teachers, registrars, and IT administrators, each interacting with the system according to their role. This ensures controlled access to specific functionalities, such as attendance recording, student management, and system configuration.

The Front-End Interface, built with ReactJS and Tailwind CSS, serves as the primary interaction point. It provides modules for login/authentication, class management, student registration, attendance capture, and dashboard/report generation, enabling users to perform their tasks efficiently.

The Application Server acts as the central processing hub. It manages the Facial Recognition API, Attendance Logging Controller, User Authentication Handler, CSV Export Module, and Notification Module, ensuring that user actions and machine learning predictions are processed accurately and securely.

The Machine Learning Processing Component is the core of the system, employing MTCNN and FaceNet to detect and recognize faces from group images. Recognized students are automatically logged in the attendance records, streamlining the process and reducing human error.

The Supabase Cloud Database securely stores all essential data, including users, students, classes, attendance records, guardian information, and facial embeddings, providing real-time accessibility and consistency.

The Email Notification Server (Gmail API) automatically sends alerts to parents and guardians about student attendance, supporting timely communication and engagement.

The system architecture integrates user interaction, data processing, machine learning, and cloud-based storage into a cohesive framework. Each component works in concert to streamline attendance monitoring, improve administrative efficiency, and support informed decision-making in educational institutions. The design emphasizes reliability, real-time performance, and automation to reduce administrative workload while ensuring accurate tracking of student attendance.

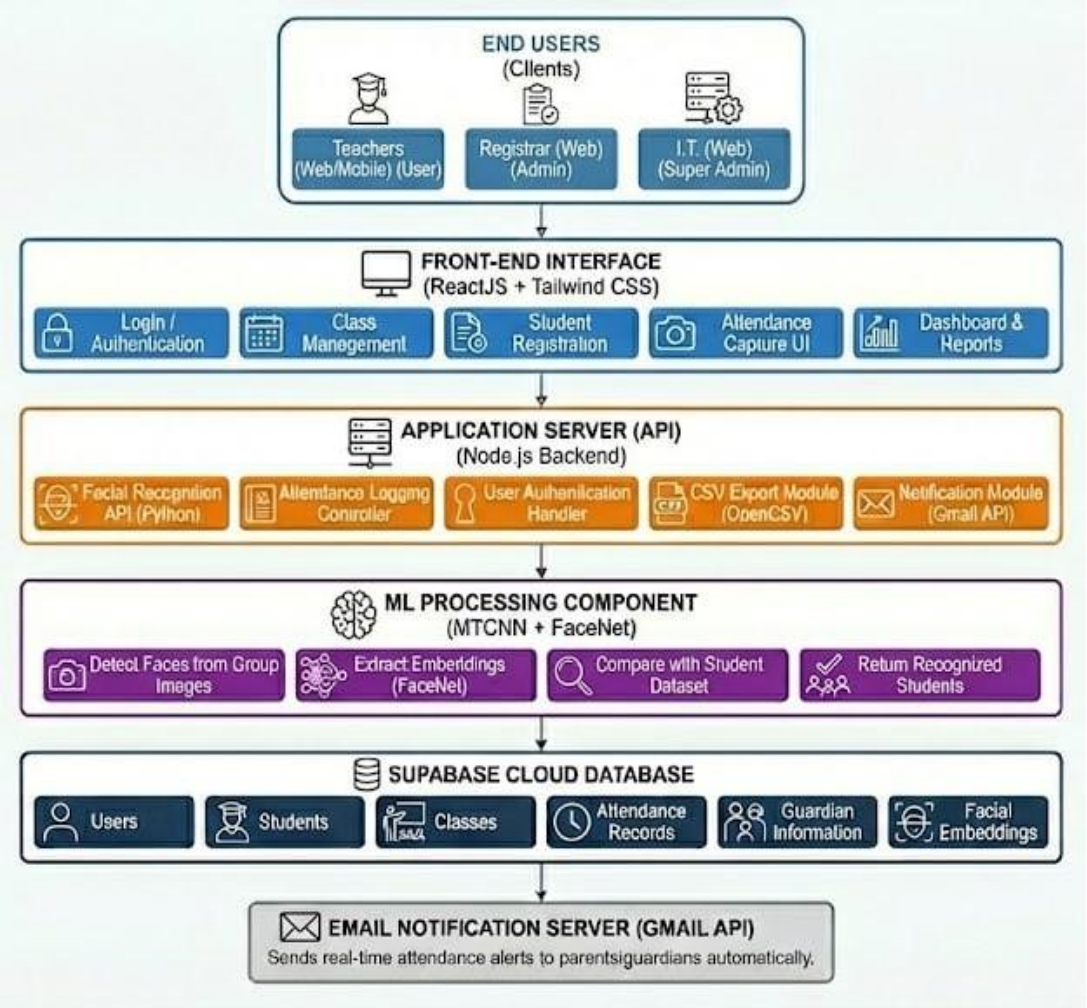


Figure 2. System Architecture

Block Diagram

Figure 3 illustrates the overall architecture of the proposed web-based attendance monitoring system using group-shot-based facial recognition. The process begins when the teacher accesses the web-based application to upload or capture a group photo of students during class sessions. The system then forwards the image to the group facial recognition module, which uses the MTCNN algorithm for face detection and the FaceNet model for facial feature extraction and student identification.

Once students are successfully recognized, the attendance processing module automatically marks them as present and records the date and time. The processed attendance data, along with student information and facial data, are stored in a centralized database. The reporting and notification module retrieves the stored records to generate attendance reports and send attendance notifications to parents or guardians. This architecture ensures efficient, accurate, and secure attendance monitoring within the school environment.

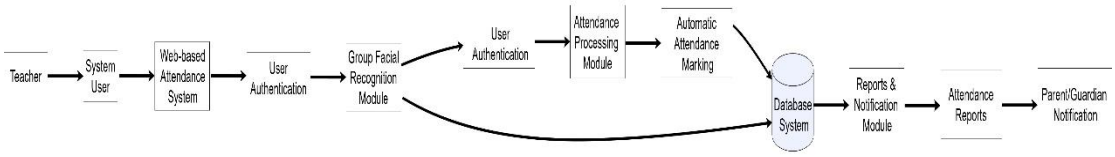


Figure 3. Block Diagram

Project Locale

This study was conducted at Cabuyao Integrated National High School in Cabuyao City, Laguna, which serves a diverse student community and is equipped with classrooms, laboratories, and administrative facilities that support both academic and extracurricular activities. Attendance is currently recorded manually, a process that is time-consuming, error-prone, and often delays communication with parents or guardians. To address these challenges, the school implemented the I AM PRESENT: Web-Based Attendance Monitoring System Through Group Shot-Based Facial Recognition. This system not only improves accuracy and streamlines record-keeping but also provides real-time updates and strengthens communication with parents, helping create a more efficient, organized, and student-centered learning environment.



Figure 4. Cabuyao Integrated National High School

Model Development

The figure 5 presents the structured workflow used to develop the MTCNN–FaceNet-based facial recognition model for the web-based attendance monitoring system. This model serves as the core component for accurate, real-time identification of students from group photos, enabling efficient attendance tracking at Cabuyao Integrated National High School. The model development process consists of several sequential stages, each ensuring that raw image data is appropriately processed, analyzed, and converted into actionable attendance records. The stages are explained as follows:

Data Collection and Preparation. The first stage involves gathering facial image data of enrolled students. Images are captured in various lighting conditions, angles, and expressions to improve recognition robustness and minimize bias. To ensure the dataset is representative, multiple photos are collected per student, including different facial expressions, hairstyles, and accessories such as glasses or masks. The collected dataset undergoes preprocessing steps, including resizing to a standard resolution, pixel normalization, histogram equalization for lighting correction, and facial-landmark-based alignment. This standardization ensures that the MTCNN–FaceNet pipeline receives high-quality, consistent inputs, reducing recognition errors and enhancing the accuracy of subsequent model stages. Additionally, data augmentation techniques, such as rotation and flipping, are applied to artificially expand the dataset and improve the model's generalization.

Face Detection with MTCNN. After data preparation, the system applies the Multi-task Cascaded Convolutional Neural Network (MTCNN) to detect faces in group photos. MTCNN efficiently identifies multiple faces within a single image by generating bounding boxes for each detected face. This stage also includes facial landmark localization to align faces for better feature extraction. Accurate face detection is critical for ensuring that each student is correctly isolated before recognition.

Feature Extraction with FaceNet. After preprocessing, the system applies the Multi-task Cascaded Convolutional Neural Network (MTCNN) to detect faces within group photos. MTCNN performs simultaneous face detection and facial landmark localization, enabling it to efficiently identify multiple faces in a single image. Detected faces are enclosed in bounding boxes and aligned using the key landmarks (eyes, nose, mouth) to ensure uniform positioning for the recognition stage. Accurate face detection is critical for minimizing false positives or missed faces, especially in crowded group images. To further enhance reliability, the system implements thresholding mechanisms to filter low-confidence detections, ensuring only high-quality faces are forwarded for feature extraction.

Student Identification. The aligned faces are processed using the FaceNet model, which converts each face into a high-dimensional embedding vector that uniquely represents the individual's facial characteristics. FaceNet embeddings capture subtle variations in facial features, enabling the system to distinguish between students with similar appearances. The generated embeddings are securely stored in the database, forming a reference library for future attendance verification. This stage also includes embedding

normalization to maintain consistency in distance calculations. The feature extraction process ensures high accuracy even in scenarios involving partial occlusions, different angles, or lighting conditions.

Attendance Marking. Students identified as present are automatically marked as present in the attendance system. Each recognized student is associated with the corresponding class and timestamp, creating a secure and accurate attendance record. This automation eliminates manual record-keeping errors and accelerates attendance monitoring during class.

System Integration and Reporting. The final stage integrates the MTCNN–FaceNet model with the web-based application. Recognized attendance data is stored in the database and used by the reporting module to generate attendance summaries and notifications for parents or guardians. This integration ensures that facial recognition directly supports real-time attendance tracking, reporting, and parental notifications.

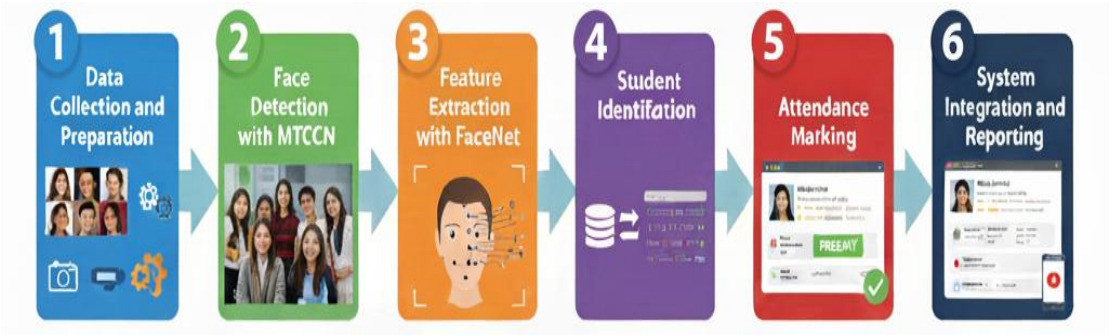


Figure 5. MTCNN Model

MTCNN Model and Dataset Description

For this study, a pre-trained MTCNN model was applied using a dataset of student facial images collected at Cabuyao Integrated National High School. The dataset includes multiple images per student, captured under various

lighting conditions, angles, and facial expressions, ensuring robust recognition. Images were preprocessed through resizing, normalization, and facial alignment to standardize inputs for the MTCNN–FaceNet pipeline. By using this curated dataset, the model can accurately detect and extract facial features from group photos, even in complex classroom scenarios. The pre-collected dataset allows the MTCNN model to detect multiple faces simultaneously and provide reliable input for downstream FaceNet-based identification.

Why MTCNN Was Used for Facial Detection

The system uses MTCNN for facial detection because it is highly efficient and accurate at identifying multiple faces in a single image. Unlike traditional methods, MTCNN uses a deep learning architecture optimized for real-time face detection and landmark localization. This makes it ideal for classroom settings where students may be at different angles or partially hidden. Compared to other detection methods, such as Haar cascades and SSD-based detectors, MTCNN provides higher accuracy, better handling of multiple faces in group photos, and integrates smoothly with FaceNet for feature extraction. These qualities make MTCNN especially suitable for a web-based, real-time attendance monitoring system.

MTCNN Model Parameters

The MTCNN facial detection model uses three cascaded networks, P-Net, R-Net, and O-Net, to detect faces and facial landmarks with high accuracy. Its parameters were configured to achieve efficient real-time performance for classroom group photos. A high confidence threshold ensures only likely face regions are detected, while an IoU threshold filters overlapping bounding boxes

to prevent duplicate detections. The minimum face size was set to capture students seated at the back. Each network stage refines bounding boxes and detects five facial landmarks: eyes, nose, and mouth corners, to align faces for FaceNet embedding extraction, improving recognition under varying poses and lighting. A pre-trained MTCNN model was used without additional training and applied to over 2,000 cleaned and preprocessed student images captured at different angles, under different lighting conditions, and with varying expressions. These configurations enable accurate and efficient real-time attendance monitoring.

Table 3. MTCNN Model Parameters

Parameter	Description	Assigned Value	Remarks
Confidence Threshold	Minimum probability for a candidate region to be considered a face	0.95	Ensures only high-confidence faces are detected, reducing false positives
IoU (Intersection-over-Union) Threshold	Threshold to filter overlapping bounding boxes	0.7	Prevents multiple detections of the same face
Minimum Face Size	Smallest detectable face in pixels	Pre-defined by MTCNN	Aligns faces for accurate feature extraction in FaceNet.
Pre-trained Model Usage	Utilization of a previously trained MTCNN network	yes	No additional on-site training required; generalizes well to classroom faces
Training Dataset Size	Number of student images used for testing and reference	2000+ images	Images captured under varying angles, lighting, and expressions; preprocessed for consistency

Software Development

The development of the software component for the project “I AM PRESENT: A Web-Based Application Optimizing Students’ Attendance Monitoring Through Group Shot-Based Facial Recognition at Cabuyao Integrated National High School” involved several key phases, including planning, designing, coding, testing, and refinement. The system was designed to automate student attendance by capturing group photos, detecting faces using the MTCNN model, identifying students via FaceNet embeddings, and generating real-time attendance reports accessible to teachers and parents.

To guide the development process, the Evolutionary Prototyping Model was utilized. This approach allowed the creation of an initial web-based prototype, which was repeatedly refined through feedback from teachers, school administrators, and pilot testing. Each iteration integrated enhancements to the user interface, improved face detection accuracy, and optimized reporting features, resulting in a robust and reliable system.

This methodology allowed the development team to quickly adapt to challenges encountered during implementation, such as varying lighting conditions, partially occluded faces, or network delays. With each cycle, the software was optimized not only to ensure accurate attendance monitoring but also to improve system responsiveness, real-time notifications, and database management.

Materials

This section outlines the hardware and software tools used in the project, highlighting the essential components required for both the development

environment and system deployment. These materials were carefully selected to meet the functional and performance requirements of the web-based attendance monitoring system.

Table 4. Hardware Requirements Used for Laptop/Computer

	Requirements Used
RAM	16GB
HDD/SSD	Samsung SSD 970 EVO Plus
CPU	500GB
GPU	AMD Ryzen 5 5500
CLOCKSPEED	NVIDIA GeForce RTX 2080 3.60Ghz

The hardware specifications provided ensure smooth development and execution of the web-based system. A minimum of 16GB of RAM enables efficient handling of multiple processes, including real-time face detection and data processing. The Samsung SSD 970 EVO Plus 500GB provides fast read/write speeds, making it well-suited for large image datasets. The AMD Ryzen 5 5500 processor and NVIDIA GeForce RTX 2080 GPU ensure high-performance computing for model inference, while a clock speed of 3.60 GHz guarantees fast processing for real-time attendance marking.

Software

This section describes the software tools used in developing the web-based I AM PRESENT system, explaining the reasons for their selection and their key features. These tools are essential to the system's development, ensuring efficiency, reliability, and accuracy in the final product. They also enable seamless integration among the facial recognition models, web interface, and cloud database, supporting real-time attendance monitoring, notifications to parents, and accurate record-keeping for teachers and administrators.

ReactJS. It is the core JavaScript library used to build the system's user interface. React enables the creation of reusable components and a dynamic frontend, making the attendance dashboard responsive and interactive for teachers, students, and administrators. Its virtual DOM ensures efficient rendering of real-time attendance updates.

TypeScript. Provides type safety for JavaScript code, reducing runtime errors and improving code maintainability. TypeScript is used across all frontend components to prevent common coding mistakes, ensuring a more stable and reliable system.

Tailwind CSS. A utility-first CSS framework for designing the user interface. Tailwind enables rapid development of responsive, consistent layouts, ensuring the dashboard and student attendance pages are visually clear across all devices.

Vite. A modern build tool that powers both development and production builds of the system. Vite provides fast hot-module reloading and optimized production builds, improving development speed and overall system performance.

Lucide React. A library of SVG icons is integrated throughout the interface. Lucide React enhances user experience by providing clear visual cues for navigation, buttons, and notifications.

Supabase. A Backend-as-a-Service (BaaS) platform providing authentication, database, and storage services. Supabase manages user

accounts, class sessions, and attendance records, allowing secure, real-time data management without maintaining a dedicated server.

PostgreSQL. A relational database hosted via Supabase, storing structured data such as student profiles, class schedules, and attendance logs. PostgreSQL ensures reliable and consistent data storage and supports queries for reports and analytics.

Deno. A secure runtime environment for executing server-side Edge Functions. Deno handles backend operations, including processing attendance data, executing facial recognition comparisons, and managing API requests.

Hono. A lightweight web framework used for routing API requests and handling middleware within Deno Edge Functions. Hono simplifies the backend architecture while ensuring low-latency communication between the frontend and backend.

Supabase KV Store. A distributed key-value storage system optimized for high-performance data, including face embeddings and student metadata. KV Store allows rapid retrieval of information during real-time facial recognition.

TensorFlow.js. A machine learning library that runs directly in the browser. TensorFlow.js powers the facial recognition system, enabling on-device face detection and embedding generation without requiring server-side computations.

Face-api.js. A specialized library built on TensorFlow.js for face detection and recognition. It extracts facial features from group photos or

camera frames, compares them against stored embeddings, and accurately identifies students.

Media Devices API & Canvas API. Native browser APIs are used to access the device camera and process video frames for facial recognition. These APIs allow real-time group photo capture and preprocessing for attendance detection.

Gmail SMTP & Nodemailer. Provide email delivery functionality to send automated notifications to parents or guardians. They ensure that attendance updates are communicated promptly and professionally.

Figma. A web-based interface design platform used to create, prototype, and test the system's user interface. Figma allows developers and designers to maintain a consistent look and feel for the web application.

Git. A version control system is used to track changes, manage source code, and collaborate efficiently among developers. Git ensures that updates are integrated smoothly and prevents code conflicts.

Visual Studio Code. A lightweight code editor supporting multiple programming languages, debugging, and extensions. VS Code facilitates frontend and backend development, allowing live debugging, Git integration, and efficient project management.

Radix UI. provides accessible, unstyled components such as dialogs, menus, and tabs. Radix UI ensures that the interface is usable by all users and supports keyboard navigation and screen readers.

Recharts. A charting library used for visualizing attendance trends and statistics. Recharts generates interactive charts to help teachers and administrators quickly assess attendance patterns.

Sonner. A notification library that displays toast messages within the application. Sonner provides instant feedback on user actions, such as successful attendance marking and error alerts.

React Hook Form. Manages frontend form validation for student registration and attendance submission. It ensures data integrity and improves user experience by providing immediate error feedback.

Date-fns. A modern JavaScript library for date and time manipulation. date-fns is used to timestamp attendance records, generate weekly/monthly reports, and efficiently calculate summary statistics.

Project Development

The system titled "I AM PRESENT: A Web-Based Application Optimizing Students' Attendance Monitoring through Group Shot-Based Facial Recognition at Cabuyao Integrated National High School" will be developed using the Agile Scrum methodology. This iterative and flexible approach enables continuous feedback from teachers, students, and school administrators, ensuring the system meets the school's specific attendance-monitoring needs.

The development process will be organized into one- to two-week sprints, with regular stand-up meetings to track progress, discuss challenges, and resolve issues promptly. Key components, such as real-time group facial recognition

using MTCNN–FaceNet, a web-based attendance dashboard, and automated parent/guardian notifications, will be developed incrementally and tested continuously for performance, accuracy, and usability.

This approach allows the team to adapt quickly to end-user feedback, including teachers' requirements for class management and parents' expectations for timely notifications. Each sprint focuses on developing functional modules such as facial detection, feature extraction, student identification, and attendance reporting integrated into the central system.

By applying Agile Scrum, the project ensures rapid, adaptive development with a focus on efficiency, accuracy, and security in student attendance monitoring. The methodology also supports iterative refinement, allowing improvements to the facial recognition model, database management, and web interface to be implemented based on user feedback and testing results.

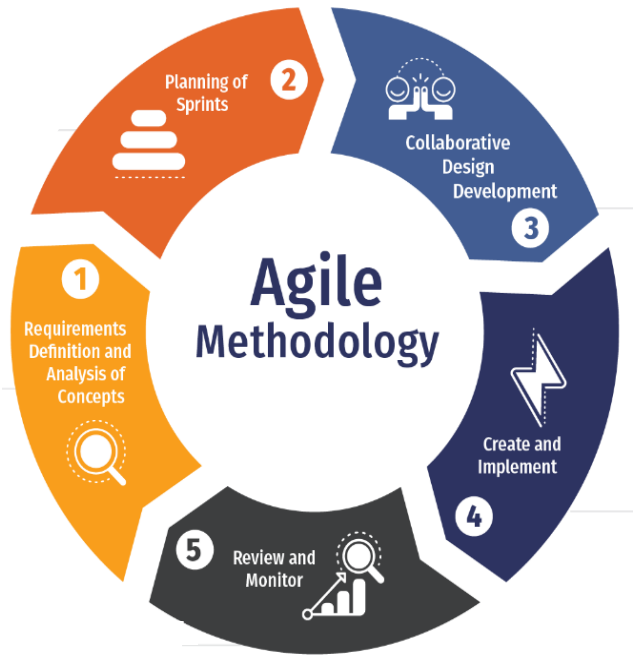


Figure 6. Agile Scrum Methodology

Scope

This study focuses on the development and implementation of a web-based attendance monitoring system that uses group-shot facial recognition for students at Cabuyao Integrated National High School. The system is intended for use by teachers, school personnel, and parents or guardians. It enables real-time attendance monitoring to improve the efficiency, accuracy, and security of recording student participation during class sessions. The system utilizes the MTCNN–FaceNet model to detect and recognize multiple students from group photos taken within a maximum distance of six (6) meters from the camera. This range is suitable for typical classroom settings and allows the system to maintain reliable recognition performance. To function effectively, the system requires images that meet basic picture quality standards. At a minimum, the captured images must clearly show students' faces with adequate lighting, proper focus, and sufficient resolution to allow accurate facial detection. The resolution must be 720 pixels and above. When higher-quality images are used such as those with better lighting, clearer focus, and higher resolution the system performs more accurately, especially when identifying multiple faces in a single image. On the other hand, images with poor lighting, low resolution, motion blur, or partially covered faces may reduce recognition accuracy, even when the distance requirement is met. After attendance is successfully recorded, the system automatically sends email notifications to parents or guardians, helping ensure transparency and timely communication. The school's ICT staff are responsible for maintaining the system, managing the database, and verifying facial recognition data. The overall performance of the system may be affected by environmental conditions such as lighting,

camera quality, student positioning, and distances beyond the six-meter limit. This study is limited to a web-based platform and a responsive system that can be used on mobile devices.

Design

The system follows a modular, user-centered design to ensure clear functionality and ease of use. It integrates several key modules that work together to facilitate attendance monitoring and reporting.

The Login Module provides secure authentication for teachers and administrators, ensuring that only authorized personnel can access sensitive student data. The Settings Module allows users to customize their preferences, such as notification settings and dashboard layout. At the core of the system is the Facial Recognition Module, which detects faces in group photos using MTCNN, extracts facial embeddings with FaceNet, and matches them against the student database for accurate identification. The Attendance Dashboard Module consolidates and displays real-time attendance records, providing teachers with an organized view of present and absent students.

The Notification Module automatically sends updates to parents or guardians about their child's attendance, helping maintain communication and accountability. Additionally, the Report Generation Module stores attendance logs and produces exportable summaries for administrative purposes.

This modular design offers a clean, user-friendly interface, efficient facial recognition, and strong data management capabilities. It supports accurate, real-time attendance tracking and detailed reports for Cabuyao Integrated National High School.

Product Backlogs

The product backlog for the I AM PRESENT system is structured to help Teachers, Admins, and Super Admins efficiently manage and monitor student attendance. Teachers primarily handle taking attendance, confirming student presence using facial recognition, and responding to system notifications. Admins are responsible for managing system settings, student profiles, and producing attendance reports. Meanwhile, Super Admins oversee the system's overall health, user roles, and data security.

The backlog features modules for secure login with lockout protection, ensuring that only authorized users can access the system. The Settings Module enables Admins and Super Admins to customize preferences such as facial recognition sensitivity, notification thresholds, and dashboard layout. The Attendance Dashboard offers Teachers and Admins real-time attendance updates, showing student photos and recognition status for quick verification. The Notification Module automatically sends alerts to parents or guardians when students are absent or when attendance data is updated.

Student Registration and Facial Recognition Modules allow Admins and Teachers to manage student profiles and verify identities in group photos using MTCNN–FaceNet technology. Manual Attendance Override lets Teachers and Admins correct recognition errors. The Attendance Report Generation and Analytics & Visualization modules offer detailed insights into attendance patterns, helping administrators monitor class performance and identify trends.

Table 5. Product Backlog for Teacher, Admin, Super Admin

	Description	User/Role	Priority	Estimated Effort Days
Login Module	Secure sign-in/sign-out with lockout: allow only 3 failed attempts, then enforce a 10-minute cooldown; record events and send email/SMS alerts on lockout.	Teacher, Admin, Super Admin	High	1 day
Settings Module	Allow customization of system preferences, such as notification thresholds, facial recognition sensitivity, and dashboard layout.	Admin, Super Admin	Medium	1.5 days
Attendance Dashboard	View real-time attendance data for each class with student photos and recognition status.	Teacher, Admin	High	1.5 days
Notification Module	Automatically notify parents/guardians via email or SMS when a student is absent or when attendance is updated.	Teacher, Admin	High	1 day
Student Registration	Add, update, or remove student profiles, including photo datasets for facial recognition.	Admin, Super Admin	High	2 days
Facial Recognition Module	Detect and identify students in group photos using MTCNN–FaceNet; flag unknown faces for manual verification.	Teacher, Admin	High	2 days
Attendance Report Generation	Generate detailed daily, weekly, or monthly attendance reports for administrative use.	Admin, Super Admin	High	2 days
Manual Attendance Override	Allow teachers or admins to manually mark attendance in case of recognition errors.	Teacher, Admin	Medium	1 day
User Management	Manage accounts and roles for Teachers, Admins, and Super Admins, including password resets and permissions.	Super Admin	High	2 days
System Logging	Maintain logs of all key actions, such as logins, attendance changes, and setting modifications for auditing.	Admin, Super Admin	Medium	1.5 days
Facial Data Management	Review and update student facial datasets to improve recognition accuracy.	Admin, Super Admin	Medium	1.5 days
Analytics & Visualization	Display attendance trends with charts and graphs to help administrators monitor class performance.	Admin, Super Admin	Medium	2 days
System Health Monitoring	Notify the admin if the system faces errors or server downtime to ensure reliability.	Super Admin	Medium	1.5 days
Secure Data Storage	Ensure all attendance records and facial embeddings are safely stored in the database with encryption.	Admin, Super Admin	High	2 days
Help & support	Provide a help section or support feature to guide teachers and admins on how to use the system.	Teacher, Admin	Low	1 day

Sprint Planning

This sprint plan details a multi-phase development approach for the teacher interface of the I AM PRESENT attendance system. Each phase progressively advances from establishing secure system access to automating attendance recording through group facial recognition. The primary objective of this sprint plan is to provide teachers with a reliable, efficient, and user-friendly tool for managing classroom attendance while reducing manual workload and errors.

The initial phase focuses on implementing secure login and role-specific access controls, ensuring that only authorized teachers can access their assigned classes and student records. This establishes a secure foundation for the system and protects sensitive attendance data. The succeeding phases concentrate on integrating group photo capture and facial recognition processing using the MTCNN–FaceNet model. These features allow teachers to record attendance quickly by capturing a single group image, automatically identifying students, and minimizing disruptions during class time.

Further development phases include attendance review and manual override functions, enabling teachers to verify recognition results and correct any inaccuracies caused by lighting conditions, occlusions, or facial changes. The final phase enhances usability by introducing attendance summaries, class-level statistics, and visual indicators that help teachers monitor student participation patterns over time. Through this structured sprint approach, the teacher interface supports accurate, efficient, and seamless daily attendance management in the classroom.

Table 6. Sprint Planning for Teacher Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Access the attendance system securely.	Develop teacher login and logout functionality, role-based access for teacher accounts, and redirect teachers to class-specific dashboards	6–8 hours/day for 2 days	Teachers can securely access their assigned classes
2	Capture attendance using group facial recognition	Integrate camera access for group photo capture, implement image submission to the facial recognition module, and Display recognition results per student	6–8 hours/day for 3 days	Teachers can take group photos and automatically record attendance
3	Review and correct attendance records	Display attendance results with student photos, enable manual attendance override for errors, and save corrected records to the database	6–8 hours/day for 2 days	Attendance records are accurate and teacher-verified
4	Monitor attendance summaries	Display class attendance statistics, daily and weekly attendance summaries, and Access student attendance status	6–8 hours/day for 2 days	Teachers can easily monitor class attendance performance

This sprint plan outlines a structured development strategy for the administrative interface of the I AM PRESENT system. Each sprint incrementally builds from managing student records and facial datasets to monitoring attendance data and generating reports. The goal is to equip administrators with complete control over student information, attendance accuracy, and communication with parents or guardians. The initial phase focuses on student registration and facial data management to support reliable facial recognition. Subsequent development integrates attendance dashboards, report generation, and notification features, enabling administrators to track attendance trends and respond to absences efficiently. The final phase emphasizes data accuracy and analytics, ensuring that

attendance records are well-maintained and helpful for academic monitoring and decision-making.

Table 7. Sprint Planning for Admin Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Manage student profiles and facial data.	Develop a student registration module, upload and update facial image datasets, and store facial embeddings securely	6–8 hours/day for 3 days	Student data and facial records are prepared for recognition
2	Monitor attendance and generate reports.	Develop an attendance dashboard for all classes. Implement report filtering by date and class. Enable export of attendance reports	6–8 hours/day for 3 days	Admin can review and export attendance data
3	Configure notifications and system settings	Create a notification settings module. Implement absence notification logic, test email/SMS notification delivery	6–8 hours/day for 2 days	Parents/guardians receive timely attendance notifications
4	Maintain attendance accuracy and analytics.	Review flagged or unknown faces, manage facial data updates, and view attendance analytics and trends	6–8 hours/day for 3 days	Attendance accuracy and data insights are improved

This sprint plan presents a phased development approach for the system-level management of the I AM PRESENT attendance monitoring platform. Each sprint focuses on strengthening system security, access control, and operational reliability. The goal is to provide super administrators with complete oversight of user accounts, permissions, and system performance.

The initial phase establishes user role management and access control, ensuring the system's secure operation. Subsequent phases integrate system logging, data security, and health monitoring features, allowing super administrators to track system activities and address technical issues promptly. The final phase ensures overall system stability and reliability, supporting continuous and secure operation of the attendance monitoring system.

Table 8. Sprint Planning for Super Admin Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Control system access and user roles	Develop a user account management module, assign and modify roles (Teacher, Admin), implement password reset, and account control	6–8 hours/day for 3 days	User roles and permissions are securely managed
2	Ensure system security and reliability.	Implement system logging for key activities. Apply encrypted data storage for records. Configure system health monitoring	6–8 hours/day for 3 days	The system operates securely with reliable monitoring
3	Monitor overall system performance.	Review system activity logs, handle error reports and downtime alerts, and optimize backend services	6–8 hours/day for 2 days	System stability and performance are maintained

Daily Scrum

The development team holds a daily 2-hour-and-30-minute Scrum meeting to stay on track with the goals of the I AM PRESENT web-based attendance monitoring system. During this time, the team discusses progress

on system development and thesis documentation, plans tasks for the day, and identifies any challenges, particularly in implementing group-shot-based facial recognition using the MTCNN–FaceNet model. These daily discussions help the team quickly address issues, make necessary adjustments to the sprint backlog, and ensure that the system continues to meet the project objectives. Through regular coordination and collaboration, the team is steadily advancing the development of an efficient, accurate, and reliable attendance monitoring system for Cabuyao Integrated National High School.

Sprint Execution

During sprint execution, the development team works closely with the project adviser and intended users, including teachers and school administrators of Cabuyao Integrated National High School, to define sprint tasks and estimate the required workload. This collaboration ensures that development efforts remain focused on building and improving the core features of the I AM PRESENT web-based attendance monitoring system. Key activities during this phase include implementing group-shot-based facial recognition using the MTCNN–FaceNet model, developing a web-based attendance interface, integrating real-time notifications for parents or guardians, and preparing the attendance report generation. Continuous coordination during sprint execution helps ensure that both technical development and documentation align with the project objectives.

Sprint Review Meeting

At the end of each sprint, a sprint review meeting is conducted to present the completed features of the system. During the review, the development team

demonstrates system functionality, including group facial recognition for attendance capture, student identification accuracy, attendance record generation, and report export features. This session allows stakeholders to evaluate whether the implemented components meet the intended needs of teachers and administrators, particularly in terms of efficiency, accuracy, and ease of use. Feedback gathered during the sprint review is used to identify necessary improvements and to refine the system for subsequent sprints.

Sprint Retrospective

Following the sprint review, the team conducts a sprint retrospective to reflect on the overall development process and teamwork. The retrospective focuses on identifying what worked well, what challenges were encountered, and what can be improved in future sprints. Key discussion areas include the effectiveness of the facial recognition model in group photos, the usability of the web-based interface, time management, and the development tools used. For example, early retrospectives may reveal the successful implementation of the face detection and recognition modules. At the same time, later sprints may focus on improving system accuracy, performance, and compliance with ISO/IEC 25010 quality standards. Insights from these retrospectives are documented to guide continuous improvement throughout the project.

Project Testing and Evaluation

The I AM PRESENT system undergoes testing and evaluation to ensure that all components function accurately, reliably, and securely. Each module, including user authentication, group shot-based facial recognition, attendance recording, report generation, and notification features, is tested individually to

verify correctness and proper integration with the web-based interface. Particular focus is placed on testing the MTCNN–FaceNet facial recognition model to assess its accuracy in identifying students from group photos under varying conditions.

System performance and usability are evaluated to confirm that teachers and administrators can operate the application efficiently during actual class sessions. Attendance reports and real-time notifications sent to parents or guardians are also tested to ensure timely and accurate delivery of information. The system is further evaluated against the ISO/IEC 25010 software quality standards, with emphasis on functionality, usability, performance, reliability, and security. The overall testing and evaluation process confirms that the system is stable, user-friendly, and suitable for implementation at Cabuyao Integrated National High School.

Software Testing

The figure below illustrates the comprehensive software testing process implemented during the development of the I AM PRESENT web-based attendance monitoring system. This testing ensured that each software module performed reliably and accurately in automating student attendance via group-shot-based facial recognition. Functional testing began with unit testing of core features, including user authentication, face detection, face recognition, attendance recording, and report generation. Integration testing validated the interaction between these modules to ensure seamless data flow from facial recognition to the attendance database and notification systems. System testing verified that the entire platform functioned as an integrated solution,

while acceptance testing confirmed that the system met the needs and expectations of teachers, administrators, and school stakeholders. Beyond functional testing, non-functional testing addressed vital system qualities, including performance during real-time attendance capture, security of sensitive student data, usability for teachers with varying technical skills, and compatibility across different web browsers and devices commonly used at Cabuyao Integrated National High School. This comprehensive testing framework ensured that I AM PRESENT is stable, secure, user-friendly, and ready for deployment to improve attendance accuracy and streamline monitoring.

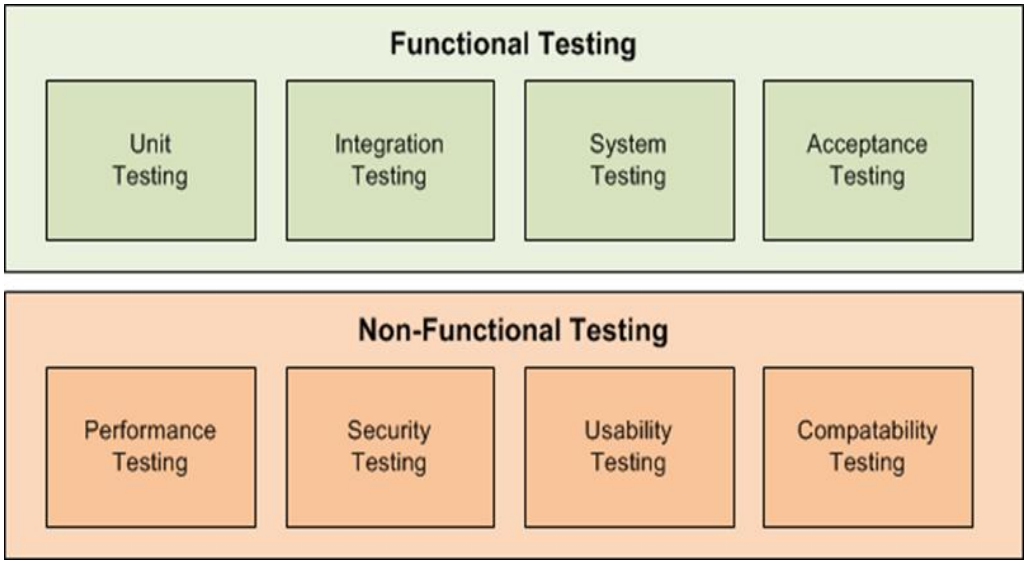


Figure 7. Levels of Testing

Unit Testing

Unit testing was conducted to verify that each component of the I AM PRESENT system, including user authentication, face detection, facial recognition, attendance recording, and report generation, functioned correctly in isolation. By testing these modules independently, the development team was able to identify and resolve errors early, ensuring that each feature

performed as intended before integration into the complete attendance monitoring platform. This process played a critical role in enhancing the system's overall reliability and stability, which aims to improve student attendance accuracy through automated group-shot-based facial recognition. Test cases were carefully designed to cover a range of scenarios, including typical use cases, boundary conditions, and unexpected inputs, to guarantee robustness. Where applicable, automated testing scripts were implemented to facilitate repeated testing and ensure consistency of results.

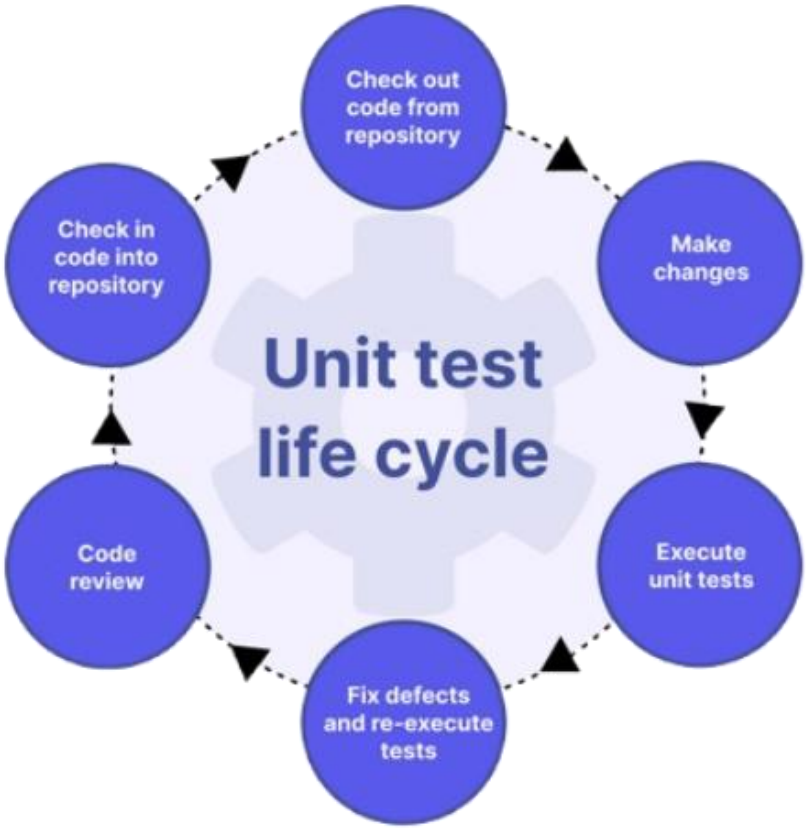


Figure 8. Unit Testing

Integration Testing

Integration testing ensured that all core modules of the I AM PRESENT system, such as face detection, facial recognition, attendance recording, notification delivery, and report generation, worked together seamlessly.

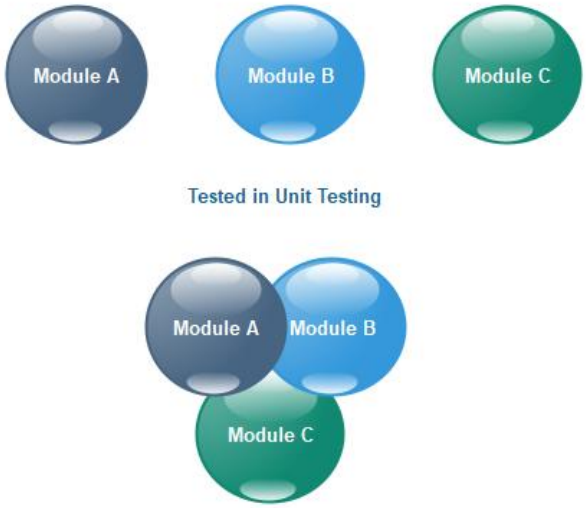


Figure 9. Integration Testing

System Testing

System testing evaluated the I AM PRESENT attendance monitoring system as a fully integrated solution. This phase tested whether all components, including group-shot facial recognition, attendance logging, notification delivery, report creation, and the web interface, worked seamlessly together within a complete workflow. End-to-end scenarios were conducted to verify accurate student identification, data processing, and timely attendance reports under real-world conditions at Cabuyao Integrated National High School. Additionally, performance, reliability, and compatibility were tested to ensure the system could support multiple users, process data in real time, and secure information without issues. The results showed that the system met its

functional requirements and was stable enough to proceed to acceptance testing and deployment.

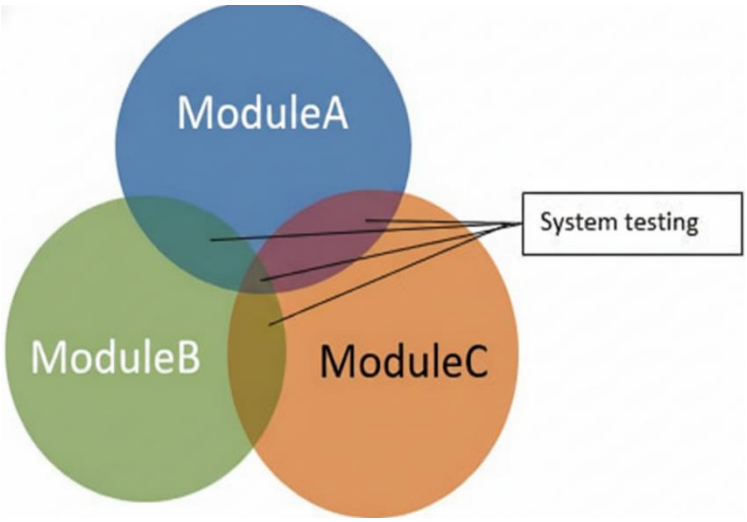


Figure 10. System Testing

Acceptance Testing

Acceptance testing was conducted to evaluate whether the I AM PRESENT attendance monitoring system fully satisfied the functional and usability requirements of its intended users, including teachers, school administrators, and parents or guardians at Cabuyao Integrated National High School. This testing phase focused on validating real-world system behavior rather than internal technical design, ensuring that the application operates as expected in an actual classroom environment.

Key system functions were tested, including group-shot-based facial recognition using MTCNN, accurate and automated attendance recording, timely email notifications for absences, and the generation of attendance reports. Teachers assessed the ease of capturing group photos and reviewing attendance records, while administrators evaluated data accuracy, system reliability, and report accessibility. Parents or guardians indirectly validated the system by receiving clear and timely attendance notifications.

The system was considered acceptable once all identified requirements were met, errors were resolved, and users confirmed that the system improved the efficiency of attendance monitoring compared to traditional methods. Successful completion of acceptance testing confirmed that the system is ready for deployment and practical use within the school.



Figure 11. User Acceptance Testing

System Flowchart

The teacher flowchart illustrates the operational workflow of the I AM PRESENT web-based attendance monitoring system, focusing on how teachers interact with the application to manage student records and attendance using group shot-based facial recognition. The process begins at the authentication stage, where the teacher logs into the system using a valid username and password. This login mechanism ensures that only authorized personnel can access attendance data and student information, addressing basic system security and data privacy requirements. If incorrect credentials

are entered, the system denies access and redirects the user back to the login interface; otherwise, successful authentication leads to the main dashboard.

Once authenticated, the teacher is directed to the Dashboard, the system's central navigation hub. From this interface, the teacher can access multiple core modules, including attendance reports, student information management, student registration, and system settings. This centralized design reduces navigation complexity and allows teachers to perform attendance-related tasks efficiently within a single platform.

In the Attendance Report module, the system displays attendance records generated through the facial recognition process. These records are retrieved from the database and presented in a structured format for easy monitoring and review. To support reporting and documentation needs, the system allows teachers to export attendance data to Excel, enabling offline access, printing, and submission to school administrators when required.

The All-Student module allows teachers to view and edit existing student information stored in the system database. This includes updating personal details or correcting records when necessary, ensuring that attendance tracking remains accurate and consistent. Any modifications made through this module are saved directly to the database, maintaining data integrity across the system.

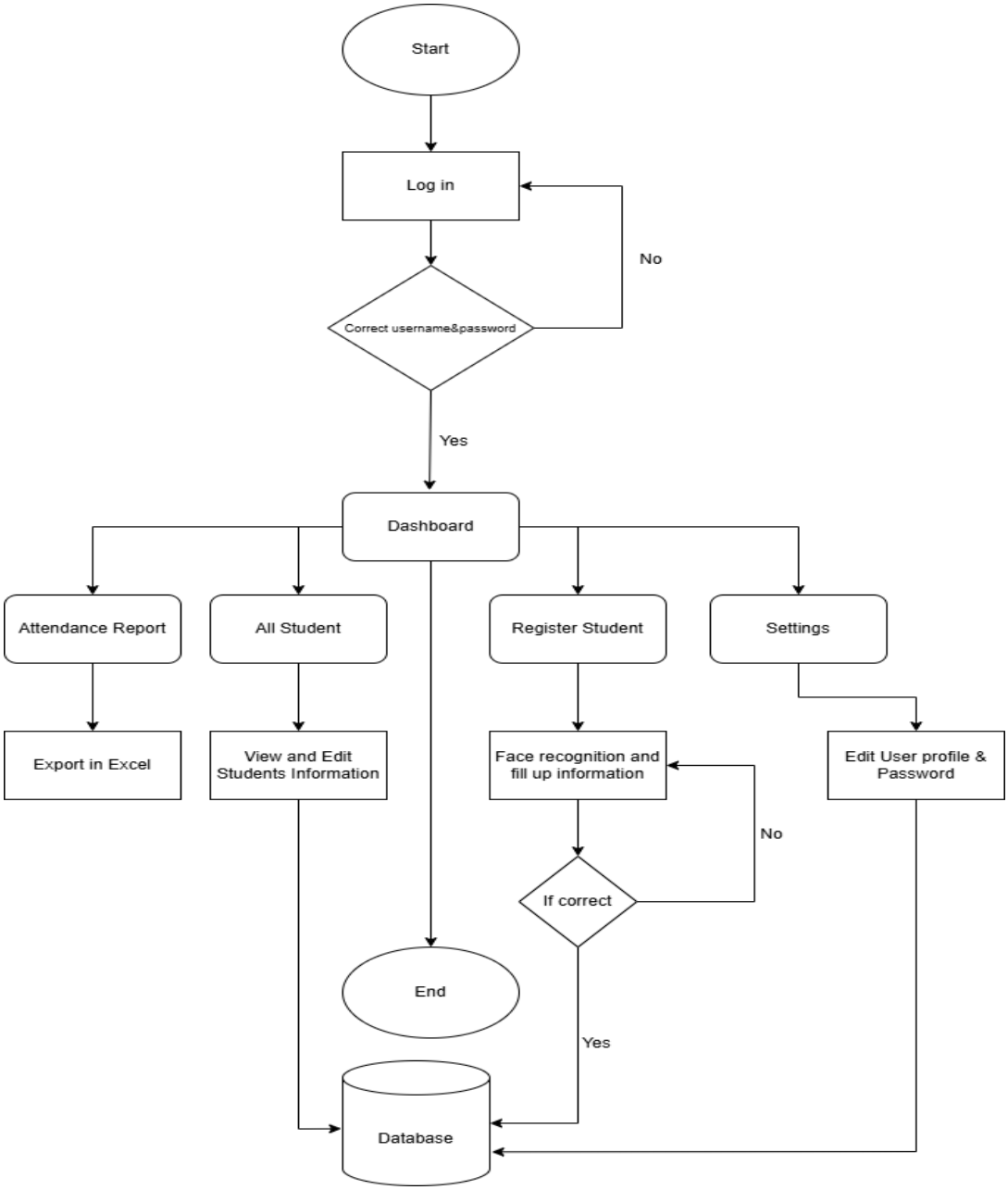


Figure 12. System Flowchart of the Teacher

The admin flowchart describes the administrator module workflow in the I AM PRESENT system. The process starts with secure login authentication, after which the administrator accesses the dashboard. From the dashboard, the administrator can manage system settings, update user profiles and passwords, maintain teacher accounts by adding or editing user information, and access attendance reports. The reports module allows the administrator to view, filter, and export attendance records generated through facial recognition.

All administrative actions and data updates are stored in the database to ensure accuracy, security, and centralized system management.

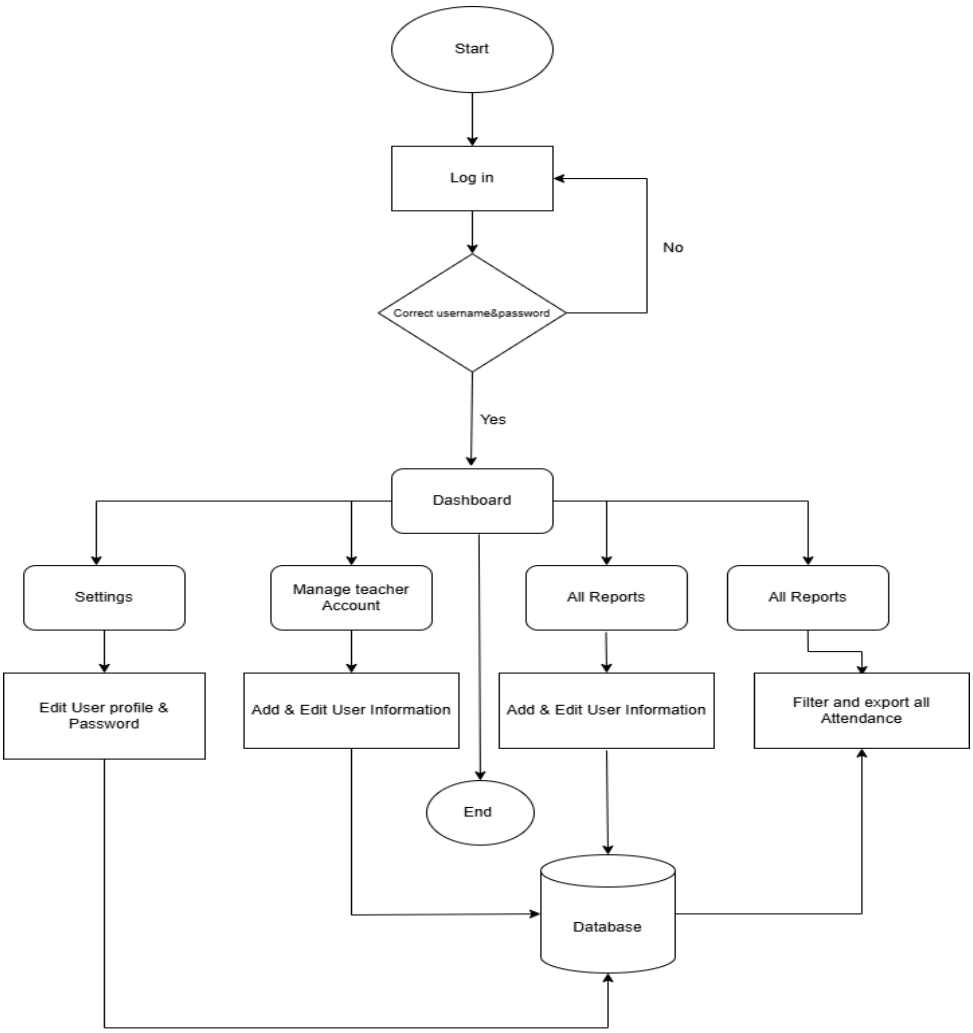


Figure 13. System Flowchart of Admin

The super admin flowchart presents the workflow of the Super Administrator module of the I AM PRESENT: A Web-Based Application Optimizing Students' Attendance Monitoring Through Group Shot-Based Facial Recognition. This module represents the highest level of system access and manages user accounts and system-level settings to ensure secure, controlled system operation.

The process begins with the login phase, where the super admin enters a valid username and password. The system verifies the credentials to prevent unauthorized access. If the credentials are incorrect, the system redirects the user back to the login interface. Once authenticated, the super admin is granted access to the Dashboard, the main control interface for super administrative functions. From the dashboard, the super admin can access the User Management module, which allows viewing and editing of user information, including administrator and teacher accounts. This ensures that user records remain accurate and that access privileges are adequately maintained. Any changes made in this module are stored in the system database to maintain data consistency and centralized user control.

The super admin can also access the Settings module, where profile information and account credentials, such as passwords, can be updated. This function supports account security and system integrity by allowing regular credential management. All updates performed through the settings module are recorded in the database. After completing the required administrative tasks, the process reaches the endpoint, concluding the super admin’s interaction with the system. Overall, the super admin flowchart demonstrates a structured and secure workflow that supports centralized user management, access control, and system security within the attendance monitoring application.

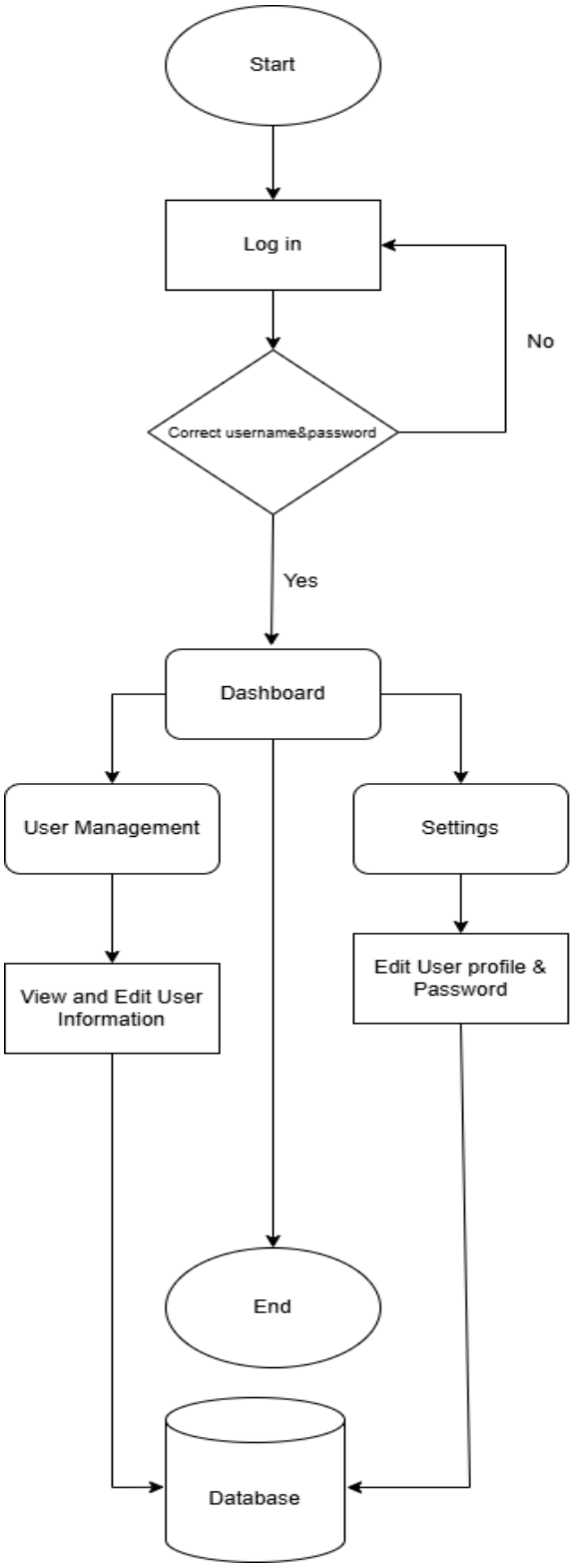


Figure 14. System Flowchart of Super Admin

Use Case Diagram

Figure 14 shows the Use Case Diagram of the I AM PRESENT Attendance Monitoring System, featuring four human actors: Teacher, Registrar (Admin), I.T. (Super Admin), and Guardian, and two external actors: ML Service and Email Service. Teachers and Registrars log in, manage classes, and register students. Teachers capture attendance via group photos with automatic facial recognition, while the Super Admin monitors activity and can export data. Guardians receive notifications through email. The system highlights secure access, automated attendance, and real-time communication.

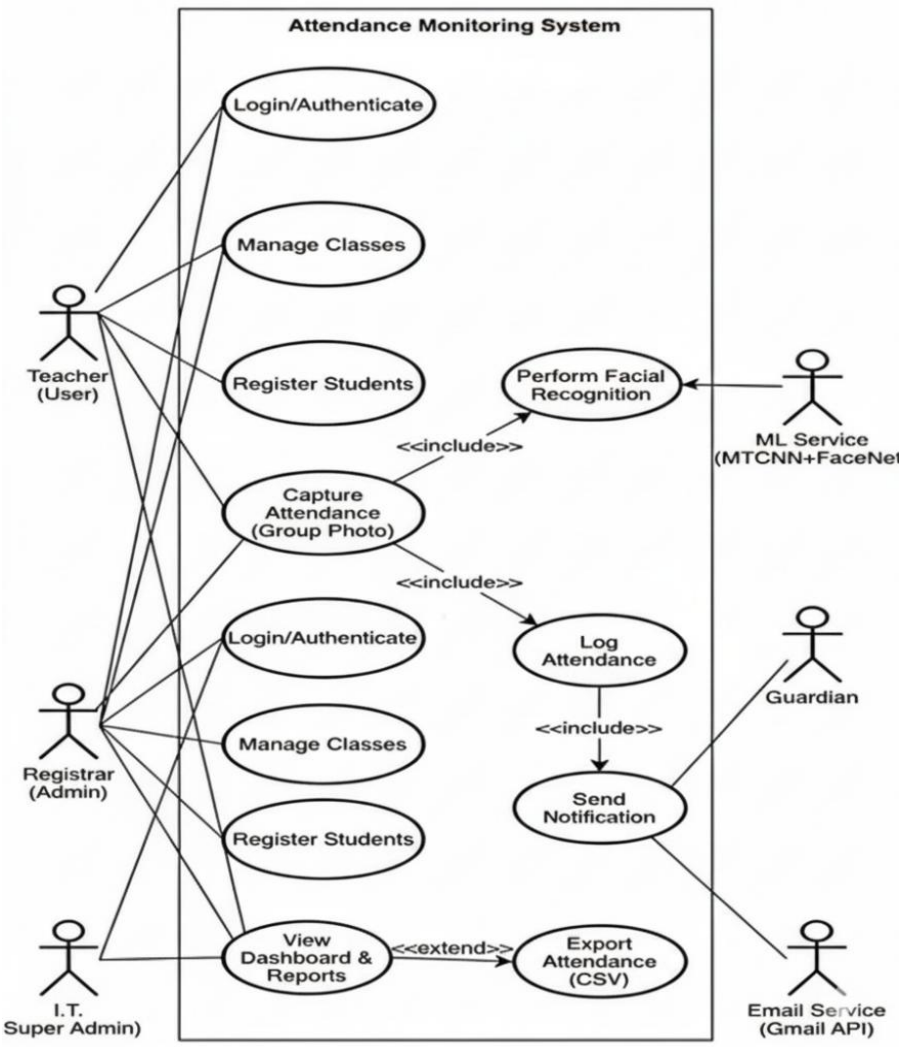


Figure 15. Used Case Diagram

Evaluation Procedure

The I AM PRESENT system was evaluated through beta testing to assess its performance, usability, and reliability in a real school setting. Feedback was collected from teachers, registrars, IT staff, and guardians using a Likert scale, focusing on key aspects such as ease of use, accuracy of attendance capture, and notification delivery. The evaluation followed the ISO 25010 software quality standard, emphasizing functionality, security, maintainability, and portability. Results highlighted areas needing improvement, including faster processing of group photos, more accurate facial recognition under varying lighting conditions, and timely email notifications. Adjustments were made based on this feedback to enhance system efficiency and user experience, confirming that the web-based IoT solution reliably supports secure, automated, and real-time student attendance monitoring.

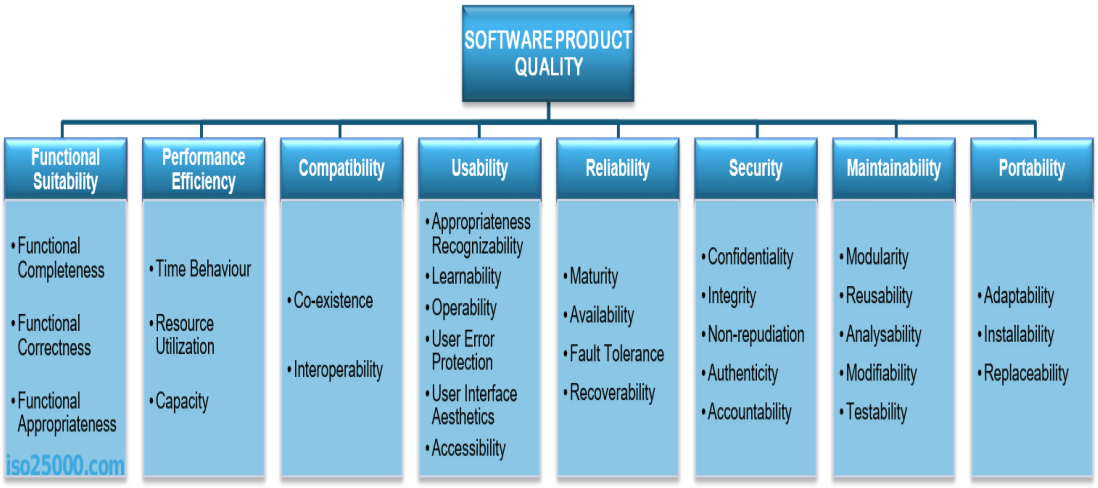


Figure 16. Quality Characteristics of ISO 25010 Standard

Population of the Study

The study on I AM PRESENT: A Web-Based Application Optimizing Students' Attendance Monitoring Through Group Shot-Based Facial Recognition at Cabuyao Integrated National High School involves two main groups: teachers and IT experts. These respondents provide the necessary perspectives to evaluate and improve the system's usability, reliability, and effectiveness in a real school environment.

Teachers contribute practical insights on the system's usability, class management features, and the accuracy of attendance tracking. Their feedback ensures that the application meets educators' day-to-day needs and supports efficient student attendance monitoring.

IT experts evaluate the system's technical aspects, including data security, system stability, and the integration of the facial recognition module with the web application. Their expertise ensures that the system functions reliably and securely under real operating conditions. By combining insights from teachers and IT experts, this study aims to develop a secure, user-centered, and efficient attendance monitoring system that enhances classroom management and automates student attendance processes.

Table 9. Respondents of the Research Project

CATEGORY OF RESPONDENTS	SAMPLE
Teachers	50
IT Expert	3
Total	**53**

CHAPTER 4

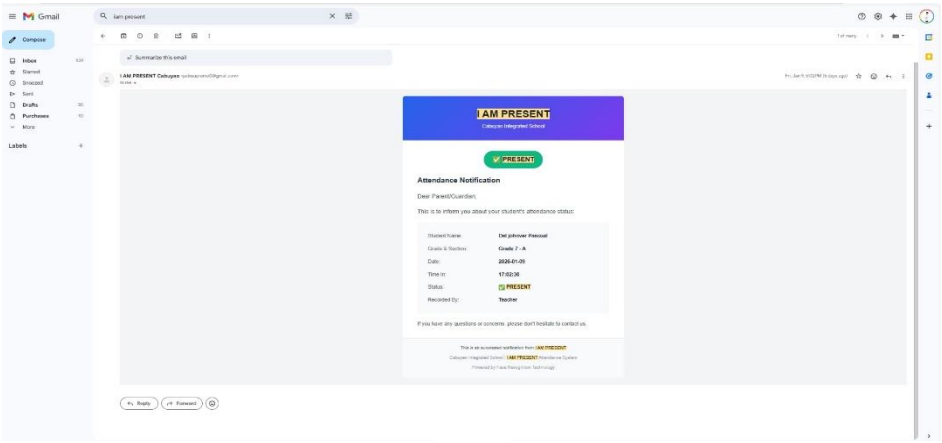
RESULTS AND DISCUSSION

This chapter presented the results and the discussion supporting the methodology used to achieve the objectives.

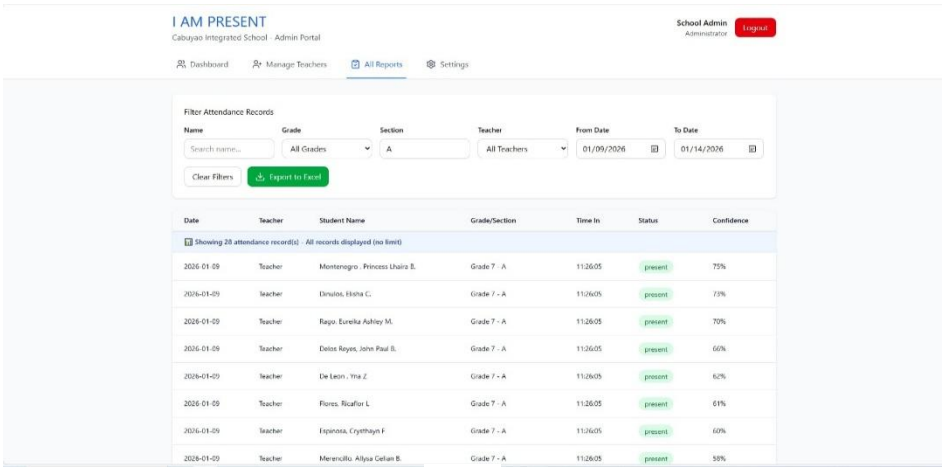
Web-Based Attendance Monitoring and Real-Time Parent/Guardian Notification

Figure 16 illustrates how the web-based attendance system supports real-time monitoring and communication with parents or guardians. (*a*) shows the automated email notification sent to parents or guardians once a student's attendance is recorded. The notification includes essential details such as the student's name, date, arrival time, and the teacher who recorded the attendance, ensuring parents are promptly informed of their child's presence at school. This feature strengthens transparency and encourages parental involvement in monitoring student attendance.

(*b*) presents the centralized attendance monitoring portal used by school administrators. Through this interface, attendance records can be viewed, searched, and filtered by grade level, section, or date for efficient tracking. The system also displays facial recognition confidence levels, allowing administrators to review and validate attendance entries when necessary. Additionally, the portal provides an option to export attendance data in Excel format to support reporting, documentation, and record-keeping. Overall, the figure demonstrates how the system streamlines attendance management while improving communication between the school and parents or guardians.



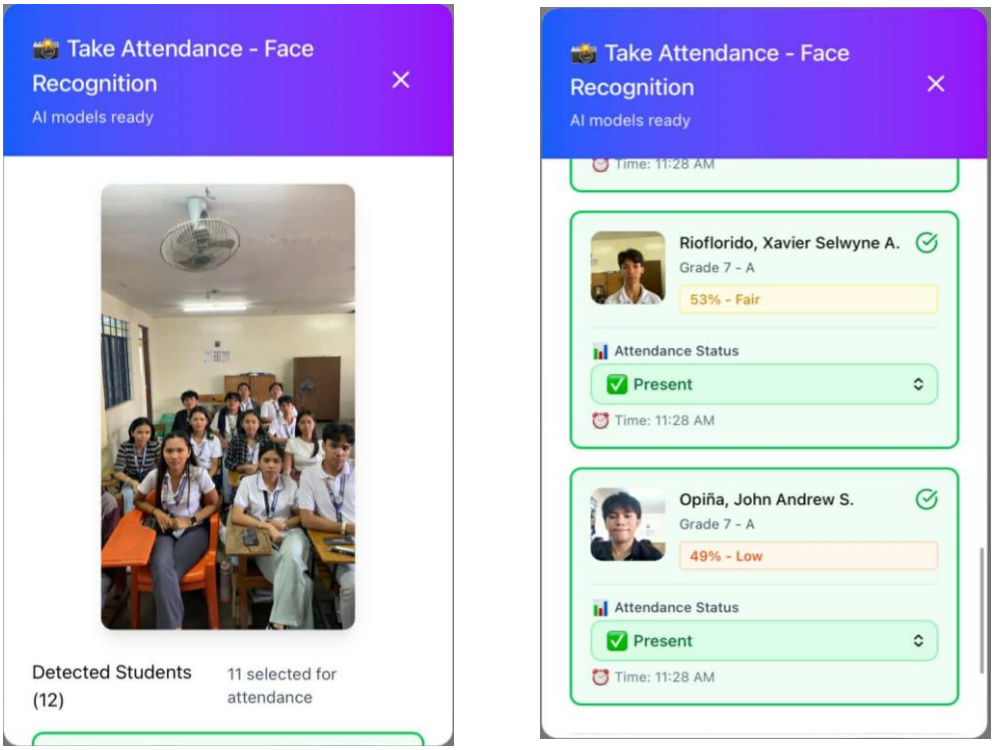
a.



b.

Figure 16. Attendance Monitoring and Email Notification

Figure 17 illustrates the implementation of the group face-detection attendance system in a classroom setting. Part (*a*) depicts how the system takes a group photo of students during class to record attendance seamlessly, without disrupting the lesson. It shows multiple faces bing detected simultaneously, even when students are seated in various positions. Part (*b*) shows the facial recognition and verification interface, where the system matches detected faces with registered student profiles stored in its database. Each profile is accompanied by a confidence score and attendance status, enabling the teacher to review and confirm the results before finalizing.

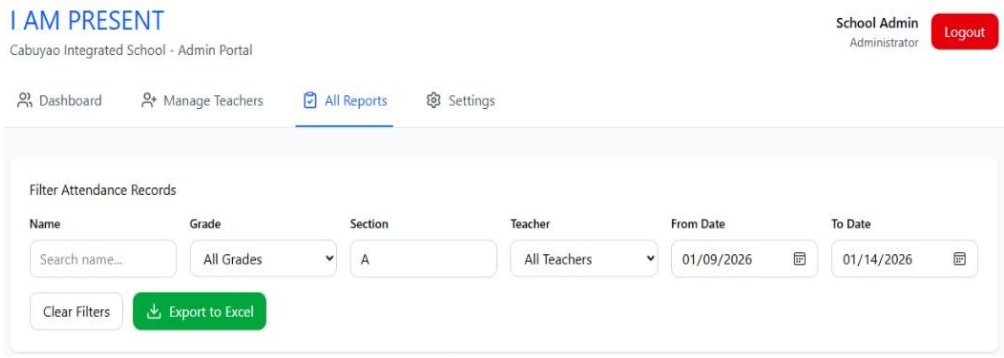


a. b.

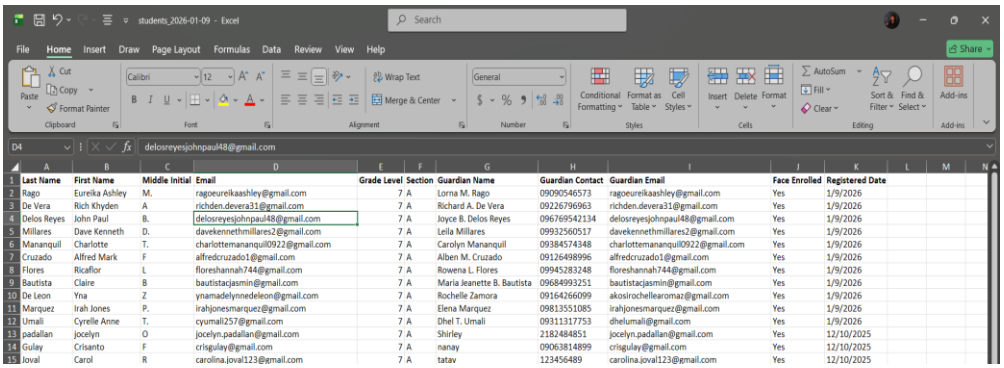
Figure 17. Group Facial Attendance

Attendance Report Generation and Exporting

Figure 18 demonstrates the system's ability to generate reports, highlighting the "Export to Excel" button that enables users to download attendance data as an Excel file. (*a*) Administrators can quickly create a digital spreadsheet with all student attendance records for a selected date range or class. The exported data includes student names, grades, and guardian contact details, aiding communication if necessary. (*b*) displays the exported Excel file, which has a clean, organized layout that clearly shows each student's attendance history. This feature helps the school maintain accurate records, prepare administrative reports, and save time on manual data entry. Automating the creation and export of attendance reports improves efficiency and supports better classroom and administrative management.



a.



Last Name	First Name	Middle Initial	Email	Grade Level	Section	Guardian Name	Guardian Contact	Guardian Email	Face Enrolled	Registered Date
Rago	Eureika Ashley	M.	ragoeureikaashley@gmail.com	7	A	Lorna M. Rago	09090546573	ragoeureikaashley@gmail.com	Yes	1/9/2026
De Vera	Rich Khyden	A	richden.devera31@gmail.com	7	A	Richard A. De Vera	09226796963	richden.devera31@gmail.com	Yes	1/9/2026
Delos Reyes	John Paul	B.	delosreyesjohnpaul48@gmail.com	7	A	Joyce B. Delos Reyes	096769542134	delosreyesjohnpaul48@gmail.com	Yes	1/9/2026
Millares	Dave Kenneth	D.	davekennethmillares2@gmail.com	7	A	Leila Millares	09932560517	davekennethmillares2@gmail.com	Yes	1/9/2026
Mananquil	Charlotte	T.	charlottemananquil0922@gmail.com	7	A	Carolyn Mananquil	09384574348	charlottemananquil0922@gmail.com	Yes	1/9/2026
Cruzado	Alfred Mark	F	alfredcruzado1@gmail.com	7	A	Alben M. Cruzado	09126498996	alfredcruzado1@gmail.com	Yes	1/9/2026
Flores	Ricaflor	L	floreshannah744@gmail.com	7	A	Rowena L. Flores	09945283248	floreshannah744@gmail.com	Yes	1/9/2026
Bautista	Claire	B	bautistacjasmin@gmail.com	7	A	Maria Jeanette B. Bautista	09684993251	bautistacjasmin@gmail.com	Yes	1/9/2026
De Leon	Yna	Z	ynamadelynnedeleon@gmail.com	7	A	Rochelle Zamora	09164266099	akosirochellearomaz@gmail.com	Yes	1/9/2026
Marquez	Irah Jones	P.	irahjonesmarquez@gmail.com	7	A	Elena Marquez	09813551085	irahjonesmarquez@gmail.com	Yes	1/9/2026
Umali	Cyrelle Anne	T.	cyumali257@gmail.com	7	A	Dhel T. Umali	09311317753	dhelumali@gmail.com	Yes	1/9/2026
padallan	jocelyn	O	jocelyn.padallan@gmail.com	7	A	Shirley	2182484851	jocelyn.padallan@gmail.com	Yes	12/10/2025
Gulay	Crisanto	F	crisgulay@gmail.com	7	A	nanay	09063814899	crisgulay@gmail.com	Yes	12/10/2025
Joval	Carol	R	carolina.joval123@gmail.com	7	A	tatav	123456489	carolina.joval123@gmail.com	Yes	12/10/2025

b.

Figure 18. Generated the Report on attendance

Absent notification in email

Figure 19 illustrates the automated absence notification generated by the I AM PRESENT system and sent to parents or guardians when a student is recorded as absent. The notification includes relevant details such as the student's name, class section, date, and attendance status, ensuring that guardians receive clear and accurate information. This real-time alert mechanism enables immediate awareness of student absences, minimizing delays commonly associated with manual attendance reporting. By automating parent notifications, the system enhances transparency, strengthens school–home communication, and encourages timely parental intervention when attendance issues arise. Moreover, integrating email-based notifications

reduces teachers' administrative workload while ensuring consistent and reliable dissemination of attendance information.

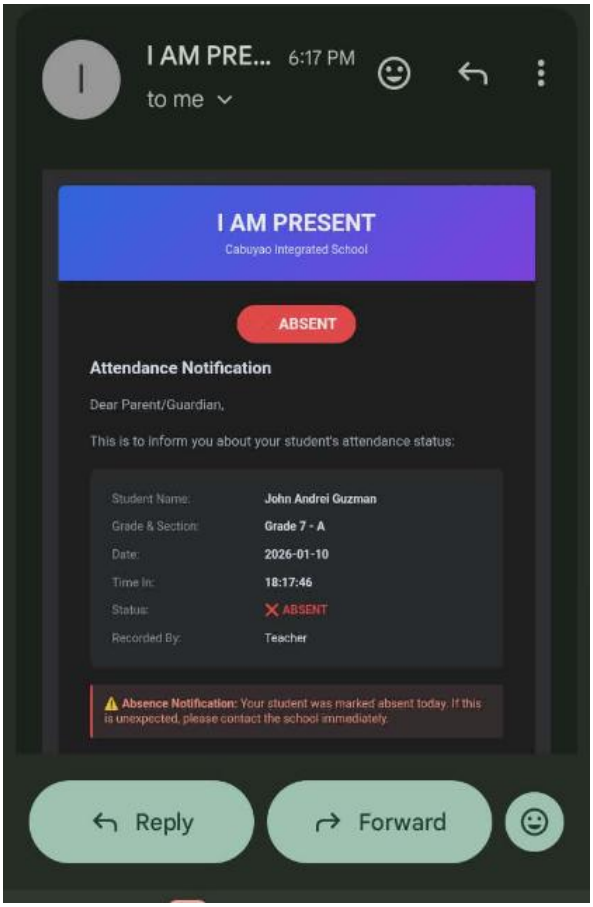


Figure 19. Absent Notification

Test and Evaluate the Software using ISO 25010

Table 10 presents the IT Specialist's assessment of the Attendance Monitoring System's Functional Suitability. The overall score was 3.47 (SD = 0.48), indicating Strongly Agree and reflecting the expert's high confidence in the system's core functionalities. Among the metrics, Functional Completeness scored the highest at 3.80 (SD = 0.45), showing the system efficiently performs all intended tasks, such as attendance tracking and providing real-time notifications to parents or guardians. Functional Correctness also received a Strongly Agree rating (3.40, SD = 0.55), confirming the system's accuracy and

dependability. In contrast, Functional Appropriateness scored slightly lower at 3.20 (SD = 0.45), falling into the Agree range. This suggests minor concerns about the system's capacity to fully support attendance monitoring objectives, though it remains largely effective. Overall, the evaluation indicates that the system is technically sound, reliable, and effective in fulfilling its primary functions, with only minor opportunities for further improvement to better meet monitoring goals.

Table 10. Functional Suitability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Functional Completeness – The system covers all the specified tasks, such as attendance monitoring and real-time notification to parents/guardians.	3.80	0.45	Strongly Agree
2. Functional Correctness – The system provides accurate sensor readings and results with the needed degree of precision.	3.40	0.55	Strongly Agree
3. Functional Appropriateness – The system helps accomplish the goal of monitoring attendance.	3.20	0.45	Agree
Composite Mean	**3.47**	**0.48**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 11 shows the IT Specialist's evaluation of the real-time group face detection attendance system's Performance Efficiency. The system achieved a high overall mean score of 3.47 (SD = 0.52), indicating a Strongly Agree rating. This suggests the system effectively performs its functions with reliability and technical competence. Notably, Capacity received the highest mean score of 3.60 (SD = 0.52), implying that the system can manage multiple data inputs, such as processing group photos and recognizing several students simultaneously, without issues. The Time Behavior indicator (Mean = 3.40, SD

= 0.52) also received a Strongly Agree rating, showing quick processing and display of attendance results for real-time student identification. Additionally, Resource Utilization (Mean = 3.40, SD = 0.52) indicates that the system operates efficiently without slowing down or overloading the web server or connected devices during attendance recording. Overall, these results demonstrate that the group facial recognition attendance system is stable, responsive, and proficient at handling its workload. Technically, these findings affirm its suitability for classroom use at Cabuyao Integrated National High School.

Table 11. Performance Efficiency based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Time Behavior – The system processes and displays real-time data quickly and without delays.	3.40	0.52	Strongly Agree
2. Resource Utilization – The system runs efficiently without slowing down or overloading the web server or connected devices.	3.40	0.52	Strongly Agree
3. Capacity – The system can handle multiple data inputs.	3.60	0.52	Strongly Agree
Composite Mean	**3.47**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 12 shows the IT Specialist's assessment of the Compatibility of the real-time group face detection attendance system. It scored a strong overall mean of 3.50 (SD = 0.55), which is interpreted as Strongly Agree, indicating good integration potential with the school’s existing IT setup. The top-rated criterion was Co-existence (Mean = 3.60, SD = 0.55), indicating confidence that the system can operate smoothly alongside other resources, such as the local network, email notifications, and database platforms, without conflicts.

Interoperability (Mean = 3.40, SD = 0.55) also received high scores, indicating that the system can efficiently exchange data among its database, web dashboard, and notification modules to support reliable attendance management. These results confirm that the system is technically prepared to operate effectively within the IT infrastructure of Cabuyao Integrated National High School. The similar standard deviation values suggest consistent evaluation across all indicators. Overall, Table 11 offers a quantitative overview of the system's compatibility, supporting its deployment for real-time classroom attendance monitoring.

Table 12. Compatibility based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Co-existence – The system runs properly while sharing resources (internet, local network, databases) with other applications.	3.60	0.55	Strongly Agree
2. Interoperability – The system can exchange data effectively between the database and the web dashboard.	3.40	0.55	Strongly Agree
Composite Mean	**3.50**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 13 presents the IT specialists' evaluation of the Interaction Capability of the real-time group face detection attendance system. The system obtained an overall mean score of 3.28 (SD = 0.27), interpreted as Strongly Agree, indicating a high level of confidence in the system's usability and interaction design. This result suggests that the system effectively supports user interaction while minimizing operational complexity, which is essential for technology adoption in school environments. High levels of agreement (Mean

= 3.40) were observed in key usability indicators, including Appropriateness, Recognizability, Learnability, User Error Protection, Inclusivity, and User Assistance. These findings imply that users can easily recognize the system's purpose, quickly learn how to use its features, and perform attendance-related tasks with minimal training. Additionally, the presence of error-prevention mechanisms and accessible assistance features reduces operational errors and supports inclusive use across varying levels of technical proficiency. Three indicators, Operability (Mean = 3.20, SD = 0.45), User Engagement (Mean = 3.00, SD = 0.00), and Self-Descriptiveness (Mean = 3.00, SD = 0.00), were rated as Agree, suggesting moderate areas for enhancement. These results indicate that while the system functions reliably during daily operations, further improvements may be made to streamline interaction flows, enhance visual or interactive elements to sustain user engagement, and provide clearer on-screen explanations of system features to reduce dependence on external guidance. Overall, the evaluation confirms that the developed attendance system demonstrates strong interaction capability and usability. Its interface design effectively supports accurate, efficient, and error-reduced attendance monitoring for teachers and school administrators. The relatively low and consistent standard deviations further indicate uniform assessments across evaluators, reinforcing the reliability of the results and the system's readiness for deployment in real classroom settings.

Table 13. Interaction Capability based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Appropriateness Recognizability – Users can easily recognize that the system is suitable for a monitoring attendance system.	3.40	0.55	Strongly Agree
2. Learnability – Users can quickly learn how to use the system to monitor the students' attendance	3.40	0.55	Strongly Agree
3. Operability – The system is easy to operate.	3.20	0.45	Agree
4. User Error Protection – The system minimizes user errors and provides clear feedback when errors occur.	3.40	0.55	Strongly Agree
5. User Engagement – The system keeps users actively engaged and motivated while monitoring attendance.	3.00	0.00	Agree
6. Inclusivity – The system can be easily used by people with different technical skills or experience levels.	3.40	0.55	Strongly Agree
7. User Assistance – The system provides helpful instructions, guides, or tooltips when users need support.	3.40	0.55	Strongly Agree
8. Self-Descriptiveness – The system clearly describes its functions and actions, so users can understand what to do without extra help.	3.00	0.00	Agree
Composite Mean	**3.28**	**0.27**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 14 shows the IT Specialist's evaluation of the Reliability of the real-time group face detection attendance system. The system received a mean score of 3.45 (SD = 0.55), interpreted as Strongly Agree, reflecting strong confidence in its ability to perform consistently and dependably during classroom attendance monitoring. The highest rating was given to Faultlessness (Mean = 3.60, SD = 0.55), indicating that the system accurately records attendance without errors in facial recognition or attendance data. This ensures that students' presence is accurately tracked, minimizing errors that can occur with manual attendance methods. The specialist also strongly agreed on Availability (Mean = 3.40),

indicating that the system is always ready and accessible when attendance needs to be taken, allowing teachers to rely on it without interruption. Fault Tolerance (Mean = 3.40, SD = 0.55) received a high rating, showing that the system continues to operate correctly even under challenging classroom conditions, such as poor lighting or temporary disruptions. Likewise, Recoverability (Mean = 3.40, SD = 0.55) confirms that the system can quickly restore data and resume normal operations after interruptions such as power outages, internet disconnections, or unexpected errors. These findings highlight that the attendance system is technically robust, dependable, and resilient, capable of handling various classroom situations without compromising performance. This reliability reduces the likelihood of downtime or errors, ensuring smooth daily operation for teachers and administrators. The low standard deviation values indicate consistent agreement from the specialist across all reliability aspects, suggesting that the system's dependable performance is not situational but stable across different scenarios.

Table 14. Reliability based on IT Specialist Response

	INDICATORS	MEAN	SD	INTERPRETATION
1.	Faultlessness – The system performs its functions accurately and consistently without producing wrong facial data or readings.	3.60	0.55	Strongly Agree
2.	Availability – The system is always accessible and ready for use whenever classroom attendance is needed.	3.40	0.55	Strongly Agree
3.	Fault Tolerance – The system continues to operate correctly even when the lighting is poorly administered.	3.40	0.55	Strongly Agree
4.	Recoverability – The system can recover data and resume regular operation automatically after power loss, internet interruption, or system error.	3.40	0.55	Strongly Agree
Composite Mean		**3.45**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 15 presents the IT Specialist's evaluation of the Security of the real-time group face detection attendance system. The system received a mean score of 3.27 (SD = 0.41), which is interpreted as Strongly Agree, indicating that the specialist viewed the system's security features positively. The highest ratings were given to Integrity (Mean = 3.60, SD = 0.55) and Accountability (Mean = 3.40, SD = 0.55). Integrity ensures that the system protects student attendance data from modification, deletion, or corruption, ensuring records remain accurate. Accountability ensures that the system maintains detailed logs of all user actions, enabling traceability when needed. Other areas, including Confidentiality (Mean = 3.20, SD = 0.45), Non-repudiation (Mean = 3.20, SD = 0.45), and Resistance (Mean = 3.20, SD = 0.45), were rated slightly lower but still positive. These scores indicate that the system limits data access to authorized users, records user actions for verification, and can withstand potential cyber threats or unauthorized access. Confidentiality also received a rating of 3.00, suggesting room for improvement in ensuring that only legitimate users can access student data. Overall, these results show that the attendance system is secure, reliable, and protective of sensitive student information, while highlighting areas where security measures could be further strengthened to make the system even more robust.

Table 15. Security based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Confidentiality – The system ensures that the data of the students is only accessible to authorized users.	3.20	0.45	Agree
2. Integrity – The system protects data from being modified, deleted, or corrupted by errors or unauthorized actions.	3.60	0.55	Strongly Agree
3. Non-repudiation – The system records all user activities, ensuring that any action taken within the system can be verified later.	3.20	0.45	Agree
4. Accountability – The system keeps detailed logs that can trace every user's action during system operation.	3.40	0.55	Strongly Agree
5. Resistance – The system can withstand potential cyber threats, hacking attempts, or unauthorized access without compromising data security.	3.20	0.45	Agree
6. Confidentiality – The system ensures that the data of the students is only accessible to authorized users.	3.00	0.00	Agree
Composite Mean	**3.27**	**0.41**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 16 shows the IT Specialist's evaluation of the real-time group face detection attendance system's maintainability. The system received a score of 3.24 (SD = 0.31), reflecting general confidence in its ease of maintenance and updating. The specialist felt most positive about Modifiability (3.60) and Modularity (3.40), indicating that updates can be made without causing errors and that components such as the database, dashboard, and ML model work independently without affecting one another. Reusability (3.20), Analyzability (3.00), and Testability (3.00) were rated slightly lower, suggesting there is room to make modules easier to reuse, improve error logs, and simplify testing.

Overall, the system is well-organized, adaptable, and designed to allow smooth updates, with consistent agreement from the specialist across all aspects.

Table 16. Maintainability based on IT Specialist Response

	INDICATORS	MEAN	SD	INTERPRETATION
1.	Modularity – Is the system composed of separate parts (Face recognition model, database, web dashboard, API) so that changes in one part do not affect the others?	3.40	0.55	Strongly Agree
2.	Reusability – Can the system's modules, such as the image processing scripts or attendance reporting features, be reused for other school management systems?	3.20	0.45	Agree
3.	Analyzability – Does the system provide clear logs or error messages that help developers identify and analyze recognition failures or database errors easily?	3.00	0.00	Agree
4.	Modifiability – Can the system be modified, updated (e.g., adding more students), or improved without causing new errors or performance problems?	3.60	0.55	Strongly Agree
5.	Testability – Can the system be tested easily to ensure that each new face registration or attendance update works as intended?	3.00	0.00	Agree
Composite Mean		**3.24**	**0.31**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 17 presents the IT Specialist's evaluation of the real-time group face detection attendance system's flexibility. It received an average score of 3.10 (SD = 0.22), indicating a generally fair perception of its flexibility. Both Adaptability and Scalability scored 3.20, suggesting the system works well on different devices and can handle more users or data with minimal issues. Install ability and Replaceability were slightly lower at 3.00, indicating minor setup and upgrade difficulties that do not significantly interfere with ongoing activities. Overall, the system shows sufficient flexibility, with room for improvement in installation and upgrade processes.

Table 17. Flexibility based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Adaptability – The system functions properly across various devices (desktops, tablets, mobiles) and web browsers without compatibility issues.	3.20	0.45	Agree
2. Scalability – The system efficiently processes increases in the number of students, facial images, and high-volume attendance records without performance degradation.	3.20	0.45	Agree
3. Install ability – The web-based application is easily accessible via a browser and requires no complex software installation or manual configuration on the user's end.	3.00	0.00	Agree
4. Replaceability – The system can be upgraded (e.g., updating the FaceNet model or database schema) without disrupting school operations or risking student data loss.	3.00	0.00	Agree
Composite Mean	**3.10**	**0.22**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 18 shows the IT Specialist's assessment of the real-time group face detection attendance system's safety. The system earned an average score of 3.12 (SD = 0.20), reflecting broad agreement that it is moderately safe. Fail Safe received the highest score at 3.40, indicating strong confidence that the system can automatically handle errors or failures, such as by recovering data or resetting itself. Other aspects, Operational Constraint (3.20), Risk Identification (3.00), Hazard Warning (3.00), and Safe Integration (3.00) were all rated in the Agree range. This demonstrates that while the system effectively manages automatic safety functions, there is room for improvement in areas such as preventing unsafe actions, identifying risks, alerting users, and securely integrating with external devices. Overall, the findings suggest the system is generally safe but could benefit from enhancements in proactive safety features.

Table 18. Safety based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
1. Operational Stability – Does the system operate consistently under normal conditions and prevent data loss during the attendance capturing process?	3.20	0.45	Agree
2. Error Identification – Can the system detect and identify possible risks, such as low-quality images, unauthorized access attempts, or software malfunctions?	3.00	0.00	Agree
3. Fail Safe – Does the system automatically activate a recovery response (like auto-saving logs or system reset) when an application error or crash occurs?	3.40	0.55	Strongly Agree
4. System Warning – Does the system alert or notify the administrator immediately when there are processing errors or database connection problems?	3.00	0.00	Agree
5. Secure Integration – Can the system connect and integrate with the school's database or web platforms securely without causing data breaches or security risks?	3.00	0.00	Agree
Composite Mean	**3.12**	**0.20**	**Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 19 shows that IT specialists strongly agree that the real-time group face detection attendance system meets key software quality standards. High ratings in Functional Suitability (3.73), Performance Efficiency (3.47), Compatibility (3.50), Reliability (3.45), Security (3.27), Maintainability (3.30), and Interaction Capability (3.28) indicate the system accurately tracks attendance, works efficiently in classrooms, integrates smoothly with existing infrastructure, remains stable, protects student data, and is easy to use. Slightly lower ratings for Flexibility (3.18) and Safety (3.12) suggest room for improvement in device adaptability and safety features. With an overall mean of 3.36 ("Strongly Agree"), the system is viewed as a reliable, high-quality

solution for modernizing attendance monitoring at Cabuyao Integrated National High School.

Table 19. Result of ISO 25010 based on IT Specialist Response

INDICATORS	MEAN	SD	INTERPRETATION
Functional Suitability	3.73	0.48	Strongly Agree
Performance Efficiency	3.47	0.48	Strongly Agree
Compatibility	3.50	0.55	Strongly Agree
Interaction Capability	3.28	0.53	Strongly Agree
Reliability	3.45	0.55	Strongly Agree
Security	3.27	0.41	Strongly Agree
Maintainability	3.30	0.31	Strongly Agree
Flexibility	3.18	0.34	Agree
Safety	3.12	0.20	Agree
Overall Mean	**3.36**	**0.43**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Results of Survey Answered by the Respondents

Table 20 presents the respondents' assessment of the Perceived Usefulness of the I am Present system. The results show an overall mean of 3.55 with a standard deviation of 0.54, interpreted as "Strongly Agree." This indicates that respondents highly recognize the system's usefulness in improving attendance monitoring. Specifically, they agree that the I am Present system helps make attendance monitoring more efficient (mean = 3.74), provides accurate and real-time recognition (mean = 3.40), helps maintain ideal conditions (mean = 3.54), and increases productivity and reduces manual checking error (mean = 3.51). Overall, the findings suggest that the I am

Present system is perceived as a valuable tool in enhancing productivity and management efficiency in monitoring classroom attendance.

Table 20. Perceived Usefulness based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. The I Am Present system helps make attendance monitoring more efficient.	3.74	0.44	Strongly Agree
2. The system provides accurate and real-time recognition	3.40	0.60	Strongly Agree
3. Using this system increases productivity and reduces manual checking errors.	3.54	0.61	Strongly Agree
4. The system allows faster access to attendance data and reporting for decision-making.	3.51	0.51	Strongly Agree
Composite Mean	**3.55**	**0.54**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 21 presents the respondents' evaluation of the Perceived Ease of Use of the I am Present system in monitoring classroom attendance. The results show an overall mean of 3.39 with a standard deviation of 0.57, interpreted as "Strongly Agree." This indicates that the respondents found the system easy to use and operate. They strongly agreed that the I am Present is simple to manage and easy to navigate (mean = 3.46), that the system is user-friendly for overseeing poultry health and the environment (mean = 3.51), and that it can quickly complete attendance tasks (mean = 3.43). Although the web dashboard was rated slightly lower (mean = 3.14, "Agree"), the overall findings suggest that the system is generally accessible, intuitive, and convenient for teachers to operate effectively.

Table 21. Perceived Ease of Use based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. The system interface is simple and easy to navigate.	3.46	0.56	Strongly Agree
2. I find it easy to learn how to use the system and understand its features.	3.51	0.61	Strongly Agree
3. It requires little effort to operate the facial recognition and reporting functions.	3.14	0.60	Agree
4. I can accomplish attendance tasks quickly using the I am Present app.	3.43	0.50	Strongly Agree
Composite Mean	**3.39**	**0.57**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 22 presents the respondents' level of openness to using a facial recognition–based attendance system. The results indicate a high level of acceptance, with all indicators interpreted as Strongly Agree. Respondents showed strong interest in trying new technologies for monitoring student attendance (Mean = 3.49, SD = 0.61) and in using innovative systems that improve the accuracy and efficiency of attendance tracking (Mean = 3.43, SD = 0.50), reflecting a positive attitude toward adopting modern technological solutions. Likewise, respondents expressed openness to using facial recognition–based attendance systems in schools (Mean = 3.37, SD = 0.55), demonstrating a willingness to transition from traditional manual attendance methods. The highest agreement was observed in the motivation to use advanced technologies for classroom management and attendance tracking (Mean = 3.57, SD = 0.56), indicating strong enthusiasm for technology-supported educational processes. The composite mean of 3.46 (SD = 0.55), interpreted as Strongly Agree, confirms that users are highly receptive to

implementing the proposed attendance system. Overall, these findings suggest a favorable environment for the adoption and successful integration of the system in classroom settings.

Table 22. Personal Innovation based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I am interested in trying new technologies for monitoring student attendance.	3.49	0.61	Strongly Agree
2. I like using innovative systems that can improve the accuracy and efficiency of attendance monitoring.	3.43	0.50	Strongly Agree
3. I consider myself open to using facial recognition–based attendance systems in school.	3.37	0.55	Strongly Agree
4. I feel motivated to use advanced technologies to support classroom management and attendance tracking.	3.57	0.56	Strongly Agree
Composite Mean	**3.46**	**0.55**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 23 presents the respondents' perceptions of social influence regarding the use of the facial recognition–based attendance system. The results indicate a high level of agreement across all indicators, with all interpreted as Strongly Agree. Respondents acknowledged that people around them believe the system can reduce attendance errors and save time during class (Mean = 3.52, SD = 0.55). They agreed that teachers and school staff who are familiar with the system consider it capable of improving the accuracy and efficiency of attendance recording (Mean = 3.58, SD = 0.49). Positive feedback from individuals who have observed the system in use also motivates respondents to adopt it (Mean = 3.44, SD = 0.50). At the same time, support from co-teachers and school administrators further encourages its utilization

(Mean = 3.37, SD = 0.46). The composite mean of 3.48 (SD = 0.50), interpreted as Strongly Agree, confirms that social influence plays a supportive role in respondents' willingness to implement the attendance system. Overall, these findings suggest that peer and administrative support contribute positively to the potential adoption of the system in the school environment.

Table 23. Social Influence based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. People around me believe that using the face detection attendance system can reduce attendance errors and save time during class.	3.54	0.61	Strongly Agree
2. Teachers and school staff who learned about the system think it can improve the accuracy and efficiency of attendance recording.	3.57	0.50	Strongly Agree
3. Positive comments from people who have seen the system in use motivate me to adopt it.	3.43	0.50	Strongly Agree
4. My co-teachers and school administrators support the use of the attendance system.	3.29	0.46	Strongly Agree
Composite Mean	**3.46**	**0.52**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 24 shows the level of Information Awareness of the respondents regarding the I AM PRESENT attendance monitoring system. The results indicate that respondents are highly aware of the system's features and benefits, with an overall mean of 3.42 and a standard deviation of 0.53, interpreted as Strongly Agree. The highest mean score (3.58) reflects a clear understanding of how the facial recognition system records attendance from group photos, showing that system demonstrations were practical. Respondents also showed strong awareness of real-time attendance recording and parent notification features. Overall, the findings suggest that users are

well-informed about the system, which supports its acceptance and effective use in the school setting.

Table 24. Information Awareness based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I am aware that the system provides real-time attendance records using facial recognition.	3.46	0.56	Strongly Agree
2. I understand how the facial recognition system automatically records student attendance from group photos.	3.71	0.46	Strongly Agree
3. I am informed that the system helps reduce attendance errors and proxy attendance.	3.40	0.50	Strongly Agree
4. I am aware that the system sends attendance notifications to parents or guardians.	3.57	0.50	Strongly Agree
Composite Mean	**3.54**	**0.50**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 25 assesses User Satisfaction among respondents, indicating a high level of technological readiness and openness to change within the group. The overall mean score for User Satisfaction is 3.46, with a standard deviation of 0.55, and falls under the 'Strongly Agree' category. This conclusion is corroborated by individual items, all of which are in the "Strongly Agree" range. The highest mean score was for recommending the system to other teachers (3.57), while the lowest was for meeting expectations regarding accuracy and reliability (3.43). Overall, these results suggest that respondents are highly satisfied with the system.

Table 25. User Satisfaction based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I am satisfied with the overall performance of the I am Present system.	3.46	0.56	Strongly Agree
2. The system meets my expectations in terms of accuracy and reliability.	3.71	0.46	Strongly Agree
3. The system provides correct and consistent attendance results.	3.40	0.50	Strongly Agree
4. I would recommend this system to other teachers	3.57	0.50	Strongly Agree
Composite Mean	**3.54**	**0.50**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 26 illustrates the high Willingness to Adopt the I am Present system among the respondents, as evidenced by an overall mean score of 3.46 (SD = 0.57), which is strongly interpreted as Strongly Agree. The results are uniformly strong across all measured items, confirming the respondents' readiness to integrate the technology into their operations. The highest mean score (3.54, SD = 0.56) indicates a strong intent to recommend the I am Present system to other teachers, suggesting high satisfaction and advocacy for its benefits. Furthermore, respondents strongly agreed with their openness to adopting the system, citing its perceived positive impact on future school activities (3.49), their confidence in using it effectively after training (3.49), and their willingness to use it to monitor classroom attendance (3.34). This collective evidence confirms that the teachers are psychologically prepared and motivated to become active users and promoters of the I am Present system.

Table 26. Willingness to Adopt based on Respondents' Response

INDICATORS	MEAN	SD	INTERPRETATION
1. I intend to use the I am Present system regularly for attendance monitoring.	3.34	0.54	Strongly Agree
2. I plan to continue using the system in future school activities.	3.49	0.56	Strongly Agree
3. I am willing to learn more about its new features and updates.	3.49	0.61	Strongly Agree
4. I am likely to recommend this system to other schools and educators.	3.54	0.56	Strongly Agree
Composite Mean	**3.46**	**0.57**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Table 27 displays respondents' evaluations of the system based on the Technology Acceptance Model (TAM). All indicators received positive feedback, with mean scores between 3.39 and 3.55, mostly indicating Strongly Agree. This suggests respondents view the system favorably in terms of usefulness, ease of use, and overall acceptance. Perceived Usefulness scored the highest at 3.55, highlighting strong confidence that the system enhances attendance monitoring. Information Awareness also scored well at 3.54, showing users understand its features and benefits clearly. Factors like Perceived Ease of Use, Personal Innovation, Social Influence, and Willingness to Adopt all received strong agreement, reflecting users' readiness and confidence. Perceived Cost had a mean of 3.54, categorized as Agree, indicating general acceptance but implying cost is a minor concern. The overall mean of 3.49 confirms respondents' strong agreement with the system's acceptability. These results indicate high user acceptance and support the system's successful potential in a school environment.

Table 27. Result of Technology Acceptance Model based on Response

INDICATORS	MEAN	SD	INTERPRETATION
Perceived Usefulness	3.55	0.54	Strongly Agree
Perceived Ease of Use	3.39	0.57	Strongly Agree
Personal Innovation	3.46	0.55	Strongly Agree
Social Influence	3.46	0.52	Strongly Agree
Information Awareness	3.54	0.50	Strongly Agree
Perceived Cost	3.54	0.50	Agree
Willingness to Adopt	3.46	0.57	Strongly Agree
Overall Mean	**3.49**	**0.54**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25: Agree; 3.26-4.0: Strongly Agree

Result of Software Testing

Each module of the system was tested through multiple iterations to evaluate performance, reliability, and correctness. The number of tests runs for each feature varied depending on the minor issues encountered during testing. Modules like Login Authentication, Student Registration, and Class Management initially experienced minor issues, including incorrect credentials, invalid input, or timestamp mismatches. These issues were identified and corrected, after which the modules passed subsequent test runs. Similarly, the Reporting and Export Module experienced a slight delay due to network latency on one run but was otherwise successful. Other features, such as Dashboard, Notification System, and Settings & Configuration, performed smoothly with minimal to no errors and required fewer test cycles. Overall, the total test runs ranged from two (2) to four (4) executions per module. After completing all iterations, all modules successfully passed their respective tests, confirming

that the system functions as intended, delivers reliable performance, and maintains consistency across repeated operations.

Table 28. Unit Testing for Web-Based Module

FEATURE TESTED	PERFORMANCE RESULT	RESPONSE TIME	NO. OF TEST RUNS	REMARKS
Login Authentication module	First test failed due to incorrect credentials; fixed and passed in subsequent runs	1 second	4 runs (1 failed, 3 passed)	Passed
Dashboard	All dashboard data loaded correctly; minor UI delay detected on first run, but resolved	1.5 seconds	3 runs (all passed)	Passed
Class Management Module	Classes added and updated successfully; second run needed due to timestamp mismatch	2 seconds	3 runs (2 passed, 1 needed re-test due to timestamp)	Passed
Student Registration Module	Student records saved correctly; first run failed due to invalid input; resolved on retry	1.8 seconds	4 runs (1 failed, 3 passed)	Passed
Notification System Module	Notifications sent correctly; no issues detected	1.6 seconds	3 runs (all passed)	Passed
Reporting and Export Module.	Reports exported successfully; one run delayed due to network latency	2.3 seconds	4 runs (1 delayed, 3 passed)	Passed
Settings & Configuration	Settings applied correctly; stable behavior across all test runs	2 seconds	2 runs (both passed)	Passed

The integration testing confirmed that all modules of the Facial Recognition Attendance System worked together smoothly. Face recognition, student record updating, dashboard display, notifications, and report generation functioned without interruption. Attendance data was accurately processed and displayed in real-time, notifications were sent promptly, and reports were generated correctly. Overall, the system proved reliable, efficient, and optimized for automated attendance management.

Integration Testing Result

Table 29. Integration Testing for Web-Based Module

FEATURE TESTED	PERFORMANCE RESULT	RESPONSE TIME	NO. OF TEST RUNS	REMARKS
Login Authentication module	First test failed due to incorrect credentials; fixed and passed in subsequent runs	1 second	4 runs (1 failed, 3 passed)	Passed
Dashboard	All dashboard data loaded correctly; minor UI delay detected on first run, but resolved	1.5 seconds	3 runs (all passed)	Passed
Class Management Module	Classes added and updated successfully; second run needed due to timestamp mismatch	2 seconds	3 runs (2 passed, 1 needed re-test due to timestamp)	Passed
Student Registration Module	Student records saved correctly; first run failed due to invalid input; resolved on retry	1.8 seconds	4 runs (1 failed, 3 passed)	Passed
Notification System Module	Notifications sent correctly; no issues detected	1.6 seconds	3 runs (all passed)	Passed
Reporting and Export Module.	Reports exported successfully; one run delayed due to network latency	2.3 seconds	4 runs (1 delayed, 3 passed)	Passed
Settings & Configuration	Settings applied correctly; stable behavior across all test runs	2 seconds	2 runs (both passed)	Passed

Load testing showed that the Facial Recognition Attendance System remained stable and functional with 10–15 concurrent users. The dashboard, face

recognition, attendance logging, notifications, and report generation operated smoothly, with only minor delays during bulk uploads or rapid report generation. Login functions remained stable, confirming the system's reliability for school deployment.

Load Testing Result

Table 30. Load Testing Result for Web-Based Module

FEATURE TESTED	LOAD CONDITION	PERFORMANCE RESULT	RESPONSE TIME	NUMBER OF TESTS	REMARKS
Dashboard Statistics	15 simultaneous users are viewing the attendance dashboard	System handled load smoothly; a slight delay in chart refresh was observed under peak usage	1.9–2.4 seconds	5 runs	Passed
Face Recognition Processing	Upload and processing of 1 group shot containing 7 students	No identification loss; slight delay in processing due to high image resolution	2.1 seconds	3 cycles	Passed
Attendance Notifications	7 face detection events triggered simultaneously	8 emails sent instantly; 2 notifications delayed due to server-side queuing	2.8 seconds	4 runs	Passed
Attendance Logs (Bulk Insert)	Simultaneous update of 200 student records in the database	Database accepted all inputs; minor retry required for 12 records due to concurrent update conflict	2.5–3.0 seconds	3 runs	Passed (with retry)
Face Matching Computation	7 FaceNet embedding comparisons triggered simultaneously	System remained stable; recognition queue increased slightly during peak matching	1.6–2.3 seconds	5 runs	Passed
User Authentication	20 teachers/admins logging in within 30 seconds	All login attempts processed; 1 manual failure due to incorrect credentials, not system load	1.8 seconds	3 runs	Passed

The system performed consistently across all major web browsers. Google Chrome provided the fastest loading times, while Microsoft Edge and

Brave remained stable with minimal delays. Mozilla Firefox handled real-time data correctly, but slowed slightly with large attendance datasets. Safari loaded all modules properly with only minor UI wrapping issues. Overall, the system is highly compatible across tested browsers and supports cross-platform accessibility.

Compatibility Testing Result

Table 31. Compatibility Testing Results for each Web-Browser Platform

PLATFORMS	EXPECTED FUNCTIONALITY	PERFORMANCE RESULT	RESPONSE TIME	NUMBER OF TESTS	REMARKS
Google Chrome	Should fully load the dashboard, attendance data, notifications, and reports	All features loaded smoothly; fastest rendering among all browsers.	1.5 seconds	5 tests	Passed
Microsoft Edge	Should display dashboard, charts, and notifications consistently	Fully functional; slight delay in chart transitions, but stable overall	1.7 seconds	5 tests	Passed
Mozilla Firefox	Should correctly display real-time data and support a responsive layout	Data loaded properly; minor slowdown when handling large attendance tables	2.0 seconds	5 tests	Passed
Brave Browser	Should load dashboard and charts normally with default security settings	Dashboard functional; chart script blocked on 1 run due to shield, resolved after allowing scripts	1.8 seconds	4 tests	Passed
Safari	Should display dashboard, logs, and reports properly with a responsive UI	All modules loaded; slight text wrapping in table/graph labels, no functional errors	1.9 seconds	5 tests	Passed

The stress testing results confirmed that every component of the system functioned effectively and met the required standards for accuracy, speed, and reliability. Each module was thoroughly examined to ensure it performed as

expected. During actual use, the system stayed stable while processing multiple face scans, recording attendance, and matching faces from the database. The system handled real-time face detection and recognition, identity matching, and attendance logging without delays or data errors. It remained responsive during long hours of continuous operation and during simultaneous access by multiple users. Database performance stayed consistent even as the number of registered faces increased.

Result of Stress Test

Table 32. Stress Testing Result

Test ID	Steps	Expected Results	Actual Results	Remarks
1	High Volume Face Detection Test	The system should detect and recognize faces accurately when many faces appear within a short time.	The system recognized all registered faces correctly with only a slight delay when multiple faces were scanned at once.	Passed
2	Concurrent Attendance Marking Test	The system should mark attendance correctly even when many users check in at the same time.	Attendance records were saved accurately while several users scanned their faces simultaneously.	Passed
3	Low Light and Angle Stress Test	The system should still recognize faces under poor lighting and different face angles.	Most faces were recognized successfully with minor accuracy drops in very low light conditions.	Passed
4	Continuous Operation Test	The system should function continuously for a long period without crashing or slowing down.	The system ran for 24 hours while processing facial scans without any interruptions or errors.	Passed

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

This chapter summarizes and discusses the main findings of the study. It highlights the study's importance and demonstrates how the results meet the objectives and challenges outlined in Chapter I. Additionally, it offers practical recommendations for system improvement and suggests future research areas to expand on the insights obtained from this study.

Summary

This study centered on developing and accessing I AM PRESENT, a web-based attendance system utilizing group-shot facial recognition to enhance the accuracy and efficiency of recording student attendance at Cabuyao Integrated National High School. The school's traditional manual attendance methods were slow, error-prone, and vulnerable to proxy attendance, prompting the need for an automated, reliable solution.

The system integrates MTCNN for face detection and FaceNet for facial recognition to identify multiple students from a single group photo taken during class sessions. Once recognized, attendance is automatically recorded in real time and securely stored in a cloud-based database. The system also includes features such as class management, student registration, attendance reporting, CSV export, and automated email notifications to parents or guardians.

To assess the quality and acceptability of the system, the researchers conducted system testing and evaluation using ISO 25010 software quality standards and the Technology Acceptance Model (TAM). IT specialists evaluated the system based on functional suitability, performance efficiency, compatibility, reliability, security, maintainability, flexibility, and safety.

Meanwhile, teachers and users assessed perceived usefulness, perceived ease of use, user satisfaction, and willingness to adopt the system.

The evaluation results showed that the system performed effectively across all tested criteria, indicating that the developed web-based facial recognition attendance system is functional, reliable, user-friendly, and suitable for classroom use.

Conclusion

Verifying System Effectiveness via Group Shot Facial Recognition for Attendance Tracking

The implementation of the I AM PRESENT system demonstrated that group-shot facial recognition is a practical approach to automating classroom attendance monitoring. System testing confirmed that multiple students could be accurately detected and identified from a single captured image, enabling efficient recording of attendance without requiring individual scanning. This validated the system's capability to address common issues associated with manual attendance methods, such as time consumption and proxy attendance.

Performance of the Facial Recognition Module in Real-Time Attendance Recording

The integration of the MTCNN and FaceNet models enabled reliable face detection and recognition under typical classroom conditions. During system operation, recognized faces were successfully matched with registered student records, triggering immediate attendance logging. This real-time processing ensured that attendance data were consistently updated and securely stored in the system database, supporting accurate record-keeping during class sessions.

Effectiveness of Automated Email Notifications in Supporting Attendance Monitoring

The automated email notification feature functioned as intended by informing parents or guardians once attendance was recorded. System testing confirmed that notifications were generated promptly after attendance submission, supporting timely awareness of student presence. This feature enhanced communication between the school and guardians and supported monitoring of student attendance status.

Reliability of the Web-Based Platform for Centralized Attendance Management and Reporting

The web-based platform provided a centralized environment for managing classes, registering students, and monitoring attendance records. Teachers and administrators could view attendance summaries and export records in CSV format, improving documentation efficiency and supporting administrative decision-making. The system's centralized design reduced manual record handling and improved data organization.

Evaluation of System Quality Based on ISO 25010 Software Quality Standards

Evaluation results based on ISO 25010 indicated that the system achieved strong performance in functional suitability, usability, performance efficiency, compatibility, reliability, and security, as assessed by IT experts. While flexibility and safety were rated as acceptable, these characteristics were identified as areas for future enhancement rather than limitations affecting current system operation. Overall, the evaluation confirmed that the system meets essential software quality requirements for classroom use.

Assessment of User Acceptance Using the Technology Acceptance Model

User acceptance evaluation using the Technology Acceptance Model revealed that teachers perceived the system as helpful and easy to use. Respondents agreed that the system reduced the workload of manual attendance and was easy to learn and operate. These findings indicate a positive level of user satisfaction and suggest strong potential for adoption in secondary school environments.

Recommendations

The I am Present system met its objectives and functioned effectively, as evidenced by the results and evaluations. However, further enhancements can improve accuracy, flexibility, and long-term usability. The following recommendations are suggested:

1. Encourage the integration of attendance analytics and reporting tools to support data-driven decision-making among teachers and school administrators. Features such as attendance trend analysis, absenteeism pattern monitoring, and performance correlation reports can help schools identify attendance-related concerns early and improve intervention strategies.
2. Expand notification channels beyond email by incorporating SMS notifications and mobile application alerts. This enhancement can ensure timely communication with parents or guardians, particularly in areas with limited or inconsistent internet access, and strengthen real-time attendance monitoring.

3. Promote further research on the educational impact of facial recognition-based attendance systems, particularly in relation to student punctuality, attendance behavior, and academic performance. Comparative studies between traditional attendance methods and automated systems may help validate long-term effectiveness and educational benefits.
4. Develop an offline or hybrid attendance capability to address internet dependency in school environments with unstable connectivity. Allowing attendance data to be stored locally and synchronized once internet access is restored can improve system reliability and continuity of operations.
5. Implement an ethical consent management module that records and manages parental and student consent for the use of facial recognition data. Providing transparency regarding biometric data collection, storage, and usage can increase trust, compliance, and acceptance among stakeholders.
6. Evaluate the feasibility of cross-school and division-level adoption of the system within the City Schools Division of Cabuyao. Standardizing digital attendance systems across public schools may improve reporting accuracy, monitoring efficiency, and data consistency at the administrative level.
7. Evaluate the inclusion of a visual validation feature that displays both the original reference image and the latest captured image during attendance verification to improve accuracy and transparency.

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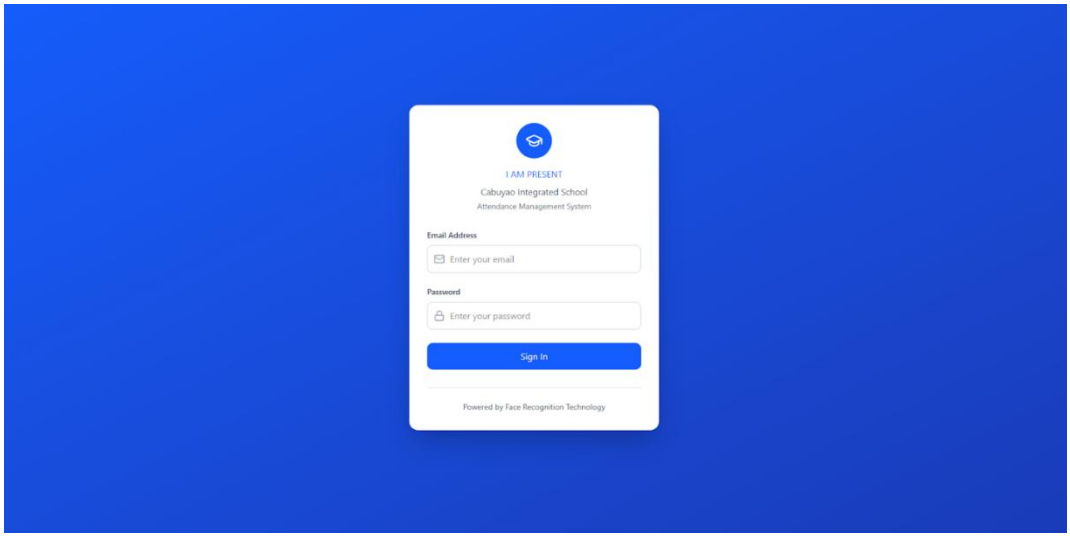
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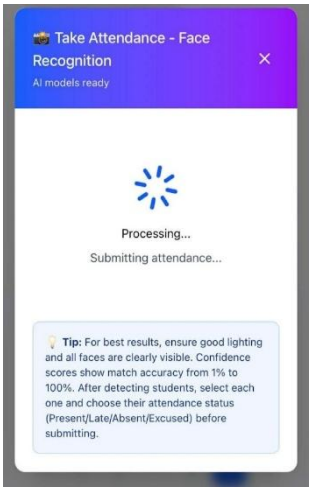
APPENDIX A

Output Screenshot

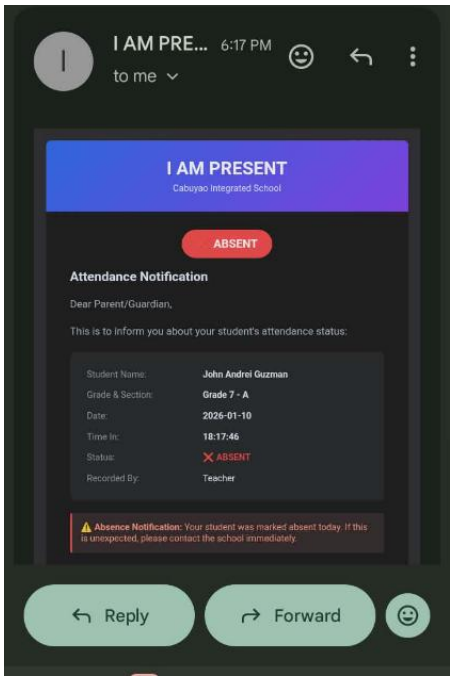


This illustration shows the User Login Interface, the main gateway to the "I AM PRESENT" Attendance Management System for Cabuyao Integrated School. The design features a centered white login card set against a vibrant blue background, with a clear visual hierarchy that directs user attention. At the top, a graduation cap icon alongside the system title immediately conveys the system's educational purpose. The interface includes two separate input fields for Email Address and Password; each paired with intuitive icons to enhance usability and reduce input errors. A prominent blue "Sign In" button functions as the main call-to-action, while the footer clearly states that the backend operates on "Powered by Face Recognition Technology." This layout underscores a professional and secure approach to handling biometric data, ensuring that only authorized users can access the sensitive attendance logs and facial data stored in the system.

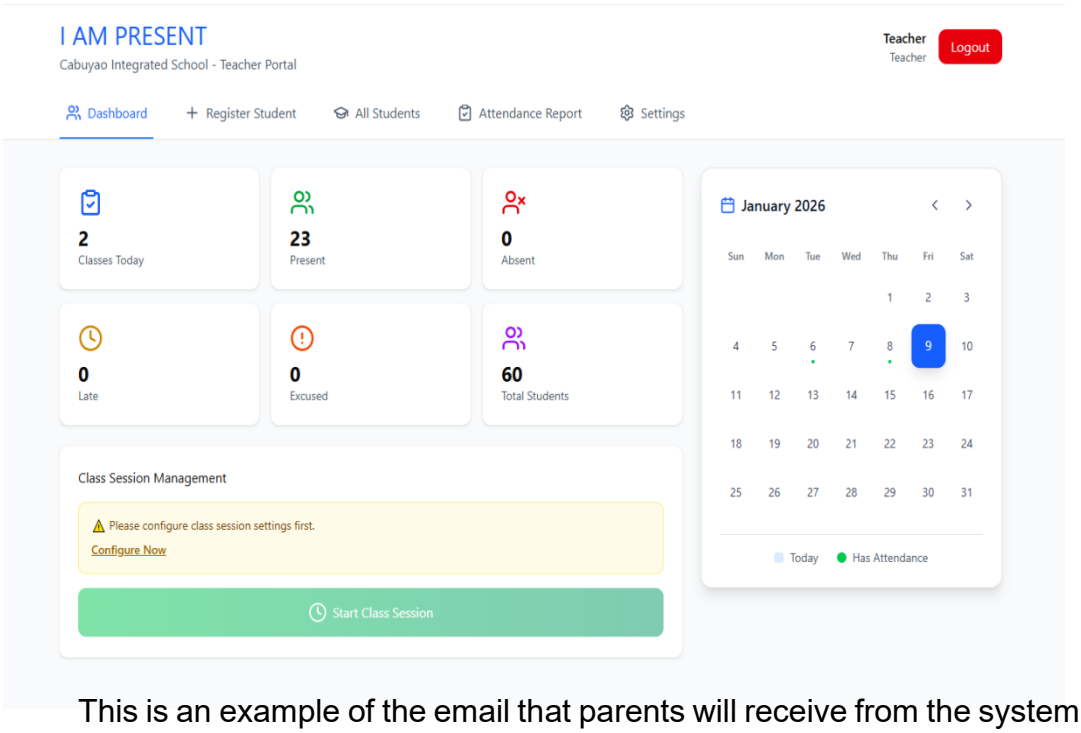
This illustration depicts a practical, modern approach to managing classroom activities using a single platform. Teachers can effortlessly view and organize student information, register new students, and store essential personal and academic details, including facial data. At the start of a class, the teacher opens the system, begins the session, and takes attendance by capturing a group photo of the students. The system recognizes each face and automatically marks attendance, saving time and minimizing errors. All records are securely stored and accessible later through clear, organized reports, simplifying the tracking of student participation over time. Additionally, there is a settings section where users can update their profile details and change passwords to ensure account security. Overall, this system streamlines classroom management by consolidating student records, attendance tracking, reports, and account settings into a single, easy-to-use platform.





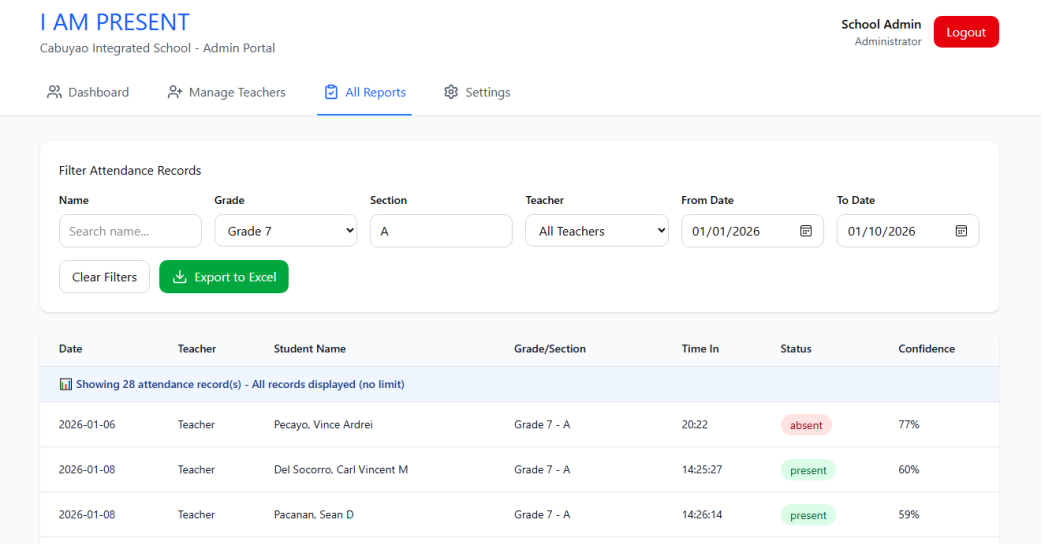


This page is designed for taking attendance efficiently using facial recognition. After capturing the students' faces, the system automatically detects and displays the corresponding names based on the recognized faces.



This is an example of the email that parents will receive from the system. The email contains important details about their child, including the student's

name, grade, and section, the date, and the time the student arrived at school. This notification keeps parents informed about their child's attendance in a clear and timely manner.

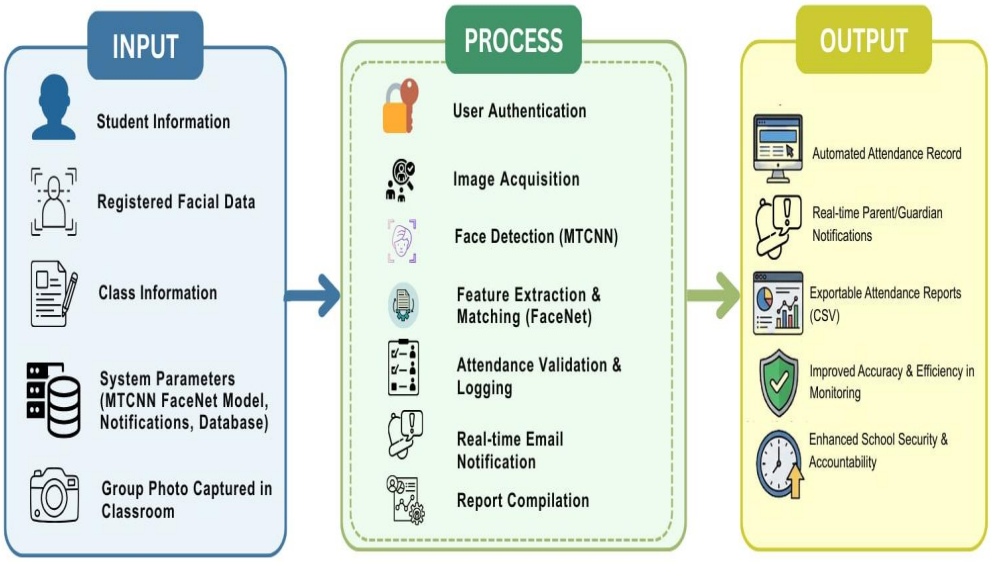


The All-Reports section in the admin dashboard allows administrators to easily view and export student attendance performance. This feature includes filtering options that make it simple to organize and select specific records before exporting. Once filtered, the data can be exported as an Excel file, making it more convenient to review, analyze, and maintain records in a clear, organized format.

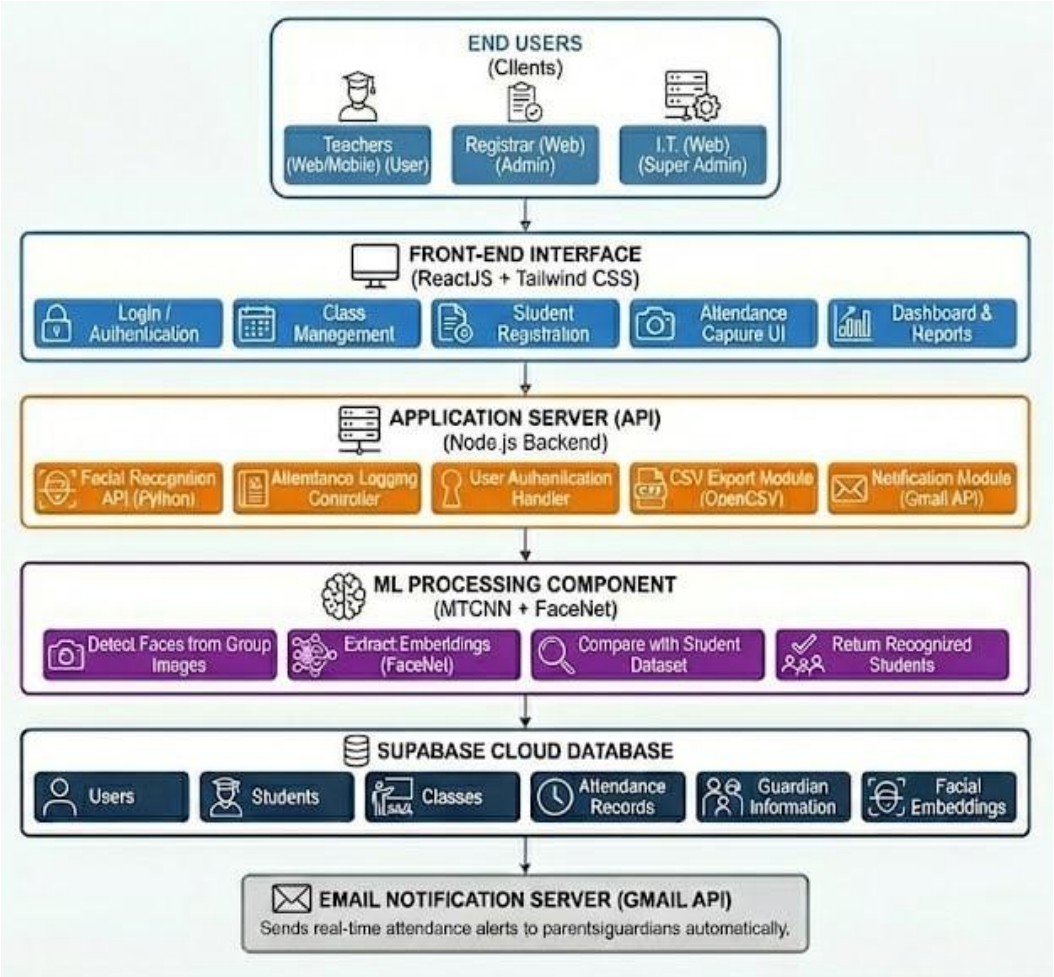
APPENDIX B

Figures

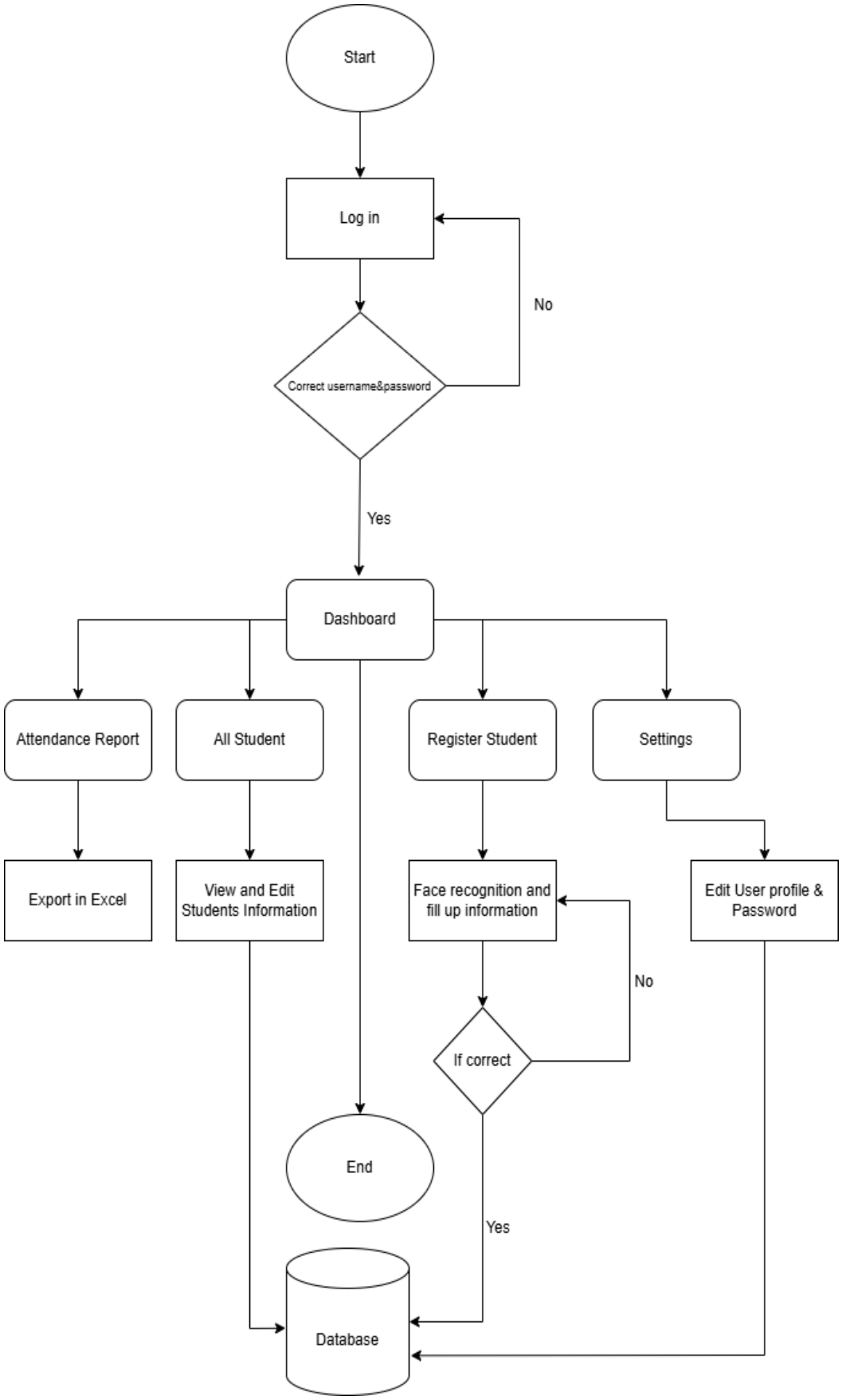
Conceptual Framework

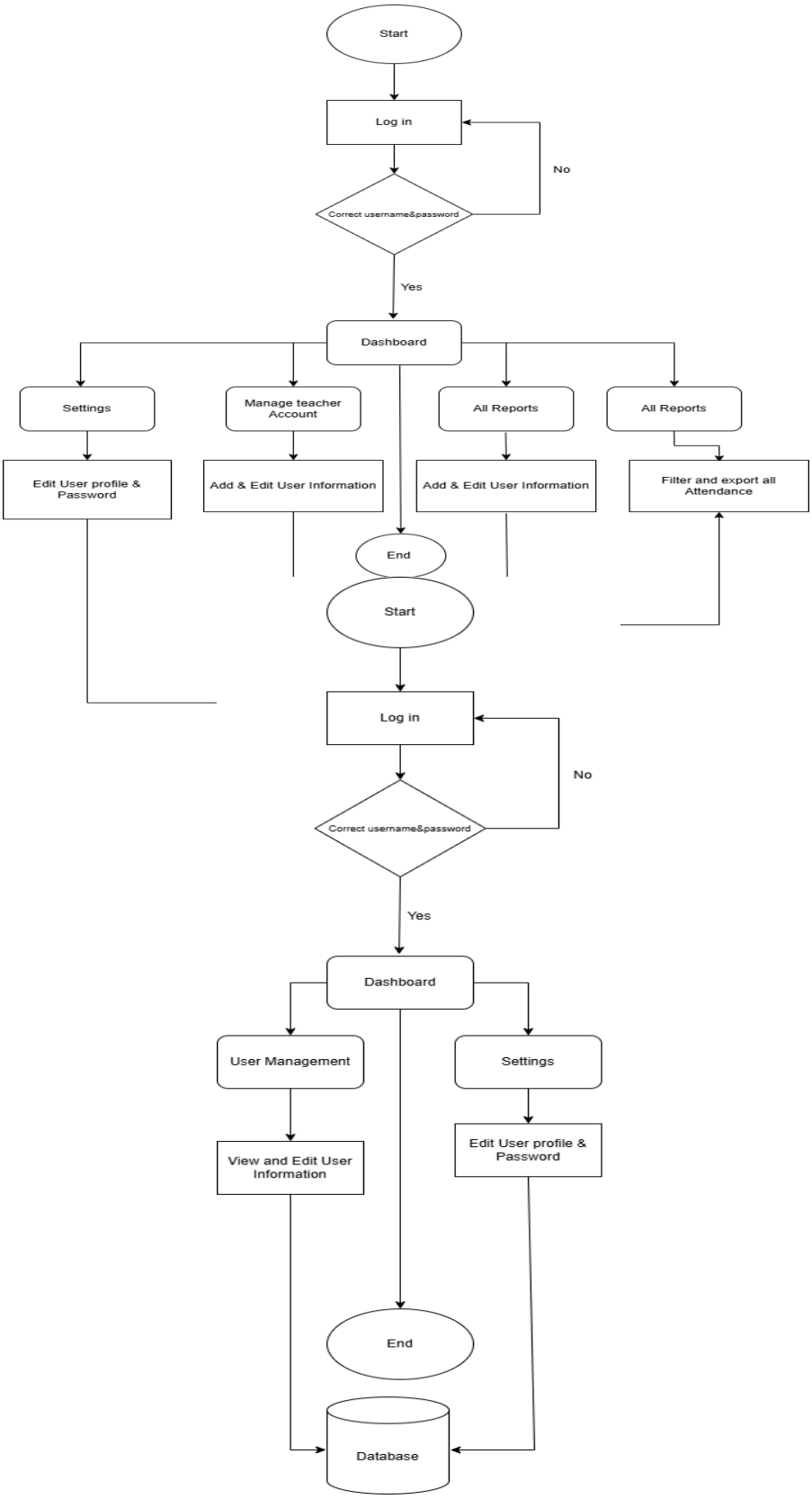


System Architecture

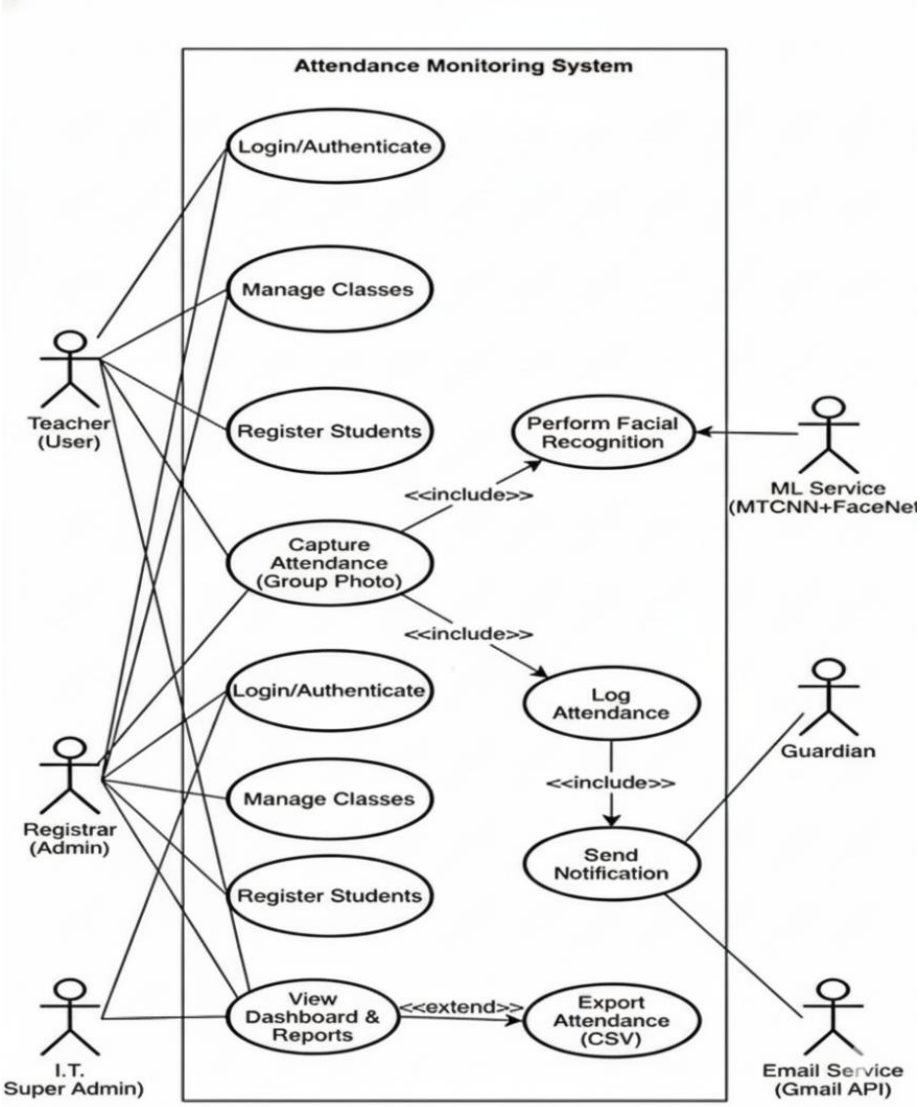


System Flow Chart



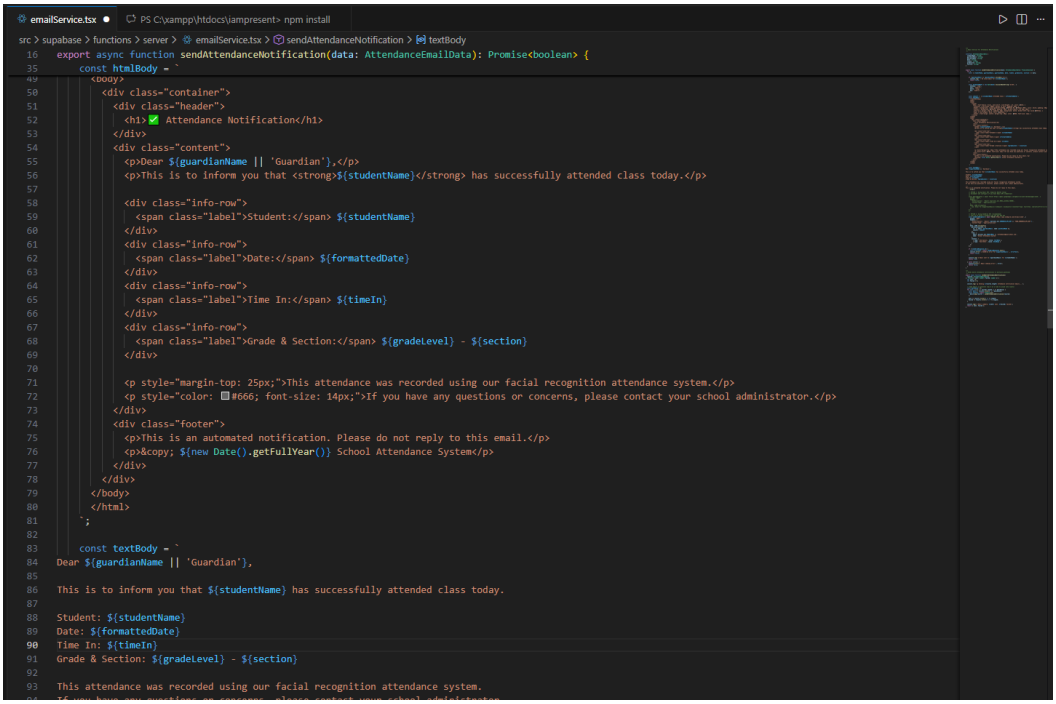

Start
Log in
No
Correct username&password
Yes
Dashboard
Settings
Manage teacher Account
All Reports
All Reports
Edit User profile & Password
Add & Edit User Information
Add & Edit User Information
Filter and export all Attendance
End
Start
Log in
No
Correct username&password
Yes
Dashboard
User Management
Settings
View and Edit User Information
Edit User profile & Password
End
Database

Use Case Diagram

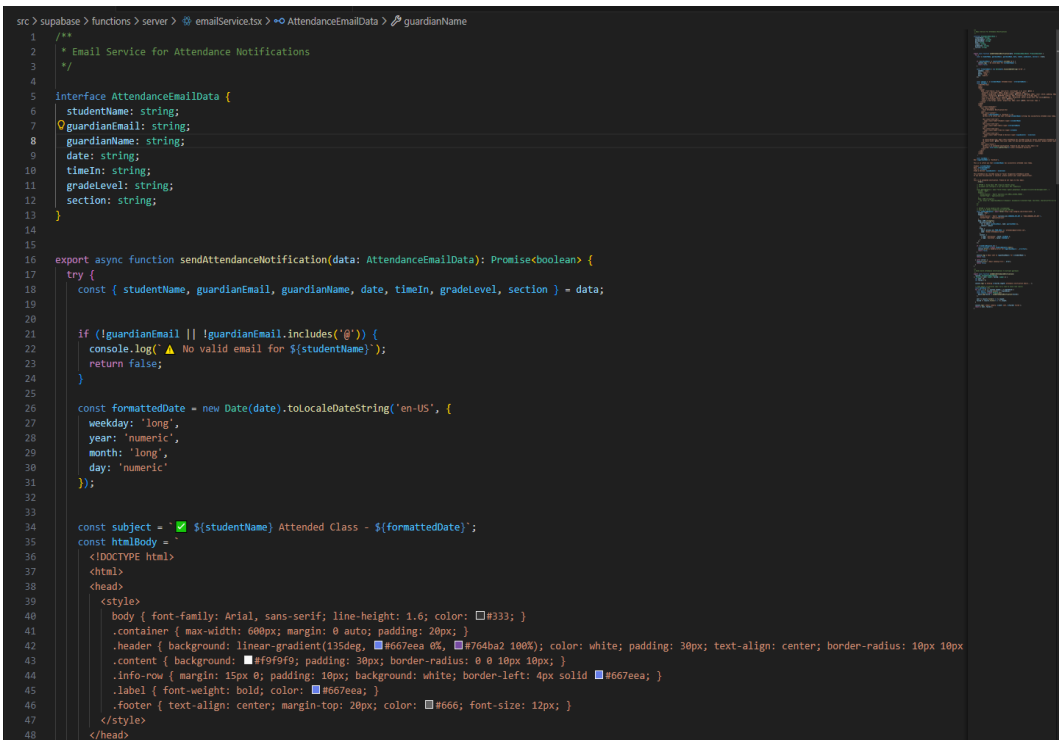


APPENDIX C

Source Code



```
emailService.tsx ● PS C:\xampp\htdocs\iampresent> npm install
src > supabase > functions > server > emailService.tsx > sendAttendanceNotification > textBody
16  export async function sendAttendanceNotification(data: AttendanceEmailData): Promise<boolean> {
35    const htmlBody = `
50      <body>
51        <div class="container">
52          <div class="header">
53            <h1>✅ Attendance Notification</h1>
54          </div>
55          <div class="content">
56            <p>Dear ${guardianName || 'Guardian'},</p>
57            <p>This is to inform you that <strong>${studentName}</strong> has successfully attended class today.</p>
58
59            <div class="info-row">
60              <span class="label">Student:</span> ${studentName}
61            </div>
62            <div class="info-row">
63              <span class="label">Date:</span> ${formattedDate}
64            </div>
65            <div class="info-row">
66              <span class="label">Time In:</span> ${timeIn}
67            </div>
68            <div class="info-row">
69              <span class="label">Grade & Section:</span> ${gradeLevel} - ${section}
70            </div>
71
72            <p style="margin-top: 25px;">This attendance was recorded using our facial recognition attendance system.</p>
73            <p style="color: #666; font-size: 14px;">If you have any questions or concerns, please contact your school administrator.</p>
74          </div>
75          <div class="footer">
76            <p>This is an automated notification. Please do not reply to this email.</p>
77            <p>&copy; ${new Date().getFullYear()} School Attendance System</p>
78          </div>
79        </div>
80      </body>
81      </html>
82    `;
83
84    const textBody = `
85  Dear ${guardianName || 'Guardian'},
86
87  This is to inform you that ${studentName} has successfully attended class today.
88
89  Student: ${studentName}
90  Date: ${formattedDate}
91  Time In: ${timeIn}
92  Grade & Section: ${gradeLevel} - ${section}
93
94  This attendance was recorded using our facial recognition attendance system.
```



```
src > supabase > functions > server > emailService.tsx > AttendanceEmailData > guardianName
1   /**
2    * Email Service for Attendance Notifications
3    */
4
5   interface AttendanceEmailData {
6     studentName: string;
7     guardianEmail: string;
8     guardianName: string;
9     date: string;
10    timeIn: string;
11    gradeLevel: string;
12    section: string;
13  }
14
15
16  export async function sendAttendanceNotification(data: AttendanceEmailData): Promise<boolean> {
17    try {
18      const { studentName, guardianEmail, guardianName, date, timeIn, gradeLevel, section } = data;
19
20
21      if (!guardianEmail || !guardianEmail.includes('@')) {
22        console.log(`⚠️ No valid email for ${studentName}`);
23        return false;
24      }
25
26      const formattedDate = new Date(date).toLocaleDateString('en-US', {
27        weekday: 'long',
28        year: 'numeric',
29        month: 'long',
30        day: 'numeric'
31      });
32
33
34      const subject = `✅ ${studentName} Attended Class - ${formattedDate}`;
35      const htmlBody = `
36        <!DOCTYPE html>
37        <html>
38        <head>
39          <style>
40            body { font-family: Arial, sans-serif; line-height: 1.6; color: #333; }
41            .container { max-width: 600px; margin: 0 auto; padding: 20px; }
42            .header { background: linear-gradient(135deg, #667eea 0%, #764ba2 100%); color: white; padding: 30px; text-align: center; border-radius: 10px 10px
43            .content { background: #f9f9f9; padding: 30px; border-radius: 0 0 10px 10px; }
44            .info-row { margin: 15px 0; padding: 10px; background: white; border-left: 4px solid #667eea; }
45            .label { font-weight: bold; color: #667eea; }
46            .footer { text-align: center; margin-top: 20px; color: #666; font-size: 12px; }
47          </style>
48        </head>
```

Real-time notification to parents/guardians.

src > components > student-registration-camera.tsx > ...

```tsx
import React, { useRef, useState, useEffect } from 'react';
import { Camera, X, CheckCircle, AlertCircle, Loader, RefreshCw } from 'lucide-react';
import { toast } from 'sonner@2.0.3';
import { enrollStudentFace, validateFaceQuality, areModelsLoaded, loadFaceRecognitionModels } from '../utils/faceRecognition';

interface StudentRegistrationCameraProps {
  onPhotoCapture: (photoData: string, faceDescriptor: Float32Array) => void;
  onClose: () => void;
}

export function StudentRegistrationCamera({ onPhotoCapture, onClose }: StudentRegistrationCameraProps) {
  const videoRef = useRef<HTMLVideoElement>(null);
  const [stream, setStream] = useState<MediaStream | null>(null);
  const [capturedPhoto, setCapturedPhoto] = useState<string>('');
  const [isProcessing, setIsProcessing] = useState(false);
  const [faceDescriptor, setFaceDescriptor] = useState<Float32Array | null>(null);
  const [validationMessage, setValidationMessage] = useState('');
  const [validationStatus, setValidationStatus] = useState<'idle' | 'success' | 'error'>('idle');
  const [modelsLoaded, setModelsLoaded] = useState(false);
  const [facingMode, setFacingMode] = useState<'user' | 'environment'>('user');

  useEffect(() => {
    initializeModels();
    startCamera();
    return () => {
      stopCamera();
    };
  }, [facingMode]);

  const initializeModels = async () => {
    if (areModelsLoaded()) {
      setModelsLoaded(true);
      return;
    }

    const loaded = await loadFaceRecognitionModels();
    setModelsLoaded(loaded);

    if (!loaded) {
      toast.error('Failed to load face recognition models');
    }
  };

  const startCamera = async () => {
    try {
      const mediaStream = await navigator.mediaDevices.getUserMedia({
        video: {
          width: { ideal: 1920 },
          height: { ideal: 1080 },
```

src > components > student-registration-camera.tsx > ...

```tsx
export function StudentRegistrationCamera({ onPhotoCapture, onClose }: StudentRegistrationCameraProps) {
  const startCamera = async () => {
    try {
      const mediaStream = await navigator.mediaDevices.getUserMedia({
        video: {
          width: { ideal: 1920 },
          height: { ideal: 1080 },
          facingMode: facingMode
        }
      });
      setStream(mediaStream);
      if (videoRef.current) {
        videoRef.current.srcObject = mediaStream;
      }
    } catch (error) {
      console.error('Camera error:', error);
      toast.error('Unable to access camera. Please check permissions.');
    }
  };

  const stopCamera = () => {
    if (stream) {
      stream.getTracks().forEach(track => track.stop());
      setStream(null);
    }
  };

  const flipCamera = () => {
    stopCamera();
    setFacingMode(prev => prev === 'user' ? 'environment' : 'user');
  };

  const capturePhoto = async () => {
    if (!videoRef.current) return;

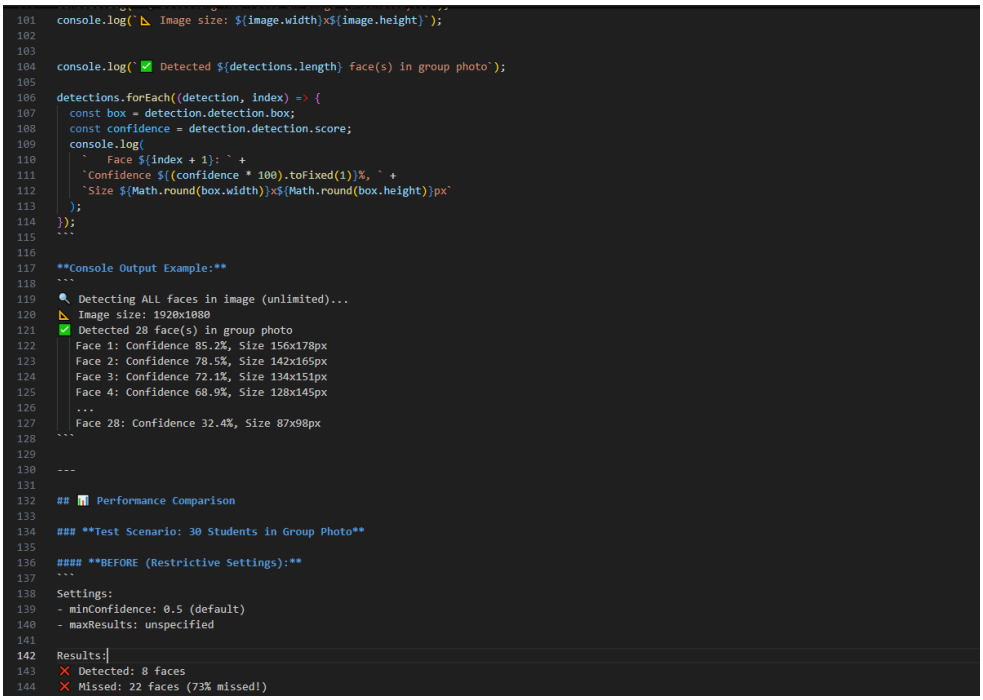
    const canvas = document.createElement('canvas');
    const video = videoRef.current;
    canvas.width = video.videoWidth;
    canvas.height = video.videoHeight;

    const ctx = canvas.getContext('2d');
    if (ctx) {
      ctx.drawImage(video, 0, 0);
      const photoData = canvas.toDataURL('image/jpeg', 0.95);
      setCapturedPhoto(photoData);
      stopCamera();

      // Process the photo
      await processPhoto(photoData);
    }
  };

  const processPhoto = async (photoData: string) => {
    setIsProcessing(true);
```

Student Registration



```
console.log(`📐 Image size: ${image.width}x${image.height}`);


console.log(`✅ Detected ${detections.length} face(s) in group photo`);

detections.forEach((detection, index) => {
  const box = detection.detection.box;
  const confidence = detection.detection.score;
  console.log(
    `   Face ${index + 1}: ` +
    `Confidence ${(confidence * 100).toFixed(1)}%, ` +
    `Size ${Math.round(box.width)}x${Math.round(box.height)}px`
  );
});
```

Console Output Example:

```
🔍 Detecting ALL faces in image (unlimited)...
📐 Image size: 1920x1080
✅ Detected 28 face(s) in group photo
   Face 1: Confidence 85.2%, Size 156x178px
   Face 2: Confidence 78.5%, Size 142x165px
   Face 3: Confidence 72.1%, Size 134x151px
   Face 4: Confidence 68.9%, Size 128x145px
   ...
   Face 28: Confidence 32.4%, Size 87x98px
```

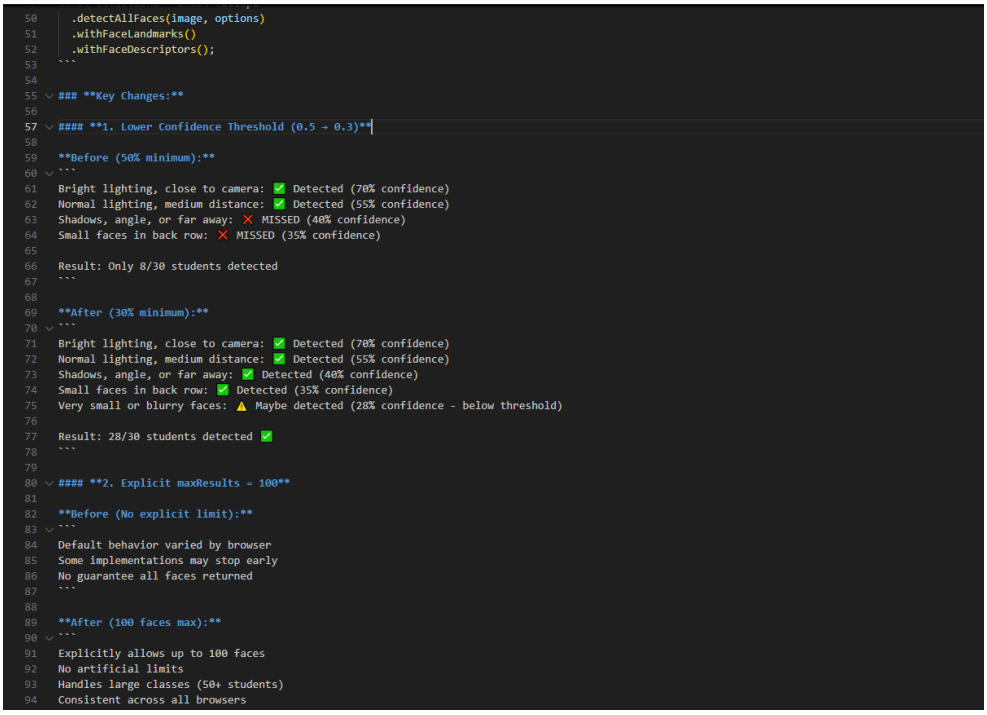
📊 Performance Comparison

Test Scenario: 30 Students in Group Photo

BEFORE (Restrictive Settings):

```
Settings:
- minConfidence: 0.5 (default)
- maxResults: unspecified

Results:
❌ Detected: 8 faces
❌ Missed: 22 faces (73% missed!)
```



```
  .detectAllFaces(image, options)
  .withFaceLandmarks()
  .withFaceDescriptors();
```

Key Changes:

1. Lower Confidence Threshold (0.5 → 0.3)

Before (50% minimum):

```
Bright lighting, close to camera: ✅ Detected (70% confidence)
Normal lighting, medium distance: ✅ Detected (55% confidence)
Shadows, angle, or far away: ❌ MISSED (40% confidence)
Small faces in back row: ❌ MISSED (35% confidence)

Result: Only 8/30 students detected
```

After (30% minimum):

```
Bright lighting, close to camera: ✅ Detected (70% confidence)
Normal lighting, medium distance: ✅ Detected (55% confidence)
Shadows, angle, or far away: ✅ Detected (40% confidence)
Small faces in back row: ✅ Detected (35% confidence)
Very small or blurry faces: ⚠️ Maybe detected (28% confidence - below threshold)

Result: 28/30 students detected ✅
```

2. Explicit maxResults = 100

Before (No explicit limit):

```
Default behavior varied by browser
Some implementations may stop early
No guarantee all faces returned
```

After (100 faces max):

```
Explicitly allows up to 100 faces
No artificial limits
Handles large classes (50+ students)
Consistent across all browsers
```

Group Face Detection

EXCEL_EXPORT_ATTENDANCE_DOCUMENTATION.md

```
 * Export attendance records to Excel file
 * Location: /components/teacher-dashboard-new.tsx (Lines 462-492)
 */
const exportToExcel = () => {
  // Step 1: Filter records by date range
  const filteredRecords = attendanceRecords.filter(record => {
    const recordDate = record.date;
    return recordDate >= dateFilter.from && recordDate <= dateFilter.to;
  });

  // Step 2: Prepare data for export
  const exportData: any[] = [];
  filteredRecords.forEach(record => {
    record.records.forEach((studentRecord: any) => {
      const student = students.find(s => s.id === studentRecord.id);
      if (student) {
        exportData.push({
          Date: record.date,
          'Last Name': student.lastName,
          'First Name': student.firstName,
          'Middle Initial': student.middleInitial,
          'Grade Level': student.gradeLevel,
          Section: student.section,
          Status: studentRecord.status,
          'Time In': studentRecord.timeIn,
          'Confidence': studentRecord.confidence
            ? `${(studentRecord.confidence * 100).toFixed(0)}%`
            : 'N/A'
        });
      }
    });
  });

  // Step 3: Create Excel worksheet
  const worksheet = XLSX.utils.json_to_sheet(exportData);

  // Step 4: Create workbook and add worksheet
  const workbook = XLSX.utils.book_new();
  XLSX.utils.book_append_sheet(workbook, worksheet, 'Attendance');

  // Step 5: Download file with dynamic filename
  XLSX.writeFile(workbook, `attendance_${dateFilter.from}_to_${dateFilter.to}.xlsx`);
};
```

EXCEL_EXPORT_ATTENDANCE_DOCUMENTATION.md

````
### UI Button Code

```typescript
/**
 * Export to Excel Button
 * Location: /components/teacher-dashboard-new.tsx (Lines 1534-1540)
 */
<button
  onClick={exportToExcel}
  className="px-4 py-2 bg-green-600 text-white rounded-lg hover:bg-green-700 transition-colors flex items-center gap-2"
>
  <Download className="size-5" />
  <span>Export to Excel</span>
</button>
```

### Import Statement

```typescript
/**
 * XLSX Library Import
 * Location: /components/teacher-dashboard-new.tsx (Line 12)
 */
import * as XLSX from 'xlsx';
```

---

## ⚙️ HOW IT WORKS

### Workflow Diagram

```
┌─────────────────────────────────────────────────────┐
│                  EXCEL EXPORT WORKFLOW              │
└─────────────────────────────────────────────────────┘

1. TEACHER ACTIONS
   ├── Opens Reports Tab
   ├── Sets date filter (From/To)
   └── Clicks "Export to Excel" button
          ↓
2. FILTER RECORDS
   ├── Get all attendance records
   │   Filter by date range
````

Generate reports
````

APPENDIX D

Communications Letter & Approvals

APPROVAL SHEET

The Capstone Project entitled "**I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL"**, prepared and submitted by **Sean D. Pacanan, Vince Ardrei Pecayo,** and **Del Johnver C. Pascual** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development, is hereby recommended for approval and acceptance.

SHERWIN B. SAPIN
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

CRISANTO B. GULAY
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted as partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology, specialized in Web and Mobile Application Development at the Laguna State Polytechnic University-Los Baños Campus.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed. D.
Chairperson, Research and Development Services

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Dear Respondent,

Good day!

We are 4th-year students of the Bachelor of Science in Information Technology under the College of Computer Studies at **Laguna State Polytechnic University – Los Baños Campus**. As part of our academic requirement, we are conducting a research study and system development entitled **"I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL."** This study aims to evaluate the quality and performance of the developed system based on the **ISO 25010 Software Product Quality Model**, focusing on functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, portability, and safety.

We respectfully seek your honest and professional evaluation as an **IT Specialist or Education Professional**, based on the indicators provided in the questionnaire. Your expertise and feedback are essential in determining the system's effectiveness and quality in optimizing attendance monitoring through facial recognition technology. Rest assured that all information gathered will be treated with the utmost confidentiality in accordance with **Republic Act No. 10173 (Data Privacy Act of 2012)** and will be used solely for academic and research

Sincerely,
Del Johnver C. Pascual Vince Ardrei Pecayo Sean D. Pacanan
09516002134

ISO 25010 Software Product Quality Evaluation Rubric

AGE: _____ Gender: _____

Category of the Respondent:

☐ I.T Specialist

Rating Scale:

1 – Strongly Disagree 2 – Disagree 3 – Agree 4 – Strongly Agree

A. Perceived Usefulness (PU) *Measures how well the system improves productivity, efficiency, and usefulness to the user.*

ISO Construct	4 – Strongly Agree	3 – Agree	2 – Disagree	1 – Strongly Disagree
General Interpretation	The software quality characteristic is highly evident; the system performs excellently and meets ISO 25010	The software quality characteristic is evident; the system performs well and meets most ISO 25010 standards.	The software quality characteristic is somewhat evident; the system partially meets the standards and	The software quality characteristic is not evident; the system fails to meet ISO 25010 standards and

	standards in all aspects.		needs improvement.	requires major improvement.
Functional Suitability	The system fully supports all attendance-monitoring functions, provides highly accurate facial recognition, and effectively assists with student reporting.	The system performs most functions correctly and supports attendance management with minor issues.	The system performs limited functions and sometimes produces inaccurate facial recognition results.	The system does not meet functional requirements or support school attendance operations properly.
Performance Efficiency	The system operates quickly, processes group shot facial recognition in real time, and uses device resources efficiently.	The system performs most of the time efficiently, with minimal delay or lag during scanning.	The system occasionally slows down or consumes too many resources when processing large groups.	The system is slow, unresponsive, or inefficient in processing and resource use.
Compatibility	The system integrates perfectly with web browsers, school databases, and cloud services without conflict.	The system connects properly with most devices and applications with few compatibility issues.	The system sometimes struggles to exchange data with the database or other school software.	The system is not compatible with other school devices or administrative systems.
Usability (Interaction Capability)	The system is user-friendly, inclusive for all technical levels, and provides clear guidance for teachers and students.	The system is generally easy to use and learn, with few usability issues.	The system is usable but lacks clear feedback, error messages, or guidance features.	The system is difficult to learn and operate, and its user interface is poor for attendance tasks.
Reliability	The system operates consistently, remains available for every class session, and recovers quickly from software errors.	The system performs reliably with occasional minor interruptions.	The system experiences moderate failures or slow recovery from processing errors.	The system is unreliable and frequently fails or loses attendance data.
Security	The system ensures student data confidentiality, the integrity of attendance records, and protection against unauthorized access.	The system protects student data and ensures secure teacher access most of the time.	The system has limited protection and may experience minor security or privacy issues.	The system is vulnerable to security threats and lacks proper data protection for student profiles.
Maintainability	The system is modular, easy to analyze for errors, and can be modified or updated without causing new bugs.	The system can be maintained and updated with minor effort from the developers.	The system is difficult to modify and may cause errors during updates to the student list.	The system is not maintainable or easily upgradable for future school years.

Portability (Flexibility)	The system is adaptable to different phones/tablets, scales for large classes, and is easy to set up for school use.	The system adapts well to most browsers/environments and can be updated smoothly.	The system needs technical adjustments to adapt to different devices or scale effectively.	The system cannot be adapted or replaced without significant disruption to school operations.
Safety	The system operates safely, identifies risks in facial recognition, and effectively protects student data and privacy.	The system provides adequate operational safety and alerts users to potential recognition problems.	The system has limited safety measures and delayed error alerts for teachers.	The system lacks safety features and poses risks to data accuracy and school operations.

_____ _____

Signature over printed name Date Signed
Respondent

LAGUNA STATE POLYTECHNIC UNIVERSITY- LOS BAÑOS CAMPUS
COLLEGE OF COMPUTER STUDIES

Dear Respondent,

Good day!

We are 4th-year students of the Bachelor of Science in Information Technology under the College of Computer Studies at **Laguna State Polytechnic University – Los Baños Campus**. We are currently conducting a research study and system development entitled **"I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL."**

This study aims to determine the acceptability of the developed technology through the **Technology Acceptance Model (TAM)**. We request your honest responses using the indicators provided below. Please refer to the description of each score to guide your evaluation. All information collected will be treated with the highest confidentiality in accordance with **Republic Act No. 10173 (Data Privacy Act of 2012)** and will be used solely for academic research purposes.
Thank you for your time and cooperation.

Sincerely,
Del Johnver C. Pascual, Vince Ardrei Pecayo, Sean D. Pacanan
09516002134

Technology Acceptance Model Rubric

AGE: _____ Gender: _____

Category of the Respondent:

☐ Teacher

Rating Scale:

1 – Strongly Disagree 2 – Disagree 3 – Agree 4 – Strongly Agree

A. Perceived Usefulness (PU)

Measures how well the system improves productivity, efficiency, and usefulness to the user.

TAM Construct	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
Perceived Usefulness (PU)	The system greatly improves attendance monitoring, reduces manual errors, and provides useful data for school records.	The system helps improve school productivity and assists in student monitoring.	The system provides minimal help in monitoring or improving attendance tracking.	The system does not contribute to better attendance management or efficiency.
Perceived Ease of Use (PEOU)	The system is user-friendly and easy to learn, including the	The system is fairly easy to use and can be operated with minimal guidance.	The system is somewhat difficult to understand or navigate during	The system is complicated and hard to use for daily attendance.

	group-shot facial-recognition feature.		attendance checking.	
Personal Innovation	The user is eager to explore and apply new AI-based systems to enhance school operations.	The user is open to using modern systems and innovative technology for the school.	The user rarely tries new technologies and prefers traditional manual attendance methods.	The user avoids new technologies and prefers not to change current attendance practices.
Social Influence	Positive feedback and support from school leaders and peers strongly influence the user to adopt the system.	The user is encouraged by fellow teachers or colleagues to try the system.	The user receives limited social encouragement or support for using the app.	The user's environment does not support or influence the use of the technology.
Information Awareness	The user has a full understanding of how group-shot facial recognition works and its benefits.	The user understands most of the system's functions and benefits.	The user has limited awareness of how the attendance system functions.	The user is not aware or informed about the system's purpose and features.
Perceived Cost (If applicable to school resources)	The system's resource requirements are reasonable and justified by its effectiveness in saving time.	The resource/time investment is acceptable considering the benefits.	The "cost" in terms of time or setup is high compared to the benefits.	The system is not worth the effort/cost and discourages adoption.
Willingness to Adopt	The user is fully ready to adopt, use, and promote the system for school-wide attendance monitoring.	The user is willing to use and recommend the system to other teachers/sections.	The user is hesitant to adopt the system for their classroom.	The user is not willing to adopt or apply the system.

Signature over printed name

Date Signed

Republic of the Philippines

Laguna State Polytechnic University

Province of Laguna

PARENT/GUARDIAN CONSENT FORM

Project Title: I AM PRESENT: A Web-Based Application Optimizing Students' Attendance Monitoring Through Group Shot-Based Facial Recognition at Cabuyao Integrated National High School

Researchers: Fourth Year BS Information Technology Students, Laguna State Polytechnic University (LSPU) Los Baños Campus.

Testing Venue: Cabuyao Integrated National High School

Purpose of the Study

This study aims to develop and test a web-based attendance monitoring system using group photo capture and facial recognition technology. The project is conducted for academic purposes only as part of the researchers' capstone requirement.

Information to Be Collected

Student: Full Name, Email Address, Facial Image (for attendance identification) Parent/Guardian: Full Name, Email Address, Contact Number.

Confidentiality and Data Privacy

All collected data will be treated with strict confidentiality and used solely for system testing and academic research. This study complies with the Data Privacy Act of 2012 (RA 10173). Data will be securely stored and accessed only by authorized researchers.

Voluntary Participation

Participation is voluntary. Parents or guardians may withdraw consent at any time without penalty.

Consent Statement

I confirm that I am the parent/legal guardian of the student named below and voluntarily give permission for my child to participate in this study.

Student Name: Lady Emery V. Tena

Parent/Guardian Name: Emerzon A. Tena
Relationship: Father
Email Address: emerytena71@gmail.com
Contact No.: 09632575750
Parent/Guardian Signature: _____
Date: January 8, 2026

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

January 7, 2026

YOLLY D. VALIENTE
Principal IV
Cabuyao Integrated National High School
Cabuyao City, Laguna

Dear Madam,

Warmest greetings!

We are students of Laguna State Polytechnic University - Los Banos Campus under the College of Computer Studies, currently conducting a research study on the project entitled **"I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL"** this 1st semester AY 2025-26. We are on the stage of data gathering

In this regard, we would like to respectfully request permission from your good office to allow us to **gather data and conduct an experiment comparing the manual and automated attendance procedures in terms of speed and accuracy**, with **Grade 12 Senior High School** students as the participants of the study.

Please be assured that all information provided will be treated with the utmost confidentiality, in accordance with RA no. 10173 (Data Privacy Act of 2012), and will be used strictly for academic purposes only.

We hope for you unwavering support and approval of this humble request. Thank you very much, and may Godspeed!

Very respectfully yours

PACANAN, SEAN D.

PECAYO, VINCE ARDREI

PASCUAL, DEL JOHNVER C.

Noted by:

SHERWIN B. SAPIN, EdD
CCS Professor / Research Adviser

Approved by:

YOLLY D. VALIENTE
Principal IV
CINHS

APPENDIX E

Plagiarism Result

Grammarly Result

I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGH SCHOOL

by Ronah joy Ramirez

General metrics

95,229	12,909	735	51 min 38 sec	1 hr 39 min
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

1	1	✓
Issues left	Critical	Advanced

Plagiarism

✓ This text seems 100% original. Grammarly found no matching text on the Internet or in ProQuest's databases.

Turnitin Result

I AM PRESENT

ORIGINALITY REPORT

12%	8%	7%	8%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Laguna State Polytechnic University Student Paper	2%
2	pdfcoffee.com Internet Source	1%
3	Submitted to Our Lady of Fatima University Student Paper	1%
4	Submitted to Manuel S. Enverga University Foundation-Candelaria Student Paper	1%
5	Gene Marck B. Catedrilla. "Mobile-Based Navigation Assistant for Visually Impaired Person with Real-time Obstacle Detection Using YOLO-based Deep Learning Algorithm", 2022 5th Asia Conference on Machine Learning and Computing (ACMLC), 2022 Publication	<1%
6	Submitted to Jose Rizal University Student Paper	<1%
7	Submitted to Our Lady of Fatima University-Valenzuela Student Paper	<1%
8	www.coursehero.com Internet Source	<1%

APPENDIX F

Documentation

System Testing





Defense



APPENDIX G

Publishable Paper

I AM PRESENT: A WEB-BASED APPLICATION OPTIMIZING STUDENTS' ATTENDANCE MONITORING THROUGH GROUP SHOT-BASED FACIAL RECOGNITION AT CABUYAO INTEGRATED NATIONAL HIGHSCHOOL

Sean D. Pacanan, Del Johnver C. Pascual, Vince Ardrei Pecayo
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Article history:

Received by
Revised by
Accepted by

Keywords:

Facial Recognition
Attendance Monitoring
Group shot Recognition
Student Attendance

ABSTRACT

This study presents I AM PRESENT, a web-based attendance monitoring system that uses group-shot–based facial recognition to improve the efficiency and accuracy of student attendance recording at Cabuyao Integrated National High School. The system employs MTCNN for multi-face detection and FaceNet for facial feature extraction and identification from a single classroom group image. Attendance is recorded in real time, securely stored in a cloud database, and automatically communicated to parents or guardians through email notifications. The system also supports report generation and CSV export for administrative use. Evaluation based on ISO 25010 software quality standards and the Technology Acceptance Model (TAM) indicates that the system is reliable, user-friendly, and effective in reducing manual effort, minimizing errors, and preventing proxy attendance. The results demonstrate that group-based facial recognition is a practical and efficient solution for modernizing attendance monitoring in secondary schools.

I. INTRODUCTION

Accurate student attendance monitoring is essential to academic performance and school safety, yet many secondary schools continue to rely on manual methods such as roll calls and written records. These approaches are time-consuming, prone to human error, and vulnerable to proxy attendance. At Cabuyao Integrated National High School, traditional attendance practices also delay record processing and limit timely communication with parents or guardians. Recent advances in computer vision and deep learning, particularly facial recognition technologies, have enabled automated and reliable identity verification across various domains, including education. Models such as MTCNN and FaceNet provide robust face detection and recognition capabilities even in group and real-time scenarios. Integrating these technologies into attendance systems offers the potential to improve accuracy, efficiency, and accountability.

This study addresses existing attendance-monitoring challenges by developing I AM PRESENT, a web-based system that records attendance via group-shot facial recognition, automates reporting, and provides real-time parental notifications. The system aims to reduce classroom disruption, improve record reliability, and enhance communication between schools and families while complying with recognized software quality standards.

II. RELATED LITERATURE

Facial recognition technology has gradually evolved from basic geometric and statistical techniques to more advanced deep learning–based approaches, leading to noticeable improvements in accuracy and reliability in real-world conditions [1]. Earlier methods, including Eigenfaces and Fisher faces, were effective only in controlled environments and often struggled with changes in lighting, facial expressions, and head position, which limited their practical use [2].

With the introduction of Convolutional Neural Networks (CNNs), facial recognition systems became more capable of learning complex facial features directly from image data [3]. One widely used model for face detection is the Multi-task Cascaded Convolutional Neural Network (MTCNN), which is known for its ability to detect multiple faces and facial landmarks within a single image. This capability makes it particularly suitable for group image scenarios, such as classroom settings [4].

For facial recognition and verification, FaceNet has been shown to produce reliable results by converting facial images into numerical embeddings that represent unique facial features [5]. Studies have demonstrated that combining MTCNN for face detection with FaceNet for feature extraction leads to improved recognition accuracy and system efficiency, even under varying conditions [6].

In educational institutions, facial recognition–based attendance systems have been widely studied and implemented to address the limitations of manual attendance monitoring. Existing research indicates that these systems help reduce time spent on attendance checking, prevent proxy attendance, and improve the overall accuracy of attendance records [7]. Additionally, automated attendance systems allow teachers and administrators to access attendance data in real time, which supports better monitoring of student participation [8].

Despite the advantages of facial recognition technology, several studies point out challenges that must be considered. Factors such as poor lighting, partial face occlusion, and changes in facial appearance can affect recognition accuracy [9]. Moreover, ethical concerns related to data privacy, user consent, and the secure handling of biometric information are frequently discussed, especially when such systems are applied in school environments involving minors [10].

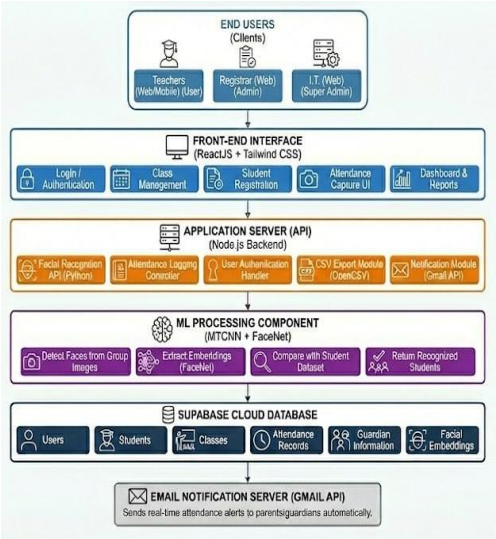
III. METHODOLOGY

Project Design

The proposed I AM PRESENT system adopts a modular architecture that integrates web-based interfaces, machine learning, cloud storage, and automated notifications to optimize student attendance monitoring. As illustrated in Figure 2, the architecture consists of the End Users Layer, Front-End Interface, Application Server, Machine Learning Processing Component, Cloud Database, and Email Notification Server. The End Users Layer includes teachers, registrars, and IT administrators, providing role-based access to attendance recording, student management, and system administration. The Front-End Interface, developed using ReactJS and Tailwind CSS, enables user authentication, class and student management, group image attendance capture, and dashboard-based reporting. The Application Server functions as

the core processing unit, handling user authentication, attendance logging, facial recognition requests, data export, and notification services. The Machine Learning Processing Component uses MTCNN for multi-face detection and FaceNet for facial recognition, enabling automatic identification of students in group images and real-time attendance recording. All system data, including user profiles, student records, attendance logs, and facial embeddings, is securely stored in a cloud-based database to ensure consistency and accessibility. An Email Notification Server automatically sends attendance alerts to parents and guardians, enhancing school–parent communication. Overall, the integrated architecture supports real-time processing, reliability, and automation, significantly reducing administrative workload while ensuring accurate and efficient student attendance monitoring.

Figure 1. System Architecture



Project Development

The I AM PRESENT web-based attendance monitoring system was developed using the Agile Scrum methodology, as shown in Figure 3, to ensure a flexible, organized, and efficient approach. By dividing the project into short, focused sprints, the team was able to concentrate on specific tasks in each phase, such as user login, student face registration, group photo capture, facial recognition with MTCNN and FaceNet, and attendance reporting. After each sprint, the system was reviewed and tested to identify any issues, make improvements, and confirm that the features met the needs of both teachers and administrators. This iterative process allowed for continuous refinement and ensured that the final system was reliable, easy to use, and well-suited to improving attendance monitoring at Cabuyao Integrated National High School.



Figure 2. Agile Scrum Methodology

IV. RESULTS AND DISCUSSION

Web-Based Attendance Monitoring and Real-Time Parent/Guardian Notification.

Figure 3 illustrates the classroom implementation of the group face-detection attendance system. Figure 3(a) shows a single group image captured during class sessions, allowing the system to record attendance without interrupting ongoing instruction. Multiple student faces are detected simultaneously despite variations in seating position and orientation. Figure 3(b) presents the facial recognition and verification interface, where detected faces are matched against pre-registered student profiles stored in the system database. Each identified student is displayed with a corresponding confidence score and attendance status, enabling teachers to review and validate recognition results before final submission. This verification step ensures accuracy while maintaining efficiency in the attendance monitoring process.

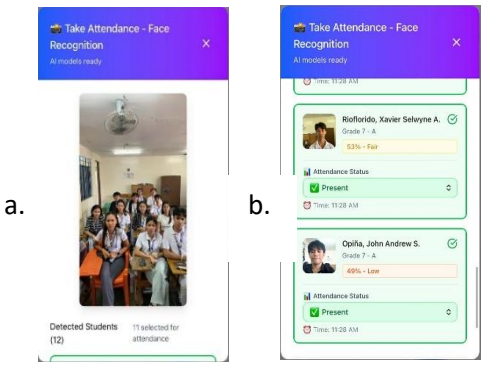


Figure 3. Attendance and Email Notification

Figure 4 illustrates the classroom implementation of the group face-detection attendance system. (a) shows the capture of a single group image during class sessions, allowing attendance to be recorded without disrupting the teaching process. The system detects multiple student faces simultaneously despite variations in seating positions and orientations. (b) presents the facial recognition and verification interface, in which detected faces are matched against pre-registered student profiles stored in the database. Each identified student is displayed with a corresponding confidence score and attendance status, enabling teachers to review and confirm the recognition results before final submission, thereby ensuring both accuracy and efficiency in attendance monitoring.

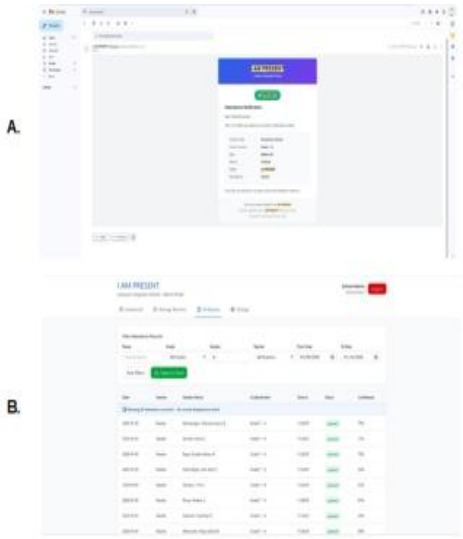


Figure 4. Attendance and Email Notification

Attendance Report Generation and Exporting

Figure 5 demonstrates the system's report generation capability, highlighting the Export to Excel feature for downloading attendance records. Figure 18(a) shows how administrators can generate attendance reports for a selected class or date range with a single action. The exported file contains structured data such as student names, grade levels, attendance status, and guardian contact information, which supports record-keeping and communication when necessary. Figure 18(b) presents the generated Excel file, which displays a clean, organized layout that clearly summarizes each student's attendance history. This automated reporting function reduces manual data entry, improves record accuracy, and supports efficient administrative and classroom management.

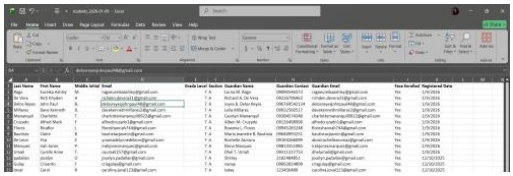


Figure 5. Export Attendance

Absent notification in email

Figure 6 illustrates the automated absence notification generated by the I AM PRESENT system and sent to parents or guardians when a student is marked absent. The notification contains essential information, including the student's name, class section, date, and attendance status, to ensure clarity and accuracy. This real-time alert mechanism enables immediate awareness of student absences, reducing delays associated with manual reporting. By automating parental notifications, the system enhances transparency, strengthens school–home communication, and supports timely parental responses, while also reducing teachers' administrative workload through consistent, reliable information dissemination.

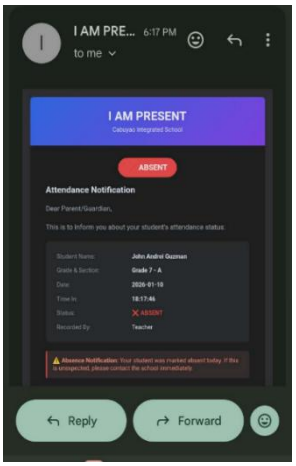


Figure 6. Absent Notification

Results of ISO 25010 and Technology and Acceptance Model.

The table presents the results of a survey conducted with three IT experts to evaluate the acceptability of the developed system. Based on the evaluation, the system demonstrates readiness for deployment to users. The overall mean rating of the system is 3.36, which falls under the "Strongly Agree" category. This indicates that the system meets

the quality criteria outlined in ISO 25010, confirming its functionality, efficiency, compatibility, reliability, security, maintainability, and interaction capabilities. At the same time, flexibility and safety are rated as "Agree."

Table 1. Results of ISO 25010

Indicators	Mean	Standard Deviation	Interpretation
Functional Suitability	3.73	0.48	Strongly Agree
Performance Efficiency	3.47	0.48	Strongly Agree
Compatibility	3.50	0.55	Strongly Agree
Interaction Capability	3.28	0.53	Strongly Agree
Reliability	3.45	0.55	Strongly Agree
Security	3.27	0.41	Strongly Agree
Maintainability	3.30	0.31	Strongly Agree
Flexibility	3.18	0.34	Agree
Safety	3.12	0.20	Agree
Overall Mean	**3.36**	**0.43**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25 Agree; 3.26-4.0: Strongly Agree

Table 2 presents the Technology Acceptance Model (TAM) assessment results from 50 teacher respondents. All indicators received mean scores ranging from 3.39 to 3.55, with most interpreted as "Strongly Agree." The highest-rated factor is Perceived Usefulness (3.55), indicating that users recognize the system as beneficial and supportive of their tasks. Other indicators, including Perceived Ease of Use, Personal Innovation, Social Influence, Information Awareness, and Willingness to Adopt, also reflect strong positive perceptions, suggesting that the system is user-friendly, encourages innovation, and is socially acceptable within the teaching environment. Although Perceived Cost has the lowest mean (3.54) and is interpreted as "Agree," it still shows that teachers consider the system affordable or justified relative to its benefits. Overall, the computed mean of 3.49 confirms that teachers strongly accept the system across all TAM dimensions, highlighting both its practicality and potential for widespread adoption in educational settings.

Indicators	Mean	Standard Deviation	Interpretation
Perceived Usefulness	3.55	0.54	Strongly Agree
Perceived Ease of Use	3.39	0.57	Strongly Agree
Personal Innovation	3.46	0.55	Strongly Agree
Social Influence	3.46	0.52	Strongly Agree
Information Awareness	3.54	0.50	Strongly Agree
Perceived Cost	3.54	0.50	Agree
Willingness to Adopt	3.46	0.57	Strongly Agree
Overall Mean	**3.49**	**0.54**	**Strongly Agree**

Legend: 1.0-1.75: Strongly Disagree; 1.76-2.50: Disagree; 2.51-3.25 Agree; 3.26-4.0: Strongly Agree

V. SUMMARY, CONCLUSION, AND RECOMMENDATION

This study developed and evaluated "I AM PRESENT," a web-based attendance system utilizing group-shot facial recognition to enhance accuracy and efficiency at Cabuyao Integrated National High School. The system leverages the MTCNN and FaceNet algorithms to automatically detect and identify multiple students in a single class photograph, securely stores records in a cloud-based database, and provides real-time notifications to parents. Evaluation based on ISO 25010 standards and the Technology Acceptance Model (TAM) demonstrated that the system is reliable, user-friendly, and highly accepted by both IT specialists and educators. The results indicate that automated facial recognition significantly reduces the need for manual intervention in classroom management. The study concludes that this technology is an effective solution for automating school attendance and recommends future enhancements, including attendance analytics, expanded notification channels, offline capabilities, and ethical data management frameworks, to improve academic administrative processes further.

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APPENDIX H

Curriculum Vitae



About Me

I am a 4th year Bachelor of Science in Information Technology student at Laguna State Polytechnic University, focused on developing my technical skills and applying technology to practical solutions. I have supporting work experience in administrative and production environments, which has strengthened my teamwork, adaptability, and attention to detail. I am motivated to continue learning and to contribute effectively in a professional IT setting.

Contact

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Skills

- Basic Web Development (HTML, CSS, JavaScript)
- Database Fundamentals and Data Handling
- System Documentation and Data Encoding
- Basic Troubleshooting and Technical Support
- Understanding of Software Development Process
- User Interface (UI) Awareness and Design Basics
- Attention to Detail and Logical Thinking
- Teamwork and Collaboration in Project-Based Tasks
- Time Management and Adaptability in Fast-Paced Environments

VINCE ARDREI PECAYO

Objectives

To obtain an On-the-Job Training position where I can apply my knowledge in Information Technology, enhance my technical and professional skills, and gain hands-on experience in a real-world work environment while contributing positively to the organization.

Education

(2018-2019)

ASIAN INSTITUTE OF COMPUTER STUDIES
Information & Communication Technology (BSIT)

(2016-2017)

MAKILING NATIONAL HIGH SCHOOL

Professional Experience

(2018)

NATIONAL ECONOMIC AND DEVELOPMENT AUTHORITY (NEDA)
Work Immersion (160 Hours)

- Data-Encoder / Finance Administrative Office
- Purok 06, Brgy Milagrosa Calamba City ,Laguna

(2019-2022)

C-PAK PTE LTD
3 years Agency Regular

- Production Support/ Material Handler
- Carmelray Industrial Park II, Brgy Milagrosa Calamba ,Laguna 4027 Philippines



About Me

A motivated 4th-year BSIT student with hands-on experience in web-based system development and database management. Skilled in developing academic and real-world projects including attendance systems and inventory systems. Passionate about learning new technologies and applying IT solutions to solve real-life problems.

Contact

+6939516002134

johnverzpascual@gmail.com

Calamba hills, Brgy. Mayapa, Laguna, Philippines

Skills

- Problem-Solving
- Team Collaboration
- Time Management
- Presentation Skills
- Networking
- Web-Based System Development
- Basic UI/UX Design (figma)
- System Analysis and Design

Language

- English
- Filipino

DEL JOHNVER PASCUAL

Objectives

To gain practical experience in the IT industry by applying my academic knowledge and technical skills in a professional working environment.

Education

(2019 -2021)

CAMP VICENTE LIM INTEGRATED SCHOOL

Science, Technology, Engineering, and Mathematics (STEM)

(2022 -2026)

LSPU LOS BAÑOS CAMPUS

Bachelor of Information and Technology (BSIT)

Experience

(2024-2025)

BAKERY

- Deliver baked bread
- Making bread

Trainings

- TESDA Creative Web Design 2024
- IBM skill build 2025



About Me

A detail-oriented 4th-year BSIT student with hands-on experience in web application development and database design. Proficient in building academic and real-world systems, including attendance and inventory management applications. Passionate about adopting new technologies and delivering efficient IT solutions to practical problems.

Contact

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Skills

- Problem-Solving
- Team Collaboration
- Time Management
- Presentation Skills
- Networking
- Web-Based System Development
- Basic UI/UX Design (figma)
- System Analysis and Design

Language

- English
- Filipino

PACANAN SEAN

Objectives

To become a competent IT professional by gaining hands-on experience, expanding my technical expertise, and contributing innovative solutions to organizational challenges.

Education

(2019 -2021)

HOLY REDEEMER CALAMBA

Block 3, Lot 206B, Hacienda Hill, Palo-Alto, Calamba City, Laguna, Philippines.

(2022 -2026)

JHS - RIZAL INSTITUE INC.

Brgy. Jose Yulo Sr. Ave, Calamba, Laguna

(2022 -2026)

SHS - PALO ALTO INTEGRATED SCHOOL

Barangay Palo Alto, Calamba City, Laguna,

(2022 -2026)

LSPU LOS BAÑOS CAMPUS

Bachelor of Information and Technology (BSIT)